AI & COMPUTING

Introduction to Infinitesimal Analysis; Functions of One Real Variable

Oswald Veblen and N. J. Lennes

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Transcriber’s Notes. A large number of printer errors have been corrected. These are shaded like this , and details can be found in the source code in the syntax \ correction{corrected}{original}. In addition, the formatting of a few lem- mas, corollaries etc. has been made consistent with the others. The unusual inequality sign >= used a few times in the book in addition to has been preserved, although it may reflect the printing rather than the author’s intention. The notation || a b for intervals is not in common use today, and the reader able to run LATEX will find it easy to redefine this macro to give a modern equivalent. Similarly, the original did not mark the ends of proofs in any way and so nor does this version, but the reader who wishes can easily redefine \ qedsymbol in the source.

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INTRODUCTION TO INFINITESIMAL ANALYSIS FUNCTIONS OF ONE REAL V ARIABLE BY OSW ALD VEBLEN Preceptor in Mathematics, Princeton University And N. J. LENNES Instructor in Mathematics in the Wendell Phillips High School, Chicago FIRST EDITION FIRST THOUSAND NEW YORK JOHN WILEY & SONS London: CHAPMAN & HALL, Limited 1907

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PREF ACE A course dealing with the fundamental theorems of infinitesimal calculus in a rigorous manner is now recognized as an essential part of the training of a mathematician. It appears in the curriculum of nearly every university, and is taken by students as “Advanced Calculus” in their last collegiate year, or as part of “Theory of Functions” in the first year of graduate work. This little volume is designed as a convenient reference book for such courses; the examples which may be considered necessary being supplied from other sources. The book may also be used as a basis for a rather short theoretical course on real functions, such as is now given from time to time in some of our universities. The general aim has been to obtain rigor of logic with a minimum of elaborate machin- ery. It is hoped that the systematic use of the Heine-Borel theorem has helped materially toward this end, since by means of this theorem it is possible to avoid almost entirely the sequential division or “pinching” process so common in discussions of this kind. The definition of a limit by means of

the notion “value approached” has simplified the proofs of theorems, such as those giving necessary and sufficient conditions for the existence of limits, and in general has largely decreased the number of ε’s and δ’s. The theory of limits is developed for multiple-valued functions, which gives certain advantages in the treatment of the definite integral. In each chapter the more abstract subjects and those which can be omitted on a first reading are placed in the concluding sections. The last chapter of the book is more advanced in character than the other chapters and is intended as an introduction to the study of a special subject. The index at the end of the book contains references to the pages where technical terms are first defined. When this work was undertaken there was no convenient source in English containing a rigorous and systematic treatment of the body of theorems usually included in even an elementary course on real functions, and it was necessary to refer to the French and German treatises. Since then one treatise, at least, has appeared in English on the Theory of Functions of Real Variables. Nevertheless it is hoped that the present volume, on account of its conciseness, will supply a real want. The authors are much indebted to Professor E. H. Moore of the University of Chicago for many helpful criticisms and suggestions; to Mr. E. B. Morrow of Princeton University for reading the manuscript and helping prepare the cuts; and to Professor G. A. Bliss of Princeton, who has suggested several desirable changes while reading the proof-sheets. iii

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Chapter 1 THE SYSTEM OF REAL NUMBERS. § 1 Rational and Irrational Numbers. The real number system may be classified as follows: (1) All integral numbers, both positive and negative, including zero. (2) All numbers m n , where m and n are integers ($n \neq 0$). (3) Numbers not included in either of the above classes, such as $\sqrt{2}$ and π.1 Numbers of classes (1) and (2) are called rational or commensurable numbers, while the numbers of class (3) are called irrational or incommensurable numbers. As an illustration of an irrational number consider the square root of 2. One ordinarily says that $\sqrt{2}$ is 1 . 4+, or 1 . 41+, or 1 . 414+, etc. The exact meaning of these statements is expressed by the following inequalities: 2 (1. 4)2 < 2 < (1. 5)2, (1. 41)2 < 2 < (1. 42)2, (1. 414)2 < 2 < (1. 415)2, etc. Moreover, by the foot-note above no terminating decimal is equal to the square root of 2. Hence Horner’s Method, or the usual algorithm for extracting the square root, leads to an 1It is clear that there is no number m n such that m2 n2 = 2, for if m2 n2 = 2, then m2 = 2 n2, where m2 and 2 n2 are integral numbers, and 2 n2 is the square of the integral number m. Since in the square of an integral number every prime factor occurs an even number of times, the factor 2 must occur an even number of times both in n2 and 2 n2, which is impossible because of the theorem that an integral number has only one set of prime factors. 2a < b signifies that a is less than b. a > b signifies that a is greater than b. 1

2 INFINITESIMAL ANALYSIS. infinite sequence of rational numbers which may be denoted by a1, a 2, a 3, . . . , a n, . . . (where a1 = 1. 4, a2 = 1. 41, etc.), and which has the property that for every positive integral value of n an≤an+1, a 2 n < 2 < (an + 1 10n)2 . Suppose, now, that there is a least number a greater than every an. We easily see that if the ordinary laws of arithmetic as to equality and inequality and addition, subtraction, and multiplication hold for a and a2, then a2 is the rational number 2. For if a2 < 2, let 2−a2 = ε, whence 2 = a2 + ε. If n were so taken that 1 10n < ε 5 , we should have from the last inequality 3 2 < (an + 1 10n)2 = a2 n + 2an·1 10n + (1 10n)2 < a 2 n + 4 ε 5 + ε 5 < a 2 + ε, so that we should have both 2 = a2 + ε and 2 < a 2 + ε. On the other hand, if a2 > 2, let a2−2 = ε′or 2 + ε′= a2. Taking n such that 1 10n < ε 5 , we should have (an + 1 10n)2 < (a2 n) + ε′< 2 + ε′< a; and since an + 1 10n is greater than ak for all values of k, this would contradict the hypothesis that a is the least number greater than every number of the sequence a1, a 2, a 3, . . . We also see without difficulty that a is the only number such that a2 = 2. § 2 Axiom of Continuity. The essential step in passing from ordinary rational numbers to the number corresponding to the symbol √ 2 is thus made to depend upon an assumption of the existence of a number a bearing the unique relation just described to the sequence a1, a 2,a3 ,. . . In order to state this hypothesis in general form we introduce the following definitions: Definition.— The notation [x] denotes a set,4 any element of which is denoted by x alone, with or without an index or subscript. A set of numbers [x] is said to have an upper bound , M , if there exists a number M such that there is no number of the set greater than M . This may be denoted by M □ [x]. A set of numbers [x] is said to have a lower bound, m, if there exists a number m such that no number of the set is less than m. This we denote by m □ [x]. 3This involves the assumption that for every number, ε, however small there is a positive integer n such that 1 10n < ε 5 . This is of course obvious when ε is a rational number. If ε is an irrational number, however, the statement will have a definite meaning only after the irrational number has been fully defined. 4Synonyms of set are class, aggregate, collection, assemblage, etc.

THE SYSTEM OF REAL NUMBERS. 3 Following are examples of sets of numbers: (1) 1, 2, 3. (2) 2, 4, 6, . . . , 2k, . . . (3) 1/ 2, 1/ 22, 1/ 23, . . . , 1/ 2n, . . . (4) All rational numbers less than 1. (5) All rational numbers whose squares are less than 2. Of the first set 1, or any smaller number, is a lower bound and 3, or any larger number, is an upper bound. The second set has no upper bound, but 2, or any smaller number, is a lower bound. The number 3 is the least upper bound of the first set, that is, the smallest number which is an upper bound. The least upper and the greatest lower bounds of a set of numbers [x] are called by some writers the upper and lower limits respectively. We shall denote them by B[x] and B[x] respectively. By what precedes, the set (5) would have no least upper bound unless √ 2 were counted as a number. We now state our hypothesis of continuity in the following form: Axiom K.— If a set [r] of rational numbers having an upper bound has no rational least upper bound, then there exists one and only one number B[r] such that (a) B[r] > r′, where r′is any number of [r] or any rational number less than some number of [r]. (b) B[r] < r″, where r″is any rational upper bound of [r].5 Definition.— The number B[r] of axiom K is called the least upper bound of [r], and as it cannot be a rational number it is

called an irrational number. The set of all rational and irrational numbers so defined is called the continuous real number system . It is also called the linear continuum. The set of all real numbers between any two real numbers is likewise called a linear continuum. Theorem 1. If two sets of rational numbers [r] and [s], having upper bounds, are such that no r is greater than every s and no s greater than every r, then B[r] and B[s] are the same; that is, in symbols, B[r] = B[s]. Proof. If B[r] is rational, it is evident, and if B[r] is irrational, it is a consequence of Axiom K that B[r] > s′, 5This axiom implies that the new (irrational) numbers have relations of order with all the rational numbers, but does not explicitly state relations of order among the irrational numbers themselves. Cf. Theorem 2.

4 INFINITESIMAL ANALYSIS. where s′is any rational number not an upper bound of [s]. Moreover, if s″is rational and greater than every s, it is greater than every r. Hence B[r] < s″, where s″is any rational upper bound of [s]. Then, by the definition of B[s], B[r] = B[s], Definition.— If a number x (in particular an irrational number) is the least upper bound of a set of rational numbers [r], then the set [r] is said to determine the number x. Corollary 1. The irrational numbers i and i′determined by the two sets [r] and [r′] are equal if and only if there is no number in either set greater than every number in the other set. Corollary 2. Every irrational number is determined by some set of rational numbers. Definition.— If i and i′are two irrational numbers determined respectively by sets of rational numbers [r] and [r′] and if some number of [r] is greater than every number of [r′], then i > i′and i′< i. From these definitions and the order relations among the rational numbers we prove the following theorem: Theorem 2. If a and b are any two distinct real numbers, then a < b or b < a ; if a < b , then not b < a ; if a < b and b < c , then a < c . Proof. Let a, b, c all be irrational and let [x], [y], [z] be sets of rational numbers deter- mining a, b, c. In the two sets [x] and [y] there is either a number in one set greater than every number of the other or there is not. If there is no number in either set greater than every number in the other, then, by Theorem 1, a = b. If there is a number in [x] greater than every number in [y], then no number in [y] is greater than every number in [x]. Hence the first part of the theorem is proved, that is, either a = b or a < b or b < a , and if one of these, then neither of the other two. If a number y1 of [y] is greater than every number of [x], and a number z1 of [z] is greater than every number of [y], then z1 is greater than every number of [x]. Therefore if a < b and b < c , then a < c . We leave to the reader the proof in case one or two of the numbers a, b, and c are rational. Lemma.— If [r] is a set of rational numbers determining an irrational number, then there is no number r1 of the set [r] which is greater than every other number of the set. This is an immediate consequence of axiom K .

THE SYSTEM OF REAL NUMBERS. 5 Theorem 3. If a and b are any two distinct numbers, then there exists a rational number c such that a < c and c < b , or b < c and c < a . Proof. Suppose a < b . When a and b are both rational b−a 2 is a number of the required type. If a is rational and b irrational, then the theorem follows from the lemma and Corollary 2, page 4. If a and b are both irrational, it follows from Corollary 1, page 4. If a is irrational and b rational, then there are rational numbers less than b and greater than every number of the set [x] which determines a, since otherwise b would be the smallest rational number which is an upper bound of [x],

whereas by definition there is no least upper bound of [x] in the set of rational numbers. Corollary.— A rational number r is the least upper bound of the set of all numbers which are less than r, as well as of the set of all rational numbers less than r. Theorem 4. Every set of numbers [x] which has an upper bound, has a least upper bound. Proof. Let [r] be the set of all rational numbers such that no number of the set [r] is greater than every number of the set [x]. Then B[r] is an upper bound of [x], since if there were a number x1 of [x] greater than B[r], then, by Theorem 3, there would be a rational number less than x1 and greater than B[r], which would be contrary to the definition of [r] and B[r]. Further, B[r] is the least upper bound of [x], since if a number N less than B[r] were an upper bound of [x], then by Theorem 3 there would be rational numbers greater than N and less than B[r], which again is contrary to the definition of [r]. Theorem 5. Every set [x] of numbers which has a lower bound has a greatest lower bound. Proof. The proof may be made by considering the least upper bound of the set [y] of all numbers, such that every number of [y] is less than every number of [x]. The details are left to the reader. Theorem 6. If all numbers are divided into two sets [x] and [y] such that x < y for every x and y of [x] and [y], then there is a greatest x or a least y, but not both. Proof. The proof is left to the reader. The proofs of the above theorems are very simple, but experience has shown that not only the beginner in this kind of reasoning but even the expert mathematician is likely to make mistakes. The beginner is advised to write out for himself every detail which is omitted from the text. Theorem 4 is a form of the continuity axiom due to Weierstrass, and 6 is the so-called Dedekind Cut Axiom . Each of Theorems 4, 5, and 6 expresses the continuity of the real number system.

6 INFINITESIMAL ANALYSIS. § 3 Addition and Multiplication of Irrationals. It now remains to show how to perform the operations of addition, subtraction, multiplication, and division on these numbers. A definition of addition of irrational numbers is suggested by the following theorem: “If a and b are rational numbers and [x] is the set of all rational numbers less than a, and [y] the set of all rational numbers less than b, then [x + y] is the set of all rational numbers less than a + b.” The proof of this theorem is left to the reader. Definition.— If a and b are not both rational and [x] is the set of all rationals less than a and [y] the set of all rationals less than b, then a + b is the least upper bound of [x + y], and is called the sum of a and b. It is clear that if b is rational, [x + b] is the same set as [x + y]; for a given x + b is equal to x′+ (b−(x′−x)) = x′+ y′, where x′is any rational number such that x < x′< a; and conversely, any x + y is equal to (x−b + y) + b = x′′+ b. It is also clear that a + b = b + a, since [x + y] is the same set as [y + x]. Likewise (a + b) +c = a + (b + c), since [(x + y) +z] is the same as [x + (y + z)]. Furthermore, in case b < a , c = B[x′−y′], where a < x′< b and a < y′< b, is such that b + c = a, and in case b < a , c = B[x′−y′] is such that b + c = a; c is denoted by a−b and called the difference between a and b. The negative of a, or−a, is simply 0−a. We leave the reader to verify that if a > 0, then a + b > b , and that if a < 0, then a + b < b for irrational numbers as well as for rationals. The theorems just proved justify the usual method of adding infinite decimals. For example: π is the least upper bound of decimals like 3 . 1415, 3 . 14159, etc. Therefore π + 2 is the least upper bound of such numbers as 5 . 1415, 5 . 14159, etc. Also e is the least upper bound of 2 . 7182818, etc. Therefore π + e is the least

upper bound of 5, 5 . 8, 5 . 85, 5 . 859, etc. The definition of multiplication is suggested by the following theorem, the proof of which is also left to the reader. Let a and b be rational numbers not zero and let [x] be the set of all rational numbers between 0 and a, and [y] be the set of all rationals between 0 and b. Then if a > 0, b > 0, it follows that ab = B[xy]; a < 0, b < 0, " " " ab = B[xy]; a < 0, b > 0, " " " ab = B[xy]; a > 0, b < 0, " " " ab = B[xy]. Definition.— If a and b are not both rational and [x] is the set of all rational numbers between 0 and a, and [y] the set of all rationals between 0 and b, then if a > 0, b > 0, ab means B[xy]; if a < 0, b < 0, ab means B[xy]; if a < 0, b > 0, ab means B[xy]; if a > 0, b < 0, ab means B[xy]. If a or b is zero, then ab = 0. It is proved, just as in the case of addition, that ab = ba, that a(bc) = (ab)c, that if a is rational [ay] is the same set as [xy], that if a > 0, b > 0, ab > 0. Likewise the quotient a b

THE SYSTEM OF REAL NUMBERS. 7 is defined as a number c such that ac = b, and it is proved that in case a > 0, b > 0, then c = B [x y′] , where [y′] is the set of all rationals greater than b. Similarly for the other cases. Moreover, the same sort of reasoning as before justifies the usual method of multiplying non-terminated decimals. To complete the rules of operation we have to prove what is known as the distributive law, namely, that a(b + c) = ab + ac. To prove this we consider several cases according as a, b, and c are positive or negative. We shall give in detail only the case where all the numbers are positive, leaving the other cases to be proved by the reader. In the first place we easily see that for positive numbers e and f , if [t] is the set of all the rationals between 0 and e, and [T] the set of all rationals less than e, while [u] and [U] are the corresponding sets for f , then e + f = B[T + U] = B[t + u]. Hence if [x] is the set of all rationals between 0 and a, [y] between 0 and b, [z] between 0 and c, b + c = B[y + z] and hence a(b + c) = B[x(y + z)]. On the other hand ab = B[xy], ac = B[xz], and therefore ab + ac = B[(xy + xz)]. But since the distributive law is true for rationals, x(y+z) = xy+xz. Hence B[x(y+z)] = B[(xy+xz)] and hence a(b + c) = ab + ac. We have now proved that the system of rational and irrational numbers is not only continuous, but also is such that we may perform with these numbers all the operations of arithmetic. We have indicated the method, and the reader may detail that every rational number may be represented by a terminated decimal, ak10k + ak−110k−1 + . . . + a0 + a−1 10 + . . . + a−n 10n = akak−1 . . . a 0a−1a−2 . . . a −n, or by a circulating decimal, akak−1 . . . a 0a−1a−2 . . . a −i . . . a −ja−i . . . a −j . . . , where i and j are any positive integers such that i < j ; whereas every irrational number may be represented by a non-repeating infinite decimal, akak−1 . . . a 0a−1a−2 . . . a −n . . . The operations of raising to a power or extracting a root on irrational numbers will be considered in a later chapter (see page 41). An example of elementary reasoning with the symbol B[x] is to be found on pages 12 and 14. For the present we need only that xn, where n is an integer, means the number obtained by multiplying x by itself n times.

8 INFINITESIMAL ANALYSIS. It should be observed that the essential parts of the definitions and arguments of this section are based on the assumption of continuity which was made at the outset. A clear understanding of the irrational number and its relations to the rational number was first reached during the latter half of the last century, and then only after protracted study and much discussion. We have sketched

only in brief outline the usual treatment, since it is believed that the importance and difficulty of a full discussion of such subjects will appear more clearly after reading the following chapters. Among the good discussions of the irrational number in the English language are: H. P. Manning, Irrational Numbers and their Representation by Sequences and Series , Wiley & Sons, New York; H. B. Fine , College Algebra, Part I, Ginn & Co., Boston; Dedekind, Essays on the Theory of Number (translated from the German), Open Court Pub. Co., Chicago; J. Pierpont , Theory of Functions of Real Variables , Chapters I and II, Ginn & Co., Boston. § 4 General Remarks on the Number System. Various modes of treatment of the problem of the number system as a whole are possible. Perhaps the most elegant is the following: Assume the existence and defining properties of the positive integers by means of a set of postulates or axioms. From these postulates it is not possible to argue that if p and q are prime there exists a number a such that a·p = q or a = q p , i.e., in the field of positive integers the operation of division is not always possible. The set of all pairs of integers {m, n}, if{mk, nk}(k being an integer) is regarded as the same as {m, n}, form an example of a set of objects which can be added, subtracted, and multiplied according to the laws holding for positive integers, provided addition, subtraction, and multiplication are defined by the equations, 6 {m, n}⊗{p, q}={mp, nq} {m, n}⊕{p, q}={mq + np, nq}. The operations with the subset of pairs {m, 1}are exactly the same as the operations with the integers. This example shows that no contradiction will be introduced by adding a further axiom to the effect that besides the integers there are numbers, called fractions, such that in the extended system division is possible. Such an axiom is added and the order relations among the fractions are defined as follows: p q < m n if pn < qm. By an analogous example 7 the possibility of negative numbers is shown and an axiom 6The details needed to show that these integer pairs satisfy the algebraic laws of operation are to be found in Chapter I, pages 5–12, of Pierpont'sTheory of Real Functions . Pierpont'sexposition differs from that indicated above, in that he says that the integer pairs actually are the fractions. 7Cf. Pierpont, loc. cit., pages 12–19.

THE SYSTEM OF REAL NUMBERS. 9 assuming their existence is justified. This completes the rational number system and brings the discussion to the point where this book begins. Our Axiom K , which completes the real number system, assuming that every bounded set has a least upper bound, should, as in the previous cases, be accompanied by an example to show that no contradiction with previous axioms is introduced by Axiom K . Such an example is the set of all lower segments, a lower segment, S, being defined as any bounded set of rational numbers such that if x is a number of S, every rational number less than x is in S. For instance, the set of all rational numbers less than a rational number a is a lower segment. Of two lower segments one is always a subset of the other. We may denote that S is a subset of S′by the symbol S /olessthanS′. According to the order relation, /olessthan, every bounded set of lower segments [S] has a least upper bound, namely the lower segment, consisting of every number in any S of [S]. If S and T are lower segments whose least upper bounds are s and t, we may define S⊕T and S⊗T as those lower segments whose least upper bounds are s + t and s×t respectively. It is now easy to see that the set of lower segments contains a subset that satisfies the same conditions

as the rational numbers, and that the set as a whole satisfies axiom K . The legitimacy of axiom K from the logical point of view is thus established, since our example shows that it cannot contradict any previous theorem of arithmetic. Further axioms might now be added, if desired, to postulate the existence of imaginary numbers, e.g. of a number x for each triad of real numbers a, b, c, such that ax2 +bx+c = 0. These axioms are to be justified by an example to show that they are not in contradiction with previous assumptions. The theory of the complex variable is, however, beyond the scope of this book. § 5 Axioms for the Real Number System. A somewhat more summary way of dealing with the problem is to set down at the outset a set of postulates for the system of real numbers as a whole without distinguishing directly between the rational and the irrational number. Several sets of postulates of this kind have been published by E. V. Huntington in the 3d, 4th, and 5th volumes of the Transactions of the American Mathematical Society. The following set is due to Huntington.8 The system of real numbers is a set of elements related to one another by the rules of addition (+), multiplication (×), and magnitude or order (<) specified below. 8Bulletin of the American Mathematical Society, Vol. XII, page 228.

10 INFINITESIMAL ANALYSIS. A 1. Every two elements a and b determine uniquely an element a + b called their sum. A 2. (a + b) + c = a + (b + c). A 3. (a + b) = (b + a). A 4. If a + x = a + y, then x = y. A 5. There is an element z, such that z + z = z. (This element z proves to be unique, and is called 0.) A 6. For every element a there is an element a′, such that a + a′= 0. M 1. Every two elements a and b determine uniquely an element ab called their product; and if a = 0 and b = 0, then ab = 0. 9 M 2. (ab)c = a(bc). M 3. ab = ba. M 4. If ax = ay, and a = 0, then x = y. M 5. There is an element u, different from 0, such that uu = u. This element proves to be uniquely determined, and is called 1. M 6. For every element a, not 0, there is an element a″, such that aa″= 1. A M 1. a(b + c) = ab + ac. O 1. If a = b, then either a < b or b < a . O 2. If a < b , then a = b. O 3. If a < b and b < c , then a < c . O 4. (Continuity.) If [x] is any set of elements such that for a certain element b and every x, x < b , then there exists an element B such that— (1) For every x of [x], x < B; (2) If y < B, then there is an x1 of x such that y < x 1. A O 1. If x < y , then a + x < a + y. M O 1. If a > 0 and b > 0, then ab > 0. 9The latter part of M 1 may be omitted from the list of axioms, since it can be proved as a theorem from A 4 and A M 1.

THE SYSTEM OF REAL NUMBERS. 11 These postulates may be regarded as summarizing the properties of the real number system. Every theorem of real analysis is a logical consequence of them. For convenience of reference later on we summarize also the rules of operation with the symbol |x|, which indicates the “numerical” or “absolute” value of x. That is, if x is positive,|x|= x, and if x is negative,|x|=−x. |x|+|y|□|x + y|. (1) □ n∑ k=1 |xk|□ □□□ n∑ k=1 xk □□□, (2) where∑n k=1 xk = x1 + x2 + . . . + xn. □□|x|−|y| □□□|x−y|=|y−x|□|x|+|y|. (3) |x·y|=|x|·|y|. (4) |x| |y|= □□□□ x y □□□□. (5) If|x−y|< e 1, |y−z|< e 2, then|x−z|< e 1 + e2. (6) If [x] is any bounded set, B[x]−B[x] = B[|x1−x2|]. (7) § 6 The Number e. In the theory of the exponential and logarithmic functions (see page 76) the irrational number e plays an important rˆ ole. This number may be defined as follows: e = B[En], (1) where En = 1 + 1 1! + 1 2! + . . . + 1 n!, where [n] is the set of all positive integers, and n! = 1·2·3 . . . n. It is obvious that (1) defines a finite number and not infinity, since En = 1 + 1 1!

+ 1 2! + . . . + 1 n! < 1 + 1 + 1 2 + 1 22 + . . . + 1 2n−1 = 3−1 2n−1 .

12 INFINITESIMAL ANALYSIS. The number e may very easily be computed to any number of decimal places, as follows: E0 = 1 1 1! = 1 1 2! = . 5 1 3! = . 166666+ 1 4! = . 041666+ 1 5! = . 008333+ 1 6! = . 001388+ 1 7! = . 000198+ 1 8! = . 000024+ 1 9! = . 000002+ E9 = 2. 7182 . . . Lemma.— If k > e , then Ek > e−1 k! . Proof. From the definitions of e and En it follows that e−Ek = B [1 (k + 1)! + 1 (k + 2)! + . . . 1 (k + l)!] , where [l] is the set of all positive integers. Hence e−Ek = 1 (k + 1)!·B [1 + 1 k + 2 + 1 (k + 2)(k + 3) + . . . + 1 (k + 2) . . . (k + l)] , or e−Ek < 1 (k + 1)!·e. If k > e , this gives Ek > e−1 k! . Theorem 7. e = B [(1 + 1 n)n] , where [n] is the set of all positive integers.

THE SYSTEM OF REAL NUMBERS. 13 Proof. By the binomial theorem for positive integers (1 + 1 n)n = 1 + n (1 n) + n(n−1) 2! · (1 n)2 + . . . + (1 n)n . Hence En− (1 + 1 n)n = n∑ k=2 (1 k!−n(n−1) . . . (n−k + 1) k! nk) = n∑ k=2 nk−n(n−1) . . . (n−k + 1) k! nk , (a) < n∑ k=2 nk−(n−k + 1)k k! nk . Hence by factoring En− (1 + 1 n)n < n∑ k=2 (k−1)(nk−1 + nk−2(n−k + 1) + . . . + (n−k + 1)k−1) k! nk < n∑ k=2 (k−1)knk−1 k! nk < 1 n n∑ k=2 (k−1)k k! i.e., En− (1 + 1 n)n < 1 n (1 + n−2∑ l=1 1 l!) < e n . (b) From (a) En > (1 + 1 n)n (1) and from (b) (1 + 1 n)n > E n−e n , (2) whence by the lemma (1 + 1 n)n > e−1 n!−e n . (3) From (1) it follows that e is an upper bound of [(1 + 1 n)n] ,

14 INFINITESIMAL ANALYSIS. and from (3) it follows that no smaller number can be an upper bound. Hence B [(1 + 1 n)n] = e. § 7 Algebraic and Transcendental Numbers. The distinction between rational and irrational numbers, which is a feature of the discussion above, is related to that between algebraic and transcendental numbers. A number is algebraic if it may be the root of an algebraic equation, a0xn + a1xn−1 + . . . + an−1x + an = 0, where n and a0, a 1, . . . , a n are integers and n > 0. A number is transcendental if not algebraic. Thus every rational number m n is algebraic because it is the root of the equation nx−m = 0, while every transcendental number is irrational. Examples of transcendental numbers are, e, the base of the system of natural logarithms, and π, the ratio of the circumference of a circle to its diameter. The proof that these numbers are transcendental follows on page 14, though it makes use of infinite series which will not be defined before page 56, and the function ex, which is defined on page 44. The existence of transcendental numbers was first proved by J. Liouville , Comptes Rendus, 1844. There are in fact an infinitude of transcendental numbers between any two numbers. Cf. H. Weber , Algebra, Vol. 2, p. 822. No particular number was proved transcendental till, in 1873, C. Hermite (Crelle’s Journal, Vol. 76, p. 303) proved e to be transcendental. In 1882 E. Lindemann (Mathematische Annalen, Vol. 20, p. 213) showed that π is also transcendental. The latter result has perhaps its most interesting application in geometry, since it shows the impossibility of solving the classical problem of constructing a square equal in area to a given circle by means of the ruler and compass. This is because any construction by ruler and compass corresponds, according to analytic geometry, to the solution of a special type of algebraic equation. On this subject, see F. Klein , Famous Problems of Elementary Geometry (Ginn & Co., Boston), and Weber and Wellstein, Encyclop¨ adie der Elementarmathematik , Vol. 1,

pp. 418–432 (B. G. Teubner, Leipzig). § 8 The Transcendence of e. Theorem 8. If c, c 1, c 2, c 3, . . . , c n are integers (or zero but ≠ 0), then c + c1e + c2e2 + . . . + cnen ≠ 0. (1)

THE SYSTEM OF REAL NUMBERS. 15 Proof. The scheme of proof is to find a number such that when it is multiplied into (1) the product becomes equal to a whole number distinct from zero plus a number between +1 and −1, a sum which surely cannot be zero. To find this number N , we study the series10 for ek, where k is an integer <=n: ek = 1 + k 1! + k2 2! + k3 3! + ... Multiplying this series successively by the arbitrary factors i!·bi, we obtain the following equations: ek·1!·b1 = b1·1! + b1k (1 + k 2 + k2 2·3 + . . .) ; ek·2!·b2 = b2·2! (1 + k 1) + b2·k2 (1 + k 3 + k2 3·4 + . . .) ; ek·3!·b3 = b3·3! (1 + k 1! + k2 2!) + b3·k3 (1 + k 4 + k2 4·5 + . . .) ; . . . . ek·s!·bs = bs·s! (1 + k 1! + k2 2! + . . . + ks−1 (s−1)!) +bs·ks (1 + k s+1 + k2 (s+1)(s+2) + . . .) . □ □□□□□□□□□□□□ □□□□□□□□□□□□ (2) For the sake of convenience in notation the numbers b1 . . . b s may be regarded as the coefficients of an arbitrary polynomial φ(x) + b0 + b1x + b2x2 + . . . + bsxs , the successive derivatives of which are φ′(x) = b1 + 2·b2x + . . . + s·bs·xs−1, ... φ(m)(x) = bm·m! + bm+1·(m+1)! 1! ·x + . . . + bs·s! (s−m)!·xs−m; . . . . . The diagonal in (2) from b1·1! to bs·s! ks−1 (s−1)! is obviously φ′(k), the next lower diagonal is φ″(k), etc. Therefore by adding equations (2) in this notation we obtain ek(1! b1 + 2! b2 + . . . + s! bs) = φ′(k) + φ″(k) + . . . + φ(s)(k) + s∑ m=1 bm·km·Rkm, (3) in which Rkm = 1 + k m + 1 + k2 (m + 1)(m + 2) + ... 10Cf. pages 56 and 78.

16 INFINITESIMAL ANALYSIS. Remembering that φ(x) is perfectly arbitrary, we note that if it were so chosen that φ′(k) = 0 , φ ″(k) = 0 , . . . , φ (p−1)(k) = 0 , for every k (k = 1, 2, 3, . . . , n) then equations (2) and (3) could be written in the form ek(1!·b1 + 2!·b2 + . . . + s!·bs) = s∑ m=1 bm·km·Rkm + bp·p! + bp+1·(p + 1)!· (1 + k 1!) + . . . + bs·s! (1 + k 1! + k2 2! + . . . + ks−p (s−p)!) . (4) A choice of φ(x) satisfying the required conditions is φ(x) = (a0 + a1x + a2x2 + . . . + anxn)p·xp−1 (p−1)! = (f (x))p·xp−1 (p−1)! , (5) where f (x) = (x−1)(x−2)(x−3) . . . (x−n). Every k (k = 1, 2, . . . , n) is a p-tuple root of (5). Here p is still perfectly arbitrary, but the degree s of φ(x) is np + p−1. If φ(x) is expanded and the result compared with φ(x) = b0 + b1x + . . . + bsxs, it is plain that b0 = 0, b 1 = 0, . . . , b p−2 = 0, on account of the factor xp−1, and bp−1 = ap 0 (p−1)!, b p = Ip (p−1)!, . . . , b s = Is (p−1)!, where Ip, I p+1, . . . , I s, are all integers. The coefficient of ek in the left-hand member of (4) is therefore Np = ap 0 + Ip (p−1)!·p! + Ip+1 (p−1)!·(p + 1)! + . . . + Is (p−1)!·s! Whenever the arbitrary number p is prime and greater than a0, Np is the sum of ap 0, which cannot contain p as a factor, plus other integers each of which does contain the factor p. Np is therefore not zero and not divisible by p. Further, since (p + t)! (p−1)!·r! = p (p + 1)(p + 2) . . . (p + t) r!

THE SYSTEM OF REAL NUMBERS. 17 is an integer divisible by p when r □ t, it follows that all the coefficients of the last block of terms in (4) contain p as a factor. Since k is also an integer, (4) evidently reduces to Np·ek = pWkp + s∑ m=1 bm·km·Rkm, where Wkp is an integer or zero, and this may be abbreviated to the form Np·ek = pWkp + rkp. (6) Before completing our proof we need to show that by choosing the arbitrary prime number p sufficiently large, rkp can be made as small as

we please. If α is a number greater than n, |Rkm|= □□□□1 + k m + 1 + k2 (m + 1)(m + 2) + . . . □□□□ < □□□□1 + α m + 1 + α2 (m + 1)(m + 2) + . . . □□□□ < □□□□1 + α 1 + α2 2! + . . . □□□□ < e α for all integral values of m and of k <= n. |rkp|= □□□□□ s∑ m=1 bm·km·Rkm □□□□□□ s∑ m=1 |bm|·km·|Rk,m|. Since the number bm is the coefficient of xm in φ(x) and since each coefficient of φ(x) is numerically less than or equal to the corresponding coefficient of xp−1 (p−1)! (|a0|+|a1|x +|a2|x2 + . . . +|an|xn)p , it follows that |rkp|< e α·αp−1 (p−1)! (|a0|+|a1|α + . . . +|an|αn)p < Qp (p−1)!·eα, where Q = α(|a0|+|a1|α + . . . +|an|αn) is a constant not dependent on p. The expression Qp (p−1)! is the pth term of the series for QeQ, and therefore by choosing p sufficiently large rkp, may be made as small as we please.

18 INFINITESIMAL ANALYSIS. If now p is chosen as a prime number, greater than α and α0 and so great that for every k, rkp < 1 n·d, where d is the greatest of the numbers c, c 1, c 2, c 3, . . . , c n, the equations (6) evidently give Np(c + c1e + c2e2 + . . . + cnen) = Npc + p(c1W1p + c2W2p + . . . + cnWnp) + c1r1p + c2r2p + . . . + cnrnp, = Npc + pW + R, (8) where W is an integer or zero and R is numerically less than unity. Since Npc is not divisible by p and is not zero, while pW is divisible by p, this sum is numerically greater than or equal to zero. Hence Np(c + c1e + c2e2 + . . . + cnen′)= 0. Hence c + c1e + c2e2 + . . . + cnen′= 0, and e is a transcendental number. § 9 The Transcendence of π. The definition of the number π is derived from Euler's formula ex√−1 = cos x + √ −1 sin x; by replacing x by π, eπ√−1 =−1. (1) If π is assumed to be an algebraic number, π√−1 is also an algebraic number and is the root of an irreducible algebraic equation F (x) = 0 whose coefficients are integers. If the roots of this equation are denoted by z1, z 2, z 3, . . . , z n, then, since π√−1 is one of the z's, it follows as a consequence of (1) that (ez1 + 1)(ez2 + 1)(ez3 + 1) . . . (ezn + 1) = 0 . (2) By expanding (2) 1 + ∑ ezi + ∑ ezi+zj + ∑ ezi+zj +zk + . . . = 0.

THE SYSTEM OF REAL NUMBERS. 19 Among the exponents zero may occur a number of times e.g., (c−1) times. If then zi, z i + zj, z i + zj + zk, . . . , be designated by x1, x 2, x 3, . . . , x n, the equation becomes c + ex1 + ex2 + . . . + exn = 0, (3) where c is a positive number at least unity and the numbers xi are algebraic. These numbers, by an argument for which the reader is referred to Weber and Wellstein's Encyclop¨ adie der Elementarmathematik, p. 427 et seq., may be shown to be the roots of an algebraic equation f (x) = a0 + a1x + a2x2 + . . . + anxn = 0, (3′) the coefficients being integers and aØ= 0 and an′= 0. The rest of the argument consists in showing that equation (3) is impossible when x1, x 2, . . . , xn are roots of (3 ′). The process is analogous to that in §8. exk·1! b1 = b1·1! + b1xk (1 + xk 2 + x2 k 2·3 + . . .) , exk·2! b2 = b2·2! (1 + xk 1!) + b2x2 k (1 + xk 3 + x2 k 3·4 + . . .) , exk·3! b3 = b3·3! (1 + xk 1! + x2 k 2!) + b3x3 k (1 + xk 4 + x2 k 4·5 + . . .) , ... exk·s! bs = bs·s! (1 + xk 1! + . . . + xs−1 k (s−1)!) +bsxs k (1 + xk s + 1 + x2 k (s + 1)(s + 2) + . . .) . □ □□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□ (4) The numbers b1, . . . , b s may be regarded as the coefficients of an arbitrary polynomial φ(x) = b0 + b1x + b2x2 + . . . + bsxs, for which φ(m)(x) = bm·m! + bm+1·(m + 1)! 1! ·x + . . . + bs s! (s−m)!·xs−m. The diagonal in equations (4) from b1·1! to bs·s! xk s−1 (s−1) ! is obviously φ′(xk), and the next lower diagonal φ′′(xk), etc. Therefore, by adding equations (4), exk(1! b1 + 2! b2 + . . . + s! bs) = φ′(xk) + φ′′(xk) + . . . + φ(s)(xk) + s∑ m=1 bm·xm k Rkm, (5)

20 INFINITESIMAL ANALYSIS. in which Rkm = 1 + xk m + 1 + x2 k (m + 1)(m + 2) + . . . Remembering that φ(x) is perfectly arbitrary, let it be so chosen that φ′(xk) = 0 , φ ″(xk) = 0 , φ ‴(xk) = 0 , . . . , φ (p−1)(xk) = 0 for every xk. Equation (5) may then be written as follows: exk(1! b1 + 2! b2 + . . . + s! bs) = s∑ m=1 bm·(xk)m·R km + bp·p! + bp+1·(p + 1)! (1 + xk 1!) + . . . + bs·s! (1 + xk 1! + x2 k 2! + . . . + xs−p k (s−p)!) . (6) A choice of φ(x) satisfying the required conditions is φ(x) = anp−1 n ·xp−1 (p−1)! (a0 + a1x + a2x2 + . . . + anxn)p = anp−1 n ·xp−1 (p−1)! (f (x))p, of which every xk is a p-tuple root. If φ(x) is expanded and the result compared with φ(x) = b0 + b1x + . . . + bsxs, it is plain that b0 = 0, b1 = 0, . . . , bp−2 = 0, on account of the factor xp−1; and bp−1 = ap 0anp−1 n (p−1)!, b p = Ip·anp−1 n (p−1)! . . . , b s = Is·anp−1 n (p−1)! , where Ip, . . . , I s, are all integers. The coefficient of exk in (6) may now be written Np = anp−1 n (ap 0 + Ip (p−1)!·p! + Ip+1 (p−1)!(p + 1)! + . . . + Is (p−1)!·s!) . If the arbitrary number p is chosen as a prime number greater than a0 and an, Np becomes the sum of ap 0anp−1 n , which cannot contain p as a factor, and a number of other integers each of which is divisible by p. Np therefore is not zero and not divisible by p.

THE SYSTEM OF REAL NUMBERS. 21 Further, since, (p+t)! (p−1)!·r! is an integer divisible by p when r □ t, it follows that all of the coefficients of the last block of terms in (6) contain p as a factor. If then (6) is added by columns, Npe xk = panp−1 n [P0 + P1xk + P2x2 k + . . . + Ps−pxs−p k] + s∑ m=1 bm·xm k ·Rkm (7) where P0, P 1, . . . , P s−p are integers. It remains to show that ∑s m=1 bm·xm k ·Rkm can be made small at will by a suitable choice of the arbitrary p. As in the proof of the transcendence of e, it follows that |rkp|= □□□□□ s∑ m=1 bm·xm k ·Rkm □□□□□< Qp (p−1)!·eα, where Q =|an n|α(|a0|+|a1|α + . . . +|an|α), and α is the largest of the absolute values of xk (k = 1 , . . . , n). If now p is chosen as a prime number, greater than unity, greater than a0 . . . a n and greater than c, and so great also that|rkp|< 1 n, it follows directly from equation (7) that Np(c + ex1 + ex2 + . . . + exn) = Npc + pan np−1(P0S0 + P1S1 + . . . + Ps−pSs−p) + n∑ k=1 rkp, (8) where |rkp|= □□□□□ s∑ m=1 bm·xm k ·Rkm □□□□□< 1 n , S0 = n, and Si = xi 1 + xi 2 + xi 3 + . . . + xi n, and therefore S1 =−an−1 an , S 2 = a2 n−1 a2 n −2an−2 an , . . . , 11 and therefore it follows that annp−1S1, annp−1S2,. . . , are all whole numbers or zero. The term panp−1 n · s−p∑ i=0 PiSi is therefore an integer divisible by p, while, on the contrary, Np and c are not divisible by p. The sum of these terms is therefore a whole number □ +1 or □−1, and since n∑ k=1 rkp < 1, the entire right-hand member of (8) is not zero, and hence (3) is not zero. Therefore— Theorem 9. The number π is transcendental. 11Cf. Burnside and Panton Theory of Equations , Chapter VIII, Vol. I.

22 INFINITESIMAL ANALYSIS.

Chapter 2 SETS OF POINTS AND OF SEGMENTS. § 1 Correspondence of Numbers and Points. The system of real numbers may be set into one-to-one correspondence with the points of a straight line. That is, a scheme may be devised by which every number corresponds to one and only one point of the line and vice versa. The point 0 is chosen arbitrarily, and the points 1 , 2, 3, 4, . . . are at regular intervals to the right of 0 in the order 1 , 2, 3, 4, . . . from left to right, while the points −1,−2,−3, . . . follow at regular intervals in the order 0,−1,−2,−3, . . . from right to

left. The points which correspond to fractional numbers are at intermediate positions as follows: 1 To fix our ideas we obtain a point corresponding to a particular decimal of a finite number of digits, say 1 . 32. Divide the segment 1 2 into ten equal parts. Then divide the 1. 32 1 . 1 . 2 . 3 . 4 . 5 1 Fig. 1 segment . 3 . 4 of this division into ten equal parts. The point marked 2 by the last division is the point corresponding to 1 . 32. If the decimal is not terminating, we simply obtain an infinite sequence of points, such that any one is to the right of all that precede it, in case of a positive number, or to the left in case of a negative number. The first few points of the sequence for the number π are the points corresponding to the numbers 3, 3 . 1, 3 . 14, 3 . 141. This set of numbers is bounded, 4, for instance, being an upper bound. Hence the points corresponding to these numbers all lie to the left of the point corresponding to the number 4. To show that there exists a definite point corresponding to the least upper bound B of the set of numbers 3, 3. 1, 3 . 14, 3 . 141, etc., use is made of the following: 1It is convenient to think of numbers in this case as simply a notation for points. In view of the correspondence of points and numbers the numbers furnish a complete notation for all points. 23

24 INFINITESIMAL ANALYSIS. Postulate of Geometric Continuity.— If a set [x] of points of a line has a right bound, that is, if there exists a point B on the line such that no point of the set [x] is to the right of B, then there exists a leftmost right bound B of the set [x]. If the set has a left bound, it has a rightmost left bound. The leftmost right bound of the set of points corresponding to the numbers 3 . , 3 . 1, 3. 14, etc., is the point which corresponds to the number π. In the same manner it follows from the postulate that there is a definite point on the line corresponding to any decimal with an infinitude of digits. 2 Conversely, given any point on the line, e.g., a point P , to the right of 0, there corre- sponds to it one and only one number. This is evident since, in dividing the line according to a decimal scale, either the point in question is one of the division-points, in which case the number corresponding to the point is a terminating decimal, or in case it is not a division-point we will have an infinite set of division -points to the left of it, the point in question being the leftmost right bound of the set. If now we pick out the rightmost point of this left set in every division and note the corresponding number, we have a set of numbers whose least upper bound corresponds to the point P . The ordinary analytic geometry furnishes a scheme for setting all pairs of real num- bers into correspondence with all points of a plane, and all triples of real numbers into correspondence with all points in space. Indeed, it is upon this correspondence that the analytic geometry is based. It should be noticed that the correspondence between numbers and points on the line preserves order, that is, if we have three numbers, a, b, c, so that $a < b < c$, then the corresponding points A, B, C are under the ordinary conventions so arranged that B is to the right of A, and C to the right of B. It will be observed that we have not put this matter of the one-to-one correspondence between points and numbers into the form of a theorem. Rather than aiming at a rigorous demonstration from a body of sharply stated axioms, we have attempted to place the subject-matter before the reader in such a manner that he will understand on the one hand the necessity, and on the other the grounds, for the hypothesis. § 2 Segments and Intervals. Theorem of Borel. Definition.— A segment a b is the set of all numbers greater than a and less than b. It does not

include its end-points a and b. An interval || a b is the segment a b together with a and b. For a segment plus its end point a we use the notation | a b , and when a is absent 2It is not implied here, of course, that it is possible to write a decimal with an infinitude of digits, or to mark the corresponding points. What is meant is that if an infinite sequence of digits is determined, a definite number and a definite point are thereby determined. Thus √ 2 determines an infinite sequence of digits, that is, it furnishes the law whereby the sequence can be extended at will.

SETS OF POINTS AND OF SEGMENTS. 25 and b present | a b . All these notations imply that a < b .3 Sometimes we denote a segment or interval by a single letter. This is done in case it is not important to designate a definite segment or interval. The set of all numbers greater than a is the infinite segment a∞, and the set of all numbers less than a is the infinite segment −∞a. The infinite segments a∞and−∞a, together with the point a, are respectively the infinite intervals | a∞and | −∞a. Unless otherwise specified the expressions segment and interval will be understood to refer to segments and intervals whose end-points are finite. By means of the one-to-one correspondence of numbers and points on a line we define the length of a segment as follows: The length of a segment a b with respect to the unit segment 0 1 is the number |a−b|. This definition applies equally to all segments whether they are commensurable or incommensurable with the unit segment. Definition.— A set of segments or intervals [σ] covers a segment or interval t if every point of t is a point of some σ. On the interval | | −1 1 consider the set of points [1 2n] . The set of intervals | | −1 0, || 1 2 1, −1 0 1/ 8 1 / 4 1 / 2 1 Fig. 2 || 1 4 1 2 , . . . , | | 1 2n 1 2n−1 , . . . covers the interval | | −1 1, because every point of | | −1 1 is a point of one of the intervals. On the other hand a set of segments −1 0, 1 2 1, . . . , 1 2n 1 2n−1 , etc., does not cover the interval because it does not include the points −1, 1, 1 2 , . . . , 1 2n , . . . , or 0. In order to obtain a set of segments which does cover the interval, it is necessary to adjoin a set of segments, no matter how small, such that one includes −1, one includes 0, one includes 1, 1 2 , 1 4 , ... The segment including 0, no matter how small it is, must include an infinitude of the points 1 2n , and there are only a finite number of them which do not lie on that segment. It therefore follows that in this enlarged set there is a subset of segments, finite in number, which includes all the points of | | −1 1 . This turns out to be a general theorem, namely, that if any set of segments covers an interval, there is a finite subset of it which also covers the interval. The example we have just given shows that such a theorem is not true of the covering of an interval by a set of intervals; furthermore, it is not true of the covering of a segment either by a set of segments or by a set of intervals. 3The notation a b , | | a b , | a b , etc., to denote the presence or absence of end-points is due to G. Peano , Analisi Infinitisimali. Torino, 1893.

26 INFINITESIMAL ANALYSIS. Theorem 10 4. If an interval || a b is covered by any set [σ] of segments, it is covered by a finite number of segments σ1, . . . , σ n of [σ]. Proof. It is evident that at least a part of || a b is covered by a finite number of σ’s; for example, if σ0 is the σ or one of the σ’s which include a and if b′is any point of | a b which lies in σ0, then || a b′is covered by σ0. Let [b′] be the set of all points of | a b , such that || a b′ is covered by a finite number of σ’s. By Theorem 4 [b′] has a least upper bound B. To complete our proof we show (a) that B is in [b′], and (b) that B =

b. (a) Let a″b″be a segment of [σ] including B. Since B is the least upper bound of [b′], there is a point of [b′], b′, between a″and B. But if σ1, σ 2, . . . , σ e, be the finite set of segments covering the interval || a b′, this set together with a″b″will cover || a B , which proves that B is a point of [b′]. (b) If B = b, then B < b and the set σ1, σ2, . . . , σe, together with a″b″, would cover an interval || a c , where c is a point between B and b″; c would therefore be a point of [b′], which is contrary to the hypothesis that B is an upper bound of [b′]. Hence B = b and the theorem is proved. An immediate consequence of this theorem is the following, which may be called the theorem of uniformity . Theorem 11. If an interval || a b is covered by a set of segments [σ], then || a b may be divided into N equal intervals such that each interval is entirely within a σ. Proof. By Theorem 10 || a b is covered by a finite set of σ's,σ1, σ 2, . . . , σ n. The end points of these σ's, together with a and b, are a finite set of points. Let d be the smallest distance between any two distinct points of this set. Because of the overlapping of the σ's, any two points not in the same segment are separated by at least two end points. Therefore any two points whose distance apart is less than d must lie on the same segment of σ1, σ 2, . . . , σ n. Now let N be such that b−a N < d, then each interval of length b−a N is contained in a σ. By this argument we have also proved the following: 4This theorem is due to E. Borel , Annales de l' ´Ecole Normale Sup´ erieure, 3d series, Vol. 12 (1895), p. 51. It is frequently referred to as the Heine-Borel theorem, because it is essentially involved in the proof of the theorem of uniform continuity given by E. Heine, Die Elemente der Functionenlehre , Crelle's Journal, Vol. 74 (1872), page 188.

SETS OF POINTS AND OF SEGMENTS. 27 a σ1 σ2 σ3 d σ4 σ5 σ6 σ7 σ8 b Fig. 3. Theorem 12. If an interval || a b is covered by a set of segments, then there is a number d such that for any two numbers x1 and x2 such that a □ x1 < x 2 □ b and|x1−x2|< d , there is a segment σ of [σ] which contains both x1 and x2. In other words, any interval of length d lies entirely within some σ. The sense in which these are theorems of uniformity is the following. Any point x of || a b , being within a segment σ, can be regarded as the middle point of an interval ix of length lx which is entirely within some σ. The length lx is in general different for different points, x. Our theorem states that a value l can be found which is effective as an lx for every x, i.e., uniformly over the interval || a b . The distinction here drawn is one of the most important in rigorous analysis. It was first observed in connection with the theorem of uniform continuity; see page 70. The presence of both end points of || a b is essential, as is shown by the following example. | 0 1 is covered by the segments 1 2 2, 1 4 1, 1 8 1 2 , . . . , 1 2n 1 2n−2 , . . . , but as we take points nearer to 0, lx becomes smaller with the lower bound 0, and no l can be found which is effective for all points of | 0 1. When the end points are absent it is possible, however, to modify the notion of covering, so that our theorem remains true. This is sufficiently indicated by the following theorem, which is an immediate consequence of Theorem 10. Theorem 13. If on a segment a b there exists any set [σ] of segments such that (1) [σ] includes a segment of which a is an end point and a segment of which b is an end point. (2) Every point of the segment a b lies on one or more of the segments of the set [σ]. Then among the segments of the set [σ] there exists a finite set of segments σ1, σ2, . . . , σn which satisfies conditions (1) and (2). The theorems which we have just proved

can be generalized to space of any number of dimensions. A planar generalization of a segment is a parallelogram with sides parallel to the coordinate axes, the boundary being excluded. The planar generalization of an interval is the same with the boundary included. The theorem of Borel becomes: Theorem 14. If every point of the interior or boundary of a parallelogram P is interior to at least one parallelogram p of a set of parallelograms [p], then every point of P is interior to at least one parallelogram of a finite subset p1 . . . p n of [p].

28 INFINITESIMAL ANALYSIS. Proof. Let x = 0, x = a > 0, y = 0, y = b > 0 determine the boundary of P . Let 0 □ y1 □ b. Upon the interval i of the line y = y1, cut off by P , those parallelograms of [p] that include points of i as interior points determine a set of segments [π] such that every point of i is an interior point of one of these segments π. There is by Theorem 10 a finite subset of [π], π1 . . . π n, including every point of i, and therefore a finite subset p1 . . . p n of [p], including as interior points every point of i. Moreover, since the number of p1 . . . pn is finite, they include in their interior all the points of a definite strip, e.g., the points between the lines y = y1−e and y = y1 + e. y = 0 y = b x = 0 x = a y1 + e y1 y1−e Fig. 4. Thus for every y1 (0 □ y1 □ b) we obtain a strip of the parallelogram P such that every point of its interior is interior to one of a finite number of the parallelograms [p]. These strips intersect the y-axis in a set of segments that include every point of the interval || 0 b. There is therefore, by Theorem 10, a finite set of strips which includes every point in P . Since each strip is included by a finite number of parallelograms p, the whole parallelogram P is included by a finite subset of [p]. The generalization of Theorems 11 and 12 is left to the reader. § 3 Limit Points. Theorem of Weierstrass. Definition.— A neighborhood or vicinity of a point a in a line (or simply a line neighbor- hood of a) is a segment of this line such that a lies within the segment. We denote a line neighborhood of a point a by V (a). The symbol V∗(a) denotes the set of all points of V (a) except a itself. The symbols V (∞) and V∗(∞) are both used to denote infinite segments a +∞, and V (−∞) and V∗(−∞) to denote infinite segments −∞a.5 5This notation is taken from Pierpont’s Theory of Functions of Real Variables . It is used here, however, with a meaning slightly different from that of Pierpont.

SETS OF POINTS AND OF SEGMENTS. 29 A neighborhood of a point in a plane (or a plane neighborhood of a point) is the interior of a parallelogram within which the point lies. A neighborhood of a point (a, b) is denoted by V (a, b) if (a, b) is included and by V∗(a, b) if (a, b) is excluded. Instead of the three linear vicinities V (a), V (∞), and V (−∞) we have the following nine in the case of the plane: It follows at once from a consideration of the scheme for setting the points on the line into correspondence with all numbers that in every neighborhood of a point there is a point whose corresponding number is rational. Definition.— A point a is said to be a limit point of a set if there are points of the set, other than a, in every neighborhood of a. In case of a line neighborhood this says that there are points of the set in every V∗(a). In the planar case this is equivalent to saying that (a, b) is a limit point of the set [x, y], either if for every V∗(a) and V (b) there is an (x, y) of which x is in V∗(a) and y in V (b), or if for every V (a) and V∗(b) there is an (x, y) of which x is in V (a) and y in V∗(b).

30 INFINITESIMAL ANALYSIS. Thus 0 is a limit point of the set [1 2k] , where k takes all positive integral values. In this case the limit point is not a point of the set. On the other hand, in the set 1, 1 −1 2 , 1−1 22 ,. . . , 1−1 2k , 1 is a limit point of the set and also a point of the set. In this case 1 is the least upper bound of the set. In case of the set 1, 2, 3, the number 3 is the least upper bound without being a limit point. The fundamental theorem about limit points is the following (due to Weierstrass): Theorem 15. Every infinite bounded set [p] of points on a line has at least one limit point. Proof. Since the set [p] is bounded, every one of its points lies on a certain interval || a b . If the set [p] has no limit point, then about every point of the interval || a b there is a segment σ which contains not more than one point of the set [p]. By Theorem 10 there is a finite set of the segments [σ] such that every point of || a b and hence of [p] belongs to at least one of them, but each σ contains at most one point of the set [p], whence [p] is a finite set of points. Since this is contrary to the hypothesis, the assumption that there is no limit point is not tenable. It is customary to say that a set which has no finite upper bound has the upper bound +∞, and that one which has no finite lower bound has the lower bound −∞. In these cases, since the set has a point in every V∗(+∞) or in every V∗(−∞) +∞and−∞are also called limit points. With these conventions the theorem may be stated as follows: Theorem 16. Every infinite set of points has a limit point, finite or infinite. The theorem also generalizes in space of any number of dimensions. In the planar case we have: Theorem 17. An infinite set of points lying entirely within a parallelogram has at least one limit point. Theorem 17 is a corollary of the stronger theorem that follows: Theorem 18. If [(x, y)] is any set of number pairs and if a is a limit point of the numbers [x], there is a value of b, finite or +∞or−∞, such that for every V∗(a) and V (b) there is an (x, y) of which x is in V∗(a) and y is in V (b). Proof. Suppose there is no value b finite or +∞or−∞such as is required by the theorem. Since neither +∞nor−∞possesses the property required of b, there is a V∗(a) and a V (∞) and a V (−∞) such that for every pair (x, y) of [(x, y)] whose x lies in V∗(a) y fails to lie in either V (∞) or V (−∞). This means that there exists a pair of numbers M and m such that for every (x, y) whose x is in V∗(a) the y satisfies the condition m < y < M . Further, since there exists no b such as is required by the theorem, there is for every number k on the interval | | m M a V (k) and a V∗ k (a), such that for no (x, y) is x in V∗ k (a) and y in

SETS OF POINTS AND OF SEGMENTS. 31 V (k). This set of segments [V (k)] covers the interval | | m M , whence by Theorem 10 there is a finite subset of [V (k)], V1(k), . . . , Vn(k) which covers | | m M , and hence a finite set of corresponding V∗ k (a)’s. Let V∗(a) be a vicinity of a contained in every one of the finite set of V∗ k (a)’s and in V∗(a). Hence if the x of a pair (x, y) is in V∗(a), its y cannot lie in one of the infinite segments M∞and−∞m, or in one of the finite segments V1(k), . . . , Vn(k), i.e., no y corresponds to this x, which is contrary to the hypothesis. This argument covers the cases when a is +∞and when a is−∞. We add the definitions of a few of the technical terms that are used in point-set theory. 6 Definition.— A set of points which includes all its limit points is called a closed set. A set of points every one of which is a limit point of the set is called dense in itself .7 A set of points which is both closed and dense in itself is called perfect. A set having no finite limit point is called

discrete. A segment not including its end points is an example of a set dense in itself but not closed. If the end points are added, the set is closed and therefore perfect. The set of rational numbers is another case of a set dense in itself but not closed. Any set containing only a finite number of points is closed, according to our definition. If every point of an interval || a b is a limit point of a set [x], then [x] is everywhere dense on || a b . Such a set has a point between every two points of the interval. A set which is everywhere dense on no interval is called nowhere dense . All rational numbers between 0 and 1 form an everywhere dense set. § 4 Second Proof of Theorem 15. To make the reader familiar with a style of argument which is frequently used in proving theorems which in this book are made to depend upon Theorems 10 and 14, we adjoin the following lemma and base upon it another proof of Theorem 15. Lemma.— Hypothesis: On a straight line there is an infinite set of intervals | | a1 b1 , | | a2 b2 , . . . , | | an bn , . . . conditioned as follows: 8 6For bibliography and an exposition in English see W. H. Young and G. C. Young , The Theory of Sets of Points . Cambridge, The University Press. 7In German “in sich dicht.” 8In particular the set of segments assumed in the hypothesis may be obtained by dividing any given segment into a given number of equal segments, then one of these segments into the same number of equal segments and so on indefinitely. To show that the sequential division into a number of equal segments gives a set of segments satisfying the conditions of the hypothesis we have merely to show that such division gives a segment less than any assigned segment ae be. This is equivalent to the statement that for every number e there is an integer n, such that 1 n < e a direct consequence of Theorem 3. This involves the notion that

32 INFINITESIMAL ANALYSIS. (1) Interval | | a2 b2 lies on interval | | a1 b1 , | | a3 b3 on | | a2 b2 , etc. In general | | an bn lies on | | an−1 bn−1 . (This does not exclude the case ak = ak+1.) (2) For every length e > 0, however small, there is some n, say ne, such that|bne−ane|< e. Conclusion: There is one and only one point b which lies upon every interval | | an bn . Proof. Since the set of points a1 . . . a n . . . is bounded, we have at once, by the postulate of continuity, that this set has a leftmost right bound Ba. Similarly, the set b1 . . . b n . . . has a rightmost left bound Bb. It follows at once that Ba = Bb, for if not, we get either an a point to the right of Ba, or a b point to the left of Bb when ne is so chosen that |bne−ane|< Ba−Bb. We now give another proof for Theorem 11. Divide the interval || a b on which all points of [p] lie into two equal intervals. Then there is an infinite number of points [p] on at least one of these intervals which we call | | a1 b1 . Divide this interval into two equal parts and so on indefinitely, always selecting for division an interval which contains an infinite number of points of the set [p]. We thus obtain an infinite sequence of intervals | | a1 b1 , | | a2 b2 , . . . , | | an bn . . . which satisfies the hypothesis of the lemma. There is therefore a point B which belongs to every one of the intervals | | a1 b1 , | | a2 b2 , . . . , | | an bn . . . , and therefore there is a point of the set [p] in every neighborhood of B. It should be noticed that the intervals in this sequence may be such that all intervals after a certain one will have, say, the right extremities in common. In this case the right extremity is the point B. Such is the sequence, obtained by decimal division, representing the number 2 = 1 . 99999 . . . . no constant infinitesimal exists. It may appear at first sight that a proof of this statement is superfluous. The fact is,

however, as was first proved by Veronese, that the non-existence of constant infinitesimals is not provable without some axiom such as the continuity axiom or the so-called Archimedean Axiom.

Chapter 3 FUNCTIONS IN GENERAL. SPECIAL CLASSES OF FUNCTIONS. § 1 Definition of a Function. Definition.— A variable is a symbol which represents any one of a set of numbers. A constant is a special case of a variable where the set consists of but one number. Definition.— A variable y is said to be a single-valued function of another variable x if to every value of x there corresponds one and only one value of y. The letter x is called the independent variable and y the dependent variable.1 Definition.— A variable y is said to be a many-valued function or multiple-valued function of another variable x if to every value of x there correspond one or more values of y. The class of multiple-valued functions thus includes the class of single-valued functions. 1 1This definition of function is the culmination of a long development of the use of the word. The idea of function arose in connection with coordinate geometry, Ren´e Descartes using the word as early as 1637. From this time to that of Leibnitz "function" was used synonymously with the word "power," such as x2, x3, etc. G. W. Leibnitz regarded "function" as "any expression standing for certain lengths connected with a curve, such as coordinates, tangents, radii of curvature, normals, etc." Johann Bernoulli (1718) defined "function" as "an expression made up of one variable and any constants whatever." Leonard Euler (1734) called the expression described by Bernoulli an analytic function and introduced the notation f (x). Euler also distinguished between algebraic and transcendental functions. He wrote the first treatise on "The Theory of Functions." The problem of vibrating strings led to the consideration of trigonometric series. J. B. Fourier set the problem of determining what kind of relations can be expressed by trigonometric series. The possibility then under consideration that any relation might be so expressed led Lejeune Dirichlet to state his celebrated definition, which is the one given above. See the Encyclop¨ adie der mathematischen Wissenschaften, II A. 1, pp. 3–5; also Ball's History of Mathematics, p. 378. 33

34 INFINITESIMAL ANALYSIS. It is sometimes convenient to think of special values taken by these two variables as arranged in two tables, one table containing values of the independent variable and the other containing the corresponding values of the dependent variable. Independent Variable Dependent Variable x1 y1 x2 y2 ·· ·· ·· xn yn If y is a single-valued function of x, one and only one value of y will appear in the table for each x. It is evident that functionality is a reciprocal relation; that is, if y is a function of x, then x is a function of y. It does not follow, however, that if y is a single-valued function of x, then x is a single-valued function of y, e.g., y = x2. It is also to be noticed that such tables cannot exhibit the functional relation completely when the independent variable takes all values of the continuum, since no table contains all such values. Definition.— That y is a function of x (and hence that x is a function of y) is expressed by the equation y = f (x) or by x = f−1(y). If y and x are connected by the equation y = f (x), f−1(y) is called the inverse function of f (x). Thus y = x2 has the inverse function x =±√y. In this case, while the first function y = x2 is defined for all real values of x, the inverse function x =±√y is defined only for positive values of y. The independent variable may or may not take all values between any two

of its values. Thus n! is a function of n where n takes only integral values. Sn, the sum of the first n terms of a series, is a function of n where n takes only integral values. Again, the amount of food consumed in a city is a function of the number of people in the city, where the independent variable takes on only integral values. Or the independent variable may take on all values between any two of its values, as in the formula for the distance fallen from rest by a body in time t, s = gt2 2 . It follows from the correspondence between pairs of numbers and points in a plane that the functional relation between two variables may be represented by a set of points in a plane. The points are so taken that while one of the two numbers which correspond to a point is a value of the independent variable, the other number is the corresponding value, or one of the corresponding values, of the dependent variable. Such representations are called graphs of the function. Cases in point where the function is single-valued are: the hyperbola referred to its asymptotes as axes (y = 1 x) ; a straight line not parallel to the y axis (y = ax + b); or a broken line such that no line parallel to the y axis contains more than one of its points. In general, the graph of a single-valued function with a single-valued inverse is a set of points [(x, y)] such that no two points have the same x or the same y. Following is a graph of a function where the independent variable does not take all values between any two of its values. Consider Sn, the sum of the first n terms as a

FUNCTIONS IN GENERAL. SPECIAL CLASSES OF FUNCTIONS. 35 function of n in the series S = 1 + 1 2 + 1 22 + . . . + 1 2n−1 + ... The numbers on the x axis are the values taken by the independent variable, while the functional relation is represented by the points within the small circles. Thus it is seen that the graph of this function consists of a discrete set of points. (Fig. 6.) The definition of a function here given is very general. It will permit, for instance, a function such that for all rational values of the independent variable the value of the function is unity, and for irrational values of the independent variable the value of the function is zero. f f f f f x y Fig. 6. § 2 Bounded Functions. Since the definition of function is so general there are few theorems that apply to all functions. If the restriction that f (x) shall be bounded is introduced, we have at once a very important theorem. Definition.— A function, f (x), has an upper bound for a set of values [x] of the indepen- dent variable if there exists a finite number M such that f (x) < M for every value of x in the set [x]. The function has a lower bound m if f (x) > m for every value of x in [x]. A function which for a given set of values of x has no finite upper bound is said to be unbounded on that set, or to have an upper bound + ∞on that set, and if it has no lower bound on the set the function is said to have the lower bound −∞on the set.

36 INFINITESIMAL ANALYSIS. Theorem 19. If on an interval || a b a function has an upper bound M , then it has a least upper bound B, and there is at least one value of x, x1 on || a b such that the least upper bound of the function on every neighborhood of x1 contained in || a b is B. Proof. (1) The set of values of the function f (x) form a bounded set of numbers. By Theorem 4 the set has a least upper bound B. (2) Suppose there were no point x1 on || a b such that the least upper bound on every neighborhood of x1 contained in || a b is B. Then for every x of || a b there would be a segment σx containing x such that the least upper bound of f (x) for values of x common to σx and || a b is less than B. The set [σx] is infinite, but by Theorem 10

there exists a finite subset [σn] of the set [σx] covering || a b . Therefore, since the upper bound of f (x) is less than B on that part of every one of these segments of [σn] which lies on || a b , it follows that the least upper bound of f (x) on || a b is less than B. Hence the hypothesis that no point x1 exists is not tenable, and there is a point x1 such that the least upper bound of the function on every one of its neighborhoods which lies in || a b is B. This argument applies to multiple-valued as well as to single-valued functions. As an exercise the reader may repeat the above argument to prove the following: Corollary.— If on an interval || a b a function has an upper bound + ∞, then there is at least one value of x, x1 on || a b such that in every neighborhood of x1 the upper bound of the function is +∞. § 3 Monotonic Functions; Inverse Functions. Definitions.— If a single-valued function f (x) on an interval || a b is such that f (x1) < f (x2) whenever x1 < x 2, the function is said to be monotonic increasing on that interval. If f (x1) > f (x2) whenever x1 < x 2, the function is said to be monotonic decreasing. If there exist three values of x on the interval || a b , x1, x2, and x3 such that f (x2) > f (x1) and f (x2) > f (x3) while x1 < x 2 < x 3 or f (x2) < f (x1) and f (x2) < f (x3), while x1 < x 2 < x 3, the function is said to be oscillating on that interval. A function which is not oscillating on an interval is called non-oscillating. It should be noticed that a function is not necessarily oscillating even if it is not monotonic. That is, it may be constant on some parts of the interval. The terms monotonic and oscillating are not convenient of application to multiple- valued functions. Hence we restrict their use to single-valued functions.

FUNCTIONS IN GENERAL. SPECIAL CLASSES OF FUNCTIONS. 37 □□□□□□□□ □□□□□□□□ ###### □□□□□□□□□□□□

@@@######E E E E E E E□□□□□□□□□□□□ 0 x1 x2 x3 x y Fig. 7.

38 INFINITESIMAL ANALYSIS. Definition.— A function f (x) is said to have a finite number of oscillations on an interval || a b if there exists a finite number of points a = x0, x1, . . . , xn = b, such that on each interval | | xk−1 xk (k = 1, 2, 3, . . . , n) f (x) is non-oscillating. It is evident that if a function has only a finite number of oscillations on an interval || a b and if there is no subinterval of || a b on which the function is constant, then the interval || a b may be subdivided into a finite set of intervals on each of which the function is monotonic. Such a function may be called partitively monotonic (Abteilungsweise monoton). The function f (x) = sin 1 x , for x ≠ 0, and f (x) = 0, for x = 0, is an example of a function with an infinite number of oscillations on every neighborhood of a point. f (x) = x sin 1 x , for x ≠ 0, f (0) = 0, and f (x) = x2 sin 1 x , for x ≠ 0, f (0) = 0 have the above property and also are continuous (see page 48 for meaning of the term continuous function). There exist continuous functions which have an infinite number of oscillations on every neighborhood of every point. The first function of this type is probably the one discovered by Weierstrass,2 which is continuous over an interval and does not possess a derivative at 2According to F. Klein, this function was discovered by Weierstrass in 1851. See Klein, Anwendung

FUNCTIONS IN GENERAL. SPECIAL CLASSES OF FUNCTIONS. 39

40 INFINITESIMAL ANALYSIS. any point on this interval (see page 120). Other functions of this type have been published by Peano, Moore, and others. 3 These latter investigators have obtained the function in question in connection with space-filling curves. Theorem 20. If y is a monotonic function of x on the interval || a b , with bounds A and B, then in turn x is a single-valued monotonic function of y on | | A B , whose upper and lower bounds are b and a. Proof. It follows from the monotonic character of y as a function of x that for no two values of x does y have the same value. Hence for every value of y on | | A B there exists der Differential- und Integralrechnung auf Geometrie , p. 83 et seq. The function was first published in a paper entitled Abhandlungen aus der Functionenlehre , Du Bois Reymond , Crelle's Journal, Vol. 79, p. 29 (1874). 3G. Peano, Sur une courbe, qui remplit toute une aire plane, Mathematische Annalen , Vol. 36, pp. 157– 160 (1890). Cesaro, Sur la repr´ esentation analytique des r´ egions et des courbes qui les remplisent, Bulletin des Sciences Math´ ematiques, 2d Ser., Vol. 21, pp. 257–267. E. H. Moore , On Certain Crinkly Curves . Transactions of the American Mathematical Society, Vol. 1, pp. 73–90 (1899). See also Steinitz, Mathe- matische Annalen , Vol. 52, pp. 58–69 (1899).

FUNCTIONS IN GENERAL. SPECIAL CLASSES OF FUNCTIONS. 41 one and only one value of x. That is, x is a single-valued function of y.4 Moreover, it is clear that for any three values of y, y1, y2, y3, such that y2 is between y1 and y3, the corresponding values of x, x1, x2, x3, are such that x2 is between x1 and x3, i.e., x is a monotonic function of y, which completes the proof of the theorem. Corollary.— If a function f (x) has a finite number k of oscillations and is constant on no interval, then its inverse is at most (k + 1)-valued. For example, the inverse of y = x2 is double-valued. § 4 Rational, Exponential, and Logarithmic Functions. Definitions.— The symbol am, where m is a positive integer and a any real number whatever, means the product of m factors a. This definition gives a meaning to the symbol y = amxm + am−1xm−1 + . . . + a1x + a0, where a0 . . . a m are any real numbers and m any positive integer. In this case y is called a rational integral function of x or a polynomial in x.5 In case y = amxm + am−1xm−1 + . . . + a1·x + a0 bnxn + bn−1xn−1 + . . . + b1·x + b0 , m and n being positive integers and ak (k = 0 , . . . m) and bl (l = 0 , . . . n) being real numbers, y is called a rational function of x. If yn + yn−1R1(x) + yn−2R2(x) + . . . + yRn−1(x) + Rn(x) = 0 , where R1(x) . . . R n(x) are rational functions of x, then y is said to be an algebraic function of x. Any function which is not algebraic is transcendental. The symbol ax, where x = m n , m and n being positive integers and a any positive real number, is defined to be the nth root of the mth power of a. By elementary algebra it is easily shown that ax1·ax2 = ax1+x2 and (ax1)x2 = ax1·x2. If y = ax, then y is an exponential function of x. At present this function is defined only for rational values of x. 4It is clear that the independent variable y of the inverse function may not take on all values of a continuum even if x does take on all such values. 5The notion of polynomial finds its natural generalization in that of a power series y = c0 + c·x + c2·x2 + . . . + cnxn + . . . For conditions under which a series defines y as a function of x see Chapter IV,§3.

42 INFINITESIMAL ANALYSIS. Theorem 21. The function ax for x on the set [m n] is a monotonic increasing function if 1 < a, and a monotonic decreasing function if 0 < a < 1. Proof. (a) For integral values of x the theorem is obvious. (b) If x1 = m1 n1

and x2 = m2 n1 , where m2 n1 > m1 n1 , then ax1 < a x2 if a > 1 and ax1 > a x2 if a < 1. The proof of this follows at once from case (a), since a m1 n1 = (a 1 n1)m1 (by definition and elementary algebra) and a m2 n1 = (a 1 n1)m2 . (c) If x1 = m1 n1 and x2 = m2 n2 , where m1 n1 < m2 n2 , we have a m1 n1 = a m1·n2 n1·n2 and a m2 n2 = a m2·n1 n2·n1 , where m1·n2 <m2·n1, which reduces case (c) to case (b). This theorem makes it natural to define ax, where a > 1 and x is a positive irrational number, as the least upper bound of all numbers of the form [a m n] , where [m n] is the set of all positive rational numbers less than x, i.e., ax = B [a m n] . It is, however, equally natural to define ax as B [a p q] , where [p q] is the set of all rational numbers greater than x. We shall prove that the two definitions are equivalent. Lemma.— If [x] is the set of all positive rational numbers, then B[ax] = 1 if a > 1

 43 and B[ax] = 1 if a < 1. Proof. We prove the lemma only for the case a > 1, the argument in the other case being similar. If x is any positive rational number, m n , then the number 1 n is less than or equal to x, and since ax is a monotonic function, a 1 n <= a m n . But [1 n] is a subset of [1 n] . Hence B[ax] = B [a 1 n] , where [n] is the set of all positive integers. If B [a 1 n] were less than 1, then there would be a value, n1, of n such that a 1 n1 < 1. This implies that a < 1, which is contrary to the hypothesis. On the other hand, if B [a 1 n] > 1, there is a number of the form 1 + e, where e > 0, such that 1 + e < a 1 n for every n. Hence (1 + e)n < a for every n, but by the binomial theorem for integral exponents (1 + e)n > 1 + ne, and the latter expression is clearly greater than a if n > a e . Since B [a 1 n] cannot be either greater or less than 1, B [a 1 n] = 1. Theorem 22. If x is any real number, and [m n] the set of all rational numbers less than x, and [p q] the set of all rational numbers greater than x, then B [a m n] = B [a p q] if a > 1, B [a m n] = B [a p q] if 0 < a < 1. Proof. We give the detailed proof only in the case a > 1, the other case being similar. By the lemma, since B [p q−m n] is zero, B [a p q−a m n] = B [a p q (1−a m n−p q)] is also zero. Now if B [a m n ⁄= B [a p q] ,

 since a p q is always greater than a m n , B [a p q] −B [a m n] = ε > 0. But from this it would follow that a p q−a m n is at least as great as ε, whereas we have proved that B [a p q−a m n] = 0. Hence B [a m n] = B [a p q] if a > 1. Definition.— In case x is a positive irrational number, and [p q] is the set of all rational numbers greater than x, and [m n] is the set of all rational numbers less than x, then ax = B [a p q] = B [a m n] if a > 1 and ax = B [a p q] = B [a m n] if 0 < a < 1. Further, if x is any negative real number, then ax = 1 a−x and a0 = 1. Theorem 23. The function ax is a monotonic increasing function of x if a > 1, and a monotonic decreasing function if 0 < a < 1. In both cases its upper bound is +∞and its lower bound is zero, the function taking all values between these bounds; further, ax1·ax2 = ax1+x2 and (ax1)x2 = ax1·x2. The proof of this theorem is left as an exercise for the reader. The proof is partly contained in the preceding theorems and involves the same kind of argument about upper and lower bounds that is used in proving them. Definition.— The logarithm of x (x > 0) to the base a (a > 0) is a number y such that ay = x, or aloga x = x. That is, the function log a x is the inverse of ax. The identity ax1·ax2 = ax1+x2 gives at once loga x1 + loga x2 = log a(x1·x2),

and (ax1)x2 = ax1·x2 gives x1·loga x2 = log a xx1 2 .

FUNCTIONS IN GENERAL. SPECIAL CLASSES OF FUNCTIONS. 45 By means of Theorem 20, the logarithm log a x, being the inverse of a monotonic func- tion, is also a monotonic function, increasing if 1 < a and decreasing if 0 < a < 1. Further, the function has the upper bound + ∞and the lower bound −∞, and takes on all real values as x varies from 0 to +∞. Thus it follows that for x < a , 1 < b, B(logb x) = log b a = log b(Bx). By means of this relation it is easy to show that the function xa, (x > 0) is monotonic increasing for all values of a, a > 0, that its lower bound is zero and its upper bound is +∞, and that it takes on all values between these bounds. The proof of these statements is left to the reader. The general type of the argument required is exemplified in the following, by means of which we infer some of the properties of the function xx. If x1 < x 2, then log2 x1 < log2 x2, and x1·log2 x1 < x 2·log2 x2, and log2 xx1 1 < log2 xx2 2 . □ xx1 1 < x x2 2 . Hence xx, (x > 0) is a monotonic increasing function of x. Since the upper bound of x·log2 x = log 2 xx is +∞, the upper bound of xx is +∞. The lower bound of xx is not negative, since x > 0, and must not be greater than the lower bound of 2 x, since if x < 2, xx < 2x; since the lower bound of 2 x is zero 6 the lower bound of xx must also be zero. Further theorems about these functions are to be found on pages 51, 63, 76, 97, and 129. 6The lower bound of ax is zero by Theorem 23.

46 INFINITESIMAL ANALYSIS.

Chapter 4 THEORY OF LIMITS. § 1 Definitions. Limits of Monotonic Functions. Definition.— If a point a is a limit point of a set of values taken by a variable x, the variable is said to approach a upon the set; we denote this by the symbol x .= a. a may be finite or +∞or−∞. In particular the variable may approach a from the left or from the right, or in the case where a is finite, the variable may take values on each side of the limit point. Even when the variable takes all values in some neighborhood on each side of the limit point it may be important to consider it first as taking the values on one side and then those on the other. Definition.— A value b (b may be +∞or−∞or a finite number) is a value approached by f (x) as x approaches a if for every V∗(a) and V (b) there is at least one value of x such that x is in V∗(a) and f (x) in V (b). Under these conditions f (x) is also said to approach b as x approaches a. Definition.— If b is the only value approached as x approaches a, then b is called the limit of f (x) as x approaches a. This is also indicated by the phrase “ f (x) converges to a unique limit b as x approaches a,” or “ f (x) approaches b as a limit ,” or by the notation L x .= a f (x) = b. The function f (x) is sometimes referred to as the limitand. The set of values taken by x is sometimes indicated by the symbol for a limit, as, for example, L x>a x .=a f (x) = b or L x<a x .=a f (x) = b or L x|[x] x .=a f (x) = b. The first means that x approaches a from the right, the second that x approaches a from the left, and the third indicates that the approach is over some set [x] otherwise defined. 47

48 INFINITESIMAL ANALYSIS. Definition.— If f (x) is single-valued and converges to a finite limit as x approaches a and L x .=a f (x) = f (a), then f (x) is said to be continuous at x = a. By reference to §3, Chapter II, the reader will see that if b is a value approached by f (x) as x approaches a, then (a, b) is a limit point of the set of

points (x, f (x)). Theorem 18 therefore translates into the following important statement: Theorem 24. If f (x) is any function defined for any set [x] of which a is a (finite or +∞ or−∞) limit point, then there is at least one value (finite or +∞or−∞) approached by f (x) as x approaches a. Corollary.— If f (x) is a bounded function, the values approached by f (x) are all finite. In the light of this theorem we see that the existence of L x .=a f (x) simply means that f (x) approaches only one value, while the non-existence of L x .=a f (x) means that f (x) approaches at least two values as x approaches a. In case f (x) is monotonic (and hence single-valued), or more generally if f (x) is a non-oscillating function, these ideas are particularly simple. We have in fact the theorem: Theorem 25. If f (x) is a non-oscillating function for a set of values [x] < a , a being a limit point of [x], then as x approaches a from the left on the set [x], f (x) approaches one and only one value b, and if f (x) is an increasing function, b = Bf (x) for x on [x], whereas if f (x) is a decreasing function, b = Bf (x) for x on [x]. Proof. Consider an increasing non-oscillating function and let b = Bf (x) for x on [x].

 In view of the preceding theorem we need to prove only that no value b′= b can be a value approached. Suppose b′> b; then since Bf (x) = b, there would be no value of f (x) between b and b′, that is, there would be a V (b′) which could contain no value of f (x), whence b′> b is not a value approached. Suppose b′< b. Then take b′< b″< b, and since Bf (x) = b, there would be a value x1 of [x] such that f (x1) > b″. If x1 < x < a , then b″< f (x1) □ f (x), because f (x) cannot decrease as x increases. This defines a V∗(a) and a V (b′) such that if x is in V∗(a), f (x) cannot be in V (b′). Hence b′< b is not a value approached. A like argument applies if f (x) is a decreasing function, and of course the same theorem holds if x approaches a from the right. It does not follow that L x<a x .=a f (x) = L x>a x .=a f (x), nor that either of these limits is equal to f (a). A case in point is the following: Let the temperature of a cooling body of water be the independent variable, and the amount of heat given out in cooling from a certain fixed temperature be the dependent variable. When the water reaches the freezing-point a great amount of heat is given off without any change in temperature. If the zero temperature is approached from below, the function approaches a definite limit point k, and if the temperature approaches zero from above, the function approaches an entirely different point k′. This function, however, is multiple-valued at the □ □ □ □ □ □ □ □□□□□□□□□□ Fig. 12 Heat Temp. zero point. A case where the limit fails to exist is the following: The function y = sin 1 /x ; (see Fig. 8, page 38) approaches an infinite number of values as x approaches zero. The value of the function will be alternately 1 and −1, as x = 2 π , 2 3π , 2 5π etc., and for all values of x between any two of these the function will take all values between 1 and −1. Clearly

 every value between 1 and −1 is a value approached as x approaches zero. In like manner y = 1 x sin 1 x approaches all values between and including + ∞and−∞, cf. Fig. 13. The functions ax, log a x, xa defined in§4 of the last chapter are all monotonic and all satisfy the condition that L x>a x .=a f (x) = f (a) = L x<a x .=a f (x) at all points where the functions are defined. These functions are therefore all continuous. § 2 The Existence of Limits. Theorem 26. A necessary and

sufficient condition 1 that f (x) shall converge to a unique limit b as x approaches a, i.e., that L x .=a f (x) = b, is that for every V (b) there shall exist a V∗(a) such that for every x in V∗(a), f (x) is in V (b). Proof. (1) The condition is necessary. It is to be proved that if L x .=a f (x) = b, then for every V (b) there exists a V∗(a) such that for every x in V∗(a) the corresponding f (x) is in V (b). If this conclusion did not follow, then for some V (b) every V∗(a) would contain at least one x′such that f (x′) is not in V (b). There is thus defined a set of points [x′] of which a is a limit point. By Theorem 20 f (x) would approach at least one value b′as x approaches a on the set [x′]. But by the definition of [x′], b′is distinct from b. Hence the hypothesis would be contradicted. (2) The condition is sufficient. We need only to show that if for every V (b) there exists a V∗(a) such that for every x in V∗(a) the corresponding f (x) is in V (b), then f (x) can approach no other value than b. If b′= b, then there exists a V (b′) and a V (b) which have no point in common. Now if V ∗ (a) is such that for every x of V ∗ (a), f (x) is in V (b), then for no such x is f (x) in V (b′) and hence b′is not a value approached. The reader should observe that this proof applies also to multiple-valued functions, although worded to fit the single-valued case. It is worthy of note that in case b is a finite number, our theorem becomes: 1This means: (a) If L x .=a f (x) = b, then for every V (b) there exists a V ∗(a), as specified by the theorem. (b) If for every V (b) there exists a V ∗(a) as specified, then L x .=a f (x) = b. A condition is necessary for a certain conclusion if it can be deduced from that conclusion; a condition sufficient for a conclusion is one from which the conclusion can be deduced. A man sufficient for a task is a man who can perform the task, while a man necessary for the task is such that the task cannot be performed without him.

52 INFINITESIMAL ANALYSIS. A necessary and sufficient condition that L x .=a f (x) = b is that for every ε > 0 there exists a Vε ∗(a) such that for every x in Vε ∗(a),|f (x)−b|< ε. In case a also is finite, the condition may be stated in a form which is frequently used as the definition of a limit, namely: L x .=a f (x) = b means that for every ε > 0 there exists a δε > 0 such that if |x−a|< δε and x≠ a, then |f (x)−b|< ε.2 Theorem 27. A necessary and sufficient condition that f (x) shall converge to a finite limit as x approaches a is that for every ε > 0 there shall exist a Vε ∗(a) such that if x1 and x2 are any two values of x in Vε ∗(a), then |f (x1)−f (x2)|< ε. Proof. (1) The condition is necessary. If L x .=a f (x) = b and b is finite, then by the preceding theorem for every ε 2 > 0 there exists a V∗(a) such that if x1 and x2 are in V∗(a), then |f (x1)−b|< ε 2 and |f (x2)−b|< ε 2, from which it follows that |f (x1)−f (x2)|< ε. (2) The condition is sufficient. If the condition is satisfied, there exists a V∗(a) upon which the function f (x) is bounded. For let ε be some fixed number. By hypothesis there exists a V∗(a) such that if x and x0 are on V∗(a), then |f (x)−f (x0)|< ε. Taking x0 as a fixed number, we have that f (x0)−ε < f (x) < f (x0) + ε for every x on V∗(a). Hence there is at least one finite value, b, approached by f (x). Now for every ε > 0 there exists a V∗ ε (a) such that if x1 and x2 are any two values of x in V∗ ε (a), 2The ε subscript to δε or to Vε ∗(a) denotes that δε or Vε ∗(a) is a function of ε. It is to be noted that inasmuch as any number less than δε is effective as δε, δε is a multiple-valued function of ε.

THEORY OF LIMITS. 53 |f (x1)−f (x2)|< ε. Hence by the definition of value approached there is an xε of V∗ ε (a) for which |f (xε)−b|< ε (a) and |f (xε)−f (x)|< ε

(b) for every x of V∗ ε (a). Hence, combining (a) and (b), for every x of V∗ ε (a) we have |f (x)−b|< 2ε, and hence by the preceding theorem we have L x .=a f (x) = b. In case a as well as b is finite, Theorem 27 becomes: A necessary and sufficient condition that L x .=a f (x) shall exist and be finite is that for every ε > 0 there exists a δε > 0 such that |f (x1)−f (x2)|< ε for every x1 and x2 such that x1 = a, x 2 = a, |x1−a|< δε, |x2−a|< δε. In case a is +∞the condition becomes: For every ε > 0 there exists a Nε > 0 such that |f (x1)−f (x2)|< ε for every x1 and x2 such that x1 > N ε, x2 > N ε. The necessary and sufficient conditions just derived have the following evident corol- laries: Corollary 1. The expression L x .=a f (x) = b, where b is finite, is equivalent to the expression L x .=a (f (x)−b) = 0 , and whether b is finite or infinite L x .=a f (x) = b is equivalent to L x .=a (−f (x)) =−b.

 Corollary 2. The expressions L x .=a f (x) = 0 and L x .=a |f (x)|= 0 are equivalent. Corollary 3. The expression L x .=a f (x) = b is equivalent to L y .=0 f (y + a) = b, where y + a = x. Corollary 4. The expression L x<a x .=a f (x) = b is equivalent to L z .=+∞ f (a + 1 z) = b, where z = 1 x−a . The reader should verify these corollaries by writing down the necessary and sufficient condition for the existence of each limit. The following less obvious statement is proved in detail for the case when b is finite, the case when b is +∞or−∞being left to the reader. Corollary 5. If L x .=a f (x) = b, then L x .=a |f (x)|=|b|. Proof. By the necessary condition of Theorem 26 for every ε there exists a V∗ ε (a) such that for every x1 of V∗ ε (a) |f (x1)−b|< ε. If f (x1) and b are of the same sign, then □□|f (x1)|−|b| □□=|f (x1)−b|< ε, and if f (x1) and b are of opposite sign, then □□|f (x1)|−|b| □□<|f (x1)−b|< ε. Hence, by the sufficient condition of Theorem 26, L x .=a |f (x)| exists and is equal to |b|.

 Corollary 6. If a function f (x) is continuous at x = a, then|f (x)|is continuous at x = a. It should be noticed that L x .=a |f (x)|=|b| is not equivalent to L x .=a f (x) = b. Suppose f (x) = +1 for all rational values of x and f (x) =−1 for all irrational values of x. Then L x .=a |f (x)|= +1, but L x .=a f (x) does not exist, since both +1 and −1 are values approached by f (x) as x approaches any value whatever. Definition.— Any set of numbers which may be written [xn], where n = 0, 1, 2, . . . , κ, or n = 0, 1, 2, . . . , κ, . . . , is called a sequence. To the corollaries of this section may be added a corollary related to the definition of a limit. Corollary 7. If for every sequence of numbers [xn] having a as a limit point, L x|[xn] x .=a f (x) = b, then L x .=a f (x) = b. Proof. In case two values b and b1 were approached by f (x) as x approaches a, then, as in the first part of the proof of Theorem 26, two sequences could be chosen upon one of which f (x) approached b and upon the other of which f (x) approached b1. § 3 Application to Infinite Series. The theory of limits has important applications to infinite series. An infinite series is defined as an expression of the form ∞∑ k=1 ak = a1 + a2 + a3 + . . . + an + ... If Sn is defined as a1 + . . . + an = n∑ k=1 ak,

 n being any positive integer, then the sum of the series is defined as L n=∞ Sn = S if this limit exists. If the limit exists and is finite, the series is said to be convergent. If S is infinite or if Sn approaches more than one value as n approaches infinity, then the series is divergent. For example, S is infinite if ∞∑

k=1 ak = 1 + 1 + 1 + 1 . . . , and Sn has more than one value approached if ∞∑ k=1 ak = 1−1 + 1−1 + 1 . . . . It is customary to write Rn = S−Sn. A necessary and sufficient condition for the convergence of an infinite series is obtained from Theorem 27. (1) For every ε > 0 there exists an integer Nε, such that if n > N ε and n′> N ε then |Sn−Sn′|< ε. This condition immediately translates into the following form: (2) For every ε > 0 there exists an integer Nε, such that if n > N ε, then for every k |an + an+1 + . . . + an+k|< ε. Corollary.— If ∞∑ k=1 ak is a convergent series, then L k .=∞ ak = 0. Definition.— A series ∞∑ k=0 ak = a0 + a1 + . . . + an + . . . is said to be absolutely convergent if |a0|+|a1|+ . . . +|an|+ . . . is convergent. Since |an + an+1 + . . . + an+k|<|an|+|an+1|+ . . . |an+k|, the above criteria give

THEORY OF LIMITS. 57 Theorem 28. A series is convergent if it is absolutely convergent. Theorem 29. If ∞∑ k=0 bk is a convergent series all of whose terms are positive and ∞∑ k=0 ak is a series such that for every k,|ak|□ bk, then ∞∑ k=0 ak is absolutely convergent. Proof. By hypothesis n∑ k=0 |ak|□ n∑ k=0 bk. Hence n∑ k=0 |ak| is bounded, and being an increasing function of n, the series is convergent according to Theorem 25. This theorem gives a useful method of determining the convergence or divergence of a series, namely, by comparison with a known series. Such a known series is the geometric series a + ar + ar2 + . . . + arn + . . . , where 0 < r < 1 and a > 0. In this series n∑ k=0 ark = a1−rn+1 1−r < a 1−r , which shows that the series is convergent. Moreover, it can easily be seen to have the sum a 1−r . If r >= 1, the geometric series is evidently divergent. This result can be used to prove the “ratio-test” for convergence. Theorem 30. If there exists a number, r, 0 < r < 1, such that □□□□ an an−1 □□□□< r for every integral value of n, then the series a1 + a2 + . . . + an + . . . (1) is absolutely convergent. If □□□an an−1 □□□>= 1 for every n, the series is divergent.

58 INFINITESIMAL ANALYSIS. Proof. The series (1) may be written a1 + a1 a2 a1 + a1 a2 a1 ·a3 a2 + . . . + a1 a2 a1 . . . an an−1 (2) □□□an an−1 □□□< r , this is numerically less term by term than a1 + a1r + a1r2 + . . . + a1rn + . . . (3) and therefore converges absolutely. If □□□an an−1 □□□□ 1, an □ a1 for every n; hence, by the corollary, page 56, (1) is divergent. Nothing is said about the case when □□□□ an an−1 □□□□< 1, but L n .=∞ □□□□ an an−1 □□□□= 1. It is evident that the ratio test need be applied only to terms beyond some fixed term an, since the sum of the first n terms a1 + a2 + . . . + an may be regarded as a finite number Sn and the whole series as Sn + an+1 + an+2 + . . . , i.e., a finite number plus the infinite series an+1 + an+2 + ... § 4 Infinitesimals. Computation of Limits. Theorem 31. A necessary and sufficient condition that L x .=a f (x) = b is that for the function ε(x) defined by the equation f (x) = b + ε(x) L x .=a ε(x) = 0 . Proof. Take ε(x) = f (x)−b and apply Theorem 26. A special case of this theorem is: A necessary and sufficient condition for the convergence of a series to a finite value b is that for every ε > 0 there exists an integer Nε, such that if n > N ε then|Rn|< ε.

THEORY OF LIMITS. 59 Definition.— A function f (x) such that L x .=a f (x) = 0 is called an infinitesimal as x approaches a.3 Theorem 32. The sum, difference, or product of two infinitesimals is an infinitesimal. Proof. Let the two infinitesimals be f1(x) and f2(x). For every ε, 1 > ε > 0, there exists a V∗ 1 (a) for every x of which

|f1(x)|< ε 2, and a V∗ 2 (a) for every x of which |f2(x)|< ε 2. Hence in any V∗(a) common to V∗ 1 (a) and V∗ 2 (a) |f1(x) + f2(x)|□|f1(x)|+|f2(x)|< ε, |f1(x)−f2(x)|□|f1(x)|+|f2(x)|< ε, |f1(x)·f2(x)|=|f1(x)|·|f2(x)|< ε. From these inequalities and Theorem 26 the conclusion follows. Theorem 33. If f (x) is bounded on a certain V∗(a) and ε(x) is an infinitesimal as x approaches a, then ε(x)·f (x) is also an infinitesimal as x approaches a. Proof. By hypothesis there are two numbers m and M , such that M > f (x) > m for every x on V∗(a). Let k be the larger of |m|and|M|. Also by hypothesis there exists for every ε a Vε ∗(a) within V∗(a) such that if x is in Vε ∗(a), then |ε(x)|< ε k or k|ε(x)|< ε. But for such values of x |f (x)·ε(x)|< k·|ε(x)|< ε, and hence for every ε there is a Vε ∗(a) such that for x an Vε ∗(a) |f (x)·ε(x)|< ε. 3No constant, however small if not zero, is an infinitesimal, the essence of the latter being that it varies so as to approach zero as a limit. Cf. Goursat, Cours d'Analyse, tome I, p. 21, etc.

60 INFINITESIMAL ANALYSIS. Corollary.— If f (x) is an infinitesimal and c any constant, then c·f (x) is an infinitesimal. Theorem 34. If L x .=a f1(x) = b1 and L x .=a f2(x) = b2, b1 and b2 being finite, then L x .=a {f1(x)±f2(x)}= b1±b2, (α) L x .=a {f1(x)·f2(x)}= b1·b2; (β) and if b2≠ 0, L x .=a f1(x) f2(x) = b1 b2 (γ) Proof. According to Theorem 31, we write f1(x) = b1 + ε1(x), f2(x) = b2 + ε2(x), where ε1(x) and ε2(x) are infinitesimals. Hence f1(x) + f2(x) = b1 + b2 + ε1(x) + ε2(x), (α′) f1(x)·f2(x) = b1·b2 + b1·ε2(x) + b2·ε1(x) + ε1(x)·ε2(x). (β′) But by the preceding theorem the terms of (α′) and (β′) which involve ε1(x) and ε2(x) are infinitesimals, and hence the conclusions (α) and (β) are established. To establish (γ), observe that by Theorem 26 there exists a V∗(a) for every x of which |f2(x)−b2|<|b2|and hence upon which f2(x)≠ 0. Hence f1(x) f2(x) = b1 + ε1(x) b2 + ε2(x) = b1 b2 + b2ε1(x)−b1ε2(x) b2{b2 + ε2(x)}, the second term of which is infinitesimal according to Theorems 32 and 33. Some of the cases in which b1 and b2 are±∞are covered by the following theorems. The other cases (∞−∞, ∞ ∞, 0 0 , etc.), are treated in Chapter VI. Theorem 35. If f2(x) has a lower bound on some V∗(a), and if L x .=0 f1(x) = +∞, then L x .=0 {f2(x) + f1(x)}= +∞.

THEORY OF LIMITS. 61 Proof. Let M be the lower bound of f2(x). By hypothesis, for every number E there exists a V∗ E(a) such that for x on V∗ E(a) f1(x) > E−M. Since f2(x) > M, this gives f1(x) + f2(x) > E, which means that f1(x) + f2(x) approaches the limit +∞. Theorem 36. If L x .=a f1(x) = +∞or−∞, and if f2(x) is such that for a V∗(a), f2(x) has a lower bound greater than zero or an upper bound less than zero, then L x .=a {f1(x)·f2(x)}is definitely infinite; i.e., if f2(x) has a lower bound greater than zero and L x .=a f1(x) = +∞, then L x .=a {f1(x)·f2(x)}= +∞, etc. Proof. Suppose f2(x) has a lower bound greater than zero, say M , and that L x .=a f1(x) = +∞. Then for every E there exists a V∗ E(a) within V∗(a) such that for every x1 of V∗ E(a), f1(x1) > E M , and therefore f1(x1)·f2(x1) >= f1(x1)·M > E . Hence by the definition of limit of a function L x .=a {f1(x)·f2(x)}= +∞. If we consider the case where f2(x) has an upper bound less than zero, we have in the same manner L{f1(x)·f2(x)}=−∞. Similar statements hold for the cases in which L x .=a f1(x) =−∞. Corollary.— If f2(x) is positive and has a finite upper bound and L x .=a f1(x) = +∞, then L x .=a f1(x) f2(x) = +∞. Theorem 37. If L x .=a f (x) = +∞, then L x .=a 1 f (x) = 0 , and there is a vicinity V∗(a) upon which f (x) > 0. Conversely, if L x .=a f (x) = 0

and there is a V∗(a) upon which f (x) > 0, then L x .=a 1 f (x) = +∞. Proof. If L x .=a f (x) = +∞, then for every ε there exists a Vε ∗(a) such that if x is in Vε ∗(a), then f (x) > 1 ε and 1 f (x) < ε.

62 INFINITESIMAL ANALYSIS. □ L x .=a 1 f (x) = 0, since both f (x) and 1 f (x) are positive. Again, if L x .=a f (x) = 0, then for every ε there is a V∗ ε (a) such that for x in V∗ ε (a), |f (x)|< ε or 1 f (x) > 1 ε (f (x) being positive). Hence L x .=a 1 f (x) = +∞. Corollary 1. If f1(x) has finite upper and lower bounds on some V∗(a) and L x .=a f2(x) = +∞ or−∞, then L x .=a f1(x) f2(x) = 0. Corollary 2. If f2(x) is positive and f1(x) has a positive lower bound on some V∗(a) and L x .=a f2(x) = 0, then L x .=a f1(x) f2(x) = +∞. Theorem 38. (change of variable). If (1) L x .=a f1(x) = b1 and L x .=b1 f2(y) = b2 when y takes all valves of f1(x) corresponding to values of x on some V∗(a), and if (2) f1(x)≠ b1 for x on V∗(a), then L x .=a f2(f1(x)) = b2. Proof. (α) Since L y .=b1 f2(y) = b2, for every V (b2) there exists a V∗(b1) such that if y is in V∗(b1), f2(y) is in V (b2). Since L x .=a f1(x) = b1, for every V (b1) there exists a V∗(a) in V∗(a) such that if x is in V∗(a), f1(x) is in V (b1). But by (2) if x is in V∗(a), f1(x)≠ b. Hence (β) for every V∗(b1) there exists a V∗(a) such that for every x in V∗(a), f1(x) is in V∗(b1). Combining statements (α) and (β): for every V (b2) there exists a V∗(a) such that for every x in V∗(a) f1(x) is in V∗(b1), and hence f2(f (x)) is in V (b2). This means, according to Theorem 26, that L x .=a f2(f1(x)) = b2.

THEORY OF LIMITS. 63 Theorem 39. If L x .=a f1(x) = b and L y .=b f2(y) = f2(b), where y takes all values taken by f1(x) for x on some V∗(a), then L x .=a f2(f1(x)) = f2(b). Proof. The proof of the theorem is similar to that of Theorem 38. In this case the notation f2(b) implies that b is a finite number. Thus for every ε1 there exists a Vε1 ∗(a) entirely within V∗(a) such that if x is in Vε1 ∗(a), |f1(x)−b|< ε1. Furthermore, for every ε2 there exists a δε2 such that for every y, y≠ b,|y−b|< δε2, |f2(y)−f2(b)|< ε2. But since |f2(y)−f2(b)|= 0 when y = b, this means that for all values of y (equal or unequal to b) such that |y−b|< δε2,|f2(y)−f2(b)|< ε2. Now let ε1 = δε2; then, if x is in Vε1 ∗(a), it follows that |f1(x)−b|< δε2 and therefore that |f2(f1(x))−f2(b)|< ε2. Hence L x .=a f2(f1(x)) = f2(b). Corollary 1. If f1(x) is continuous at x = a, and f2(y) is continuous at y = f1(a), then f2(f1(x)) is continuous at x = a. Corollary 2. If k≠ 0, f (x) □ 0, and L x .=a f (x) = b, then L x .=a (f (x))k = bk, under the convention that ∞k =∞if k > 0 and∞k = 0 if k < 0. Corollary 3. If c > 0 and f (x) > 0 and b > 0 and L x .=a f (x) = b, then L x .=a logc f (x) = log c b, under the convention that log c(+∞) = +∞and log c 0 =−∞. The conclusions of the last two corollaries may also be expressed by the equations L x .=a (f (x))k = (L x .=a f (x))k and logc L x .=a f (x) = L x .=a logc f (x). Corollary 4. If L x .=a (f (x))k or L x .=a log f (x) fails to exist, then L x .=a f (x) does not exist.

64 INFINITESIMAL ANALYSIS. § 5 Further Theorems on Limits. Theorem 40. If f (x) □ b for all values of a set [x] on a certain V∗(a), then every value approached by f (x) as x approaches a is less than or equal to b. Similarly if f (x) □ b for all values of a set [x] on a certain V∗(a), then every value approached by f (x) as x approaches a is greater than or equal to b. Proof. If f (x) □ b on V∗(a), then if b′is any value greater than b, and V (b′) any vicinity of b′which does not include b, there is no value of x on V∗(a) for which f (x) is in V (b′). Hence b′is not a value approached. A similar

argument holds for the case where f (x) □ b. Corollary 1. If f (x) □ 0 in the neighborhood of x = a, then if L x .=a f (x) exist, L x .=a f (x) □ 0. Corollary 2. If f1(x) □ f2(x) in the neighborhood of x = a, then L x .=a f1(x) □ L x .=a f2(x) if both these limits exist. Proof. Apply Corollary 1 to f1(x)−f2(x). Corollary 3. If f1(x) □ f2(x) in the neighborhood of x = a, then the largest value ap- proached by f1(x) is greater than or equal to the largest value approached by f2(x). Corollary 4. If f1(x) and f2(x) are both positive in the neighborhood of x = a, and if f1(x) □ f2(x), then if L x .=a f1(x) = 0, it follows that L x .=a f2(x) = 0 . Theorem 41. If [x′] is a subset of [x], a being a limit point of [x′], and if L x .=a f (x) exists, then L x′.=a f (x′) exists and L x .=a f (x) = L x′.=a f (x′). 4 Proof. By hypothesis there exists for every V (b) a V∗(a) such that for every x of the set [x] which is in V∗(a), f (x) is in V (b). Since [x′] is a subset of [x], the same V∗(a) is evidently efficient for x on [x′]. In the statement of necessary and sufficient conditions for the existence of a limit we have made use of a certain positive multiple-valued function of ε denoted by δε. If a given value is effective as a δε, then every positive value smaller than this is also effective. 4The notation f (x′) is used to indicate that x takes the values of the set [x′].

THEORY OF LIMITS. 65 Theorem 42. For every ε for which the set of values of δε has an upper bound there is a greatest δε. Proof. Let B[δε] be the least upper bound of the set of values of δε, for a particular ε. If x is such that|x−a|< B[δε], then there is a δε such that|x−a|< δε. But if |x−a|< δε, |f (x)−b|< ε. Hence, if |x−a|< B[δε],|f (x)−b|< ε. Theorem 43. The limit of the least upper bound of a function f (x) on a variable segment a x , a < x , as the end point approaches a, is the least upper bound of the values approached by the function as x approaches a from the right. Proof. Let l be the least upper bound of the values approached by the function as x approaches a from the right, and let b(x) represent the upper bound of f (x) for all values of x on a x . Since Bf (x) on the segment a x 1 is not greater than Bf (x) on a segment a x 2 if x1 lies on a x 2, b(x) is a non-oscillating function decreasing as x decreases. Hence L x .=a b(x) exists by Theorem 21; and by Corollary 3, Theorem 40, L x .=a b(x) □ l. If L x .=a b(x) = k > l , then there are two vicinities of k, V1(k) contained in V2(k) and V2(k) not containing l. By Theorem 26 a V∗ 1 (a) exists such that if x is in V∗ 1 (a), b(x) is in V1(k). Furthermore, by the definition of b(x), if x1 is an arbitrary value of x on V∗ 1 (a), then there is a value of x in a x 1 such that f (x) is in V (k). Hence k would be a value approached by f (x) contrary to the hypothesis k > l . § 6 Bounds of Indetermination. Oscillation. It is a corollary of Theorem 43 that in the approach to any point a from the right or from the left the least upper bounds and the greatest lower bounds of the values approached by f (x) are themselves values approached by f (x). The four numbers thus indicated may be denoted by f (a + 0) = L x .=a+0 f (x) = ← L x .=a f (x), the least upper bound of the values approached from the right: f (a−0) = L x .=a−0 f (x) = → L x .=a f (x), the least upper bound of the values approached from the left: f (a + 0) = L x .=a+0 f (x) = L ← x .=a f (x), the greatest lower bound of the values approached from the right: f (a−0) = L x .=a−0 f (x) = L → x .=a f (x),

66 INFINITESIMAL ANALYSIS. the greatest lower bound of the values approached from the left. If all four of these values coincide, there is only one value approached and L x .=a f (x) exists. If f (a + 0) and f (a + 0) coincide, this value is denoted by f (a + 0) and is the same as L x>a x .=a f (x). Similarly if f (a−0) and f (a−0) coincide,

their common value, L x<a x .=a f (x), is denoted by f (a−0). The larger of f (a + 0) and f (a−0) is denoted by L x .=a f (x), and is called the upper limit of f (x) as x approaches a. Similarly L x .=a f (x), the lower limit of f (x), is the smaller of f (a + 0) and f (a−0). L x .=a f (x) and L x .=a f (x) are called the bounds of indetermination of f (x) at x = a (Unbestimmtheitsgrenzen). See the Encyclop¨ adie der mathematischen Wissenschaften, II 41. In order that a function shall be continuous at a point a it is necessary and sufficient that f (a) = f (a + 0) = f (a + 0) = f (a−0) = f (a−0). (a) The difference between the greatest and the least of these values is called the oscillation of the function at the point a. It is denoted by Oaf (x), and according to the theorem above is equivalent to the lower bound of all values of Of (x), where Of (x) = Bf (x)−Bf (x) for a segment V (a). Ob af (x) is used for the oscillation of f (x) on the segment a b . It is sometimes also used for the oscillation of f (x) on the interval || a b . The word oscillation may also be applied to the difference between the upper and lower bounds of the function on a V∗(a). Denote this by OV∗(a)f (x). The lower bound of these values may be denoted by O∗ af (x) and is the difference between the greatest and the least of the four values f (a + 0), f (a−0), f (a + 0), f (a−0). The reader will find it a useful exercise to construct examples and to enumerate the different ways in which a function may be discontinuous, according as f (a + 0) or f (a−0) exist or do not exist, and according as f (a) does or does not coincide with any of the values approached by f (x). (Compare the reference to the E. d. m. W. given above.) The principal classification used is into discontinuities of the first kind , where f (a + 0) and f (a−0) both exist, and discontinuities of the second kind , where not both f (a + 0) and f (a−0) exist. Theorem 44. If a is a limit point of [x], then a necessary and sufficient condition that b2 and b1 shall be the upper and lower bounds of indetermination of f (x), as x .= a, is that for every set of four numbers a1, a2, c1, c2, such that 5 a1 < b 1 < c 1 < c 2 < b 2 < a 2, 5If b1 =−∞, a1 = b1 replaces a1 < b 1. If b2 = +∞, a2 = b2 replaces b2 < a 2.

THEORY OF LIMITS. 67 there exists a V∗(a) such that for every x on V∗(a) a1 < f (x) < a 2, and for some x′, x′′on V∗(a) f (x′) > c 2 and f (x′′) < c 1. Proof. I. The condition is necessary. It is to be proved that if b2 and b1 are the upper and lower bounds of indetermination of f (x), as x .= a on [x], then for every four numbers a1 < b 1 < c 1 < c 2 < b 2 < a 2 there exists a V∗(a) such that:— (1) For all values of x on V∗(a), a1 < f (x) < a 2. If this conclusion does not follow, then for a particular pair of numbers a1, a2, there are values of f (x) greater than a2 or less than a1 for x on any V∗(a), and by Theorems 24 and 40 there is at least one value approached greater than b2 or less than b1. This would contradict the hypothesis, and there is therefore a V∗(a) such that for all values of x on V∗(a), a1 < f (x) < a 2. (2) For some x′, x′′on V∗(a), f (x′) > c 2 and f (x′′) < c 1. If this conclusion should not follow, then for some V∗(a) there would be no x′such that f (x′) > c 2, or no x′′such that f (x′′) < c 1, and therefore b1 and b2 could not both be values approached. II. The condition is sufficient. It is to be proved that b2 and b1 are the upper and lower bounds of the values approached. If the condition is satisfied, then for every four numbers a1, a2, c1, c2, such that a1 < b 1 < c 1 < c 2 < b 2 < a 2 there is a V∗(a) such that for all x’s on V∗(a) a1 < f (x) < a 2, and for some x′, x′′, f (x′) > c 2 and f (x′′) < c 1. By Theorem 24 there are values approached, and hence we need only to show that b2 is the least

upper and b1 the greatest lower bound of the values approached. Suppose some B > b 2 is the least upper bound of the values approached; a2 may then be so chosen that b2 < a 2 < B , so that by hypothesis for x on V∗(a) B cannot be a value approached. Again, suppose B < b 2 to be the least upper bound; c may then be chosen so that B < c 2, and hence for some value x′on each V∗(a), f (x′) < c 2. By the set of values f (x′) there is at least one value approached. This value is greater than c2 > B . Therefore B cannot be the least upper bound. Since the least upper bound may not be either less than b2 or greater than b2, it must be equal to b2. A similar argument will prove b1 to be the greatest lower bound of the values approached.

Chapter 5 CONTINUOUS FUNCTIONS. § 1 Continuity at a Point. The notion of continuous functions will in this chapter, as in the definition on page 48, be confined to single-valued functions. It has been shown in Theorem 34 that if f1(x) and f2(x) are continuous at a point x = a, then f1(x)±f2(x), f 1(x)·f2(x), f 1(x)/f 2(x), (f2(x)≠ 0) are also continuous at this point. Corollary 1 of Theorem 39 states that a continuous function of a continuous function is continuous. The definition of continuity at x = a, namely, L x .=a f (x) = f (a), is by Theorem 26 equivalent to the following proposition: For every ε > 0 there exists a δε > 0 such that if |x−a|< δε, then |f (x)−f (a)|< ε. It should be noted that the restriction x≠ a which appears in the general form of Theorem 26 is of no significance here, since for x = a,|f (x)−f (a)|= 0 < ε. In other words, we may deal with vicinities of the type V (a) instead of V∗(a). The difference of the least upper and the greatest lower bound of a function on an interval || a b has been called in Chapter IV, page 66, the oscillation of f (x) on that interval, and denoted by Ob a(x). The definition of continuity and Theorem 27, Chapter III, give the following necessary and sufficient condition for the continuity of a function f (x) at the For every ε > 0 there exists a δε > 0 such that if |x1−a|< δ ε, and |x2−a|< δ ε then|f (x1)−f (x2)|< ε 2 . This means that for all values of x1 and x2 on the segment (a−δε) (a + δε) B|f (x1)−f (x2)|□ ε 2 < ε, and this means Bf (x)−Bf (x) < ε, 69

70 INFINITESIMAL ANALYSIS. or Oa+δε a−δεf (x) < ε. Then we have Theorem 45. If f (x) is continuous for x = a, then for every ε > 0 there exists a Vε(a) such that on Vε(a) the oscillation of f (x) is less than ε. Theorem 46. If f (x) is continuous at a point x = a and if f (a) is positive, then there is a neighborhood of x = a upon which the function is positive. Proof. If there were values of x, [x′] within every neighborhood of x = a for which the function is equal to or less than zero, then by Theorem 24 there would be a value approached by f (x′) as x′approaches a on the set [x′]. That is, by Theorem 40, there would be a negative or zero value approached by f (x), which would contradict the hypothesis. § 2 Continuity of a Function on an Interval. Definition.— A function is said to be continuous on an interval || a b if it is continuous at every point on the interval. Theorem 47. If f (x) is continuous on a finite interval || a b , then for every ε > 0, || a b can be divided into a finite number of equal intervals upon each of which the oscillation of f (x) is less than ε.1 Proof. By Theorem 45 there is about every point of || a b a segment σ upon which the oscillation is less than ε. This set of segments [σ] covers || a b , and by Theorem 11 || a b can be divided into a finite number of equal intervals each of which is interior to a σ; this

gives the conclusion of our theorem. Theorem 48. (Uniform continuity.) If a function is continuous on a finite interval || a b , then for every ε > 0 there exists a δε > 0 such that for any two values of x, x1, and x2, on || a b where|x1−x2|< δε,|f (x1)−f (x2)|< ε. Proof. This theorem may be inferred in an obvious way from the preceding theorem, or it may be proved directly as follows: By Theorem 27, for every ε there exists a neighborhood Vε(x′) of every x′of || a b such that if x1 and x2 are on Vε(x′), then |f (x1)−f (x2)|< ε. The Vε(x)'s constitute a set of segments which cover || a b . Hence, by Theorem 12, there is a δε such that if |x1−x2|<δε, x1 and x2 are on the same Vε(x′) and consequently|f (x1)−f (x2)|< ε. 1The importance of this theorem in proving the properties of continuous functions seems first to have been recognized by Goursat. See his Cours d'Analyse, Vol. 1, page 161.

CONTINUOUS FUNCTIONS. 71 The uniform continuity theorem is due to E. Heine .2 The proof given by him is essentially that given above. In 1873 L¨uroth3 gave another proof of the theorem which is based on the following definition of continuity: A single-valued function is continuous at a point x = a′if for every positive ε there exists a δε, such that for every x1 and x2 on the interval | | a−δε a + δε,|f (x1)−f (x2)|< ε (Theorem 45). By Theorem 42 there exists a greatest δ for a given point and for a given ε. Denote this by Δ ε(x). If the function is continuous at every point of || a b , then for every ε there will be a value of Δ ε(x) for every point of the interval, i.e., Δ ε(x), for any particular ε, will be a single-valued function of x. The essential part of L¨uroth'sproof consists in establishing the following fact: If f (x) is continuous at every point of its interval, then for any particular value of ε the function Δ ε(x) is also a continuous function of x. From this it follows by Theorem 50 that the function Δ ε(x) will actually reach its greatest lower bound, that is, will have a minimum value; and this minimum value, like all other values of δε, will be positive. 4 This minimum value of Δ ε(x) on the interval under consideration will be effective as a δε, independent of x. The property of a continuous function exhibited above is called uniform continuity, and Theorem 48 may be briefly stated in the form: Every function continuous on an interval is uniformly continuous on that interval. 5 This theorem is used, for example, in proving the integrability of continuous functions. See page 125. Theorem 49. If a function is continuous on an interval || a b , it is bounded on that interval. Proof. By Theorem 46 the interval || a b can be divided into a finite number of intervals, such that the oscillation on each interval is less than a given positive number ε. If the number of intervals is n, then the oscillation on the interval || a b is less than nε. Since the function is defined at all points of the interval, its value being f (x1) at some point x1, it follows that every value of f (x) on || a b is less than f (x1) + nε and greater than f (x1)−nε; which proves the theorem. 2E. Heine: Die Elemente der Functionenlehre , Crelle, Vol. 74 (1872), p. 188. 3L¨uroth: Bemerkung ¨ uber Gleichm¨ assige Stetigkeit, Mathematische Annalen, Vol. 6, p. 319. 4It is interesting to note that this proof will not hold if the condition of Theorem 26 is used as a definition of continuity. On this point see N. J. Lennes : The Annals of Mathematics, second series, Vol. 6, p. 86. 5It should be noticed that this theorem does not hold if "segment" is substituted for "interval," as is shown by the function 1 x on the segment 0 1, which is continuous but not uniformly continuous. The function is defined and continuous for every value of x on this segment, but not for every value of

x on the interval | | 0 1 .

72 INFINITESIMAL ANALYSIS. Theorem 50. If a function f (x) is continuous on an interval || a b , then the function as- sumes as values its least upper and its greatest lower bound. Proof. By the preceding theorem the function is bounded and hence the least upper and greatest lower bounds are finite. By Theorem 19 there is a point k on the interval || a b such that the least upper bound of the function on every neighborhood of x = k is the same as the least upper bound on the interval || a b . Denote the least upper bound of f (x) on || a b by B. It follows from Theorem 43 that B is a value approached by f (x) as x approaches k. But since L x .=k f (x) = f (k), the function being continuous at x = k, we have that f (k) = B. In the same manner we can prove that the function reaches its greatest lower bound. Corollary.— If k is a value not assumed by a continuous function on an interval || a b , then f (x)−k or k−f (x) is a continuous function of x and assumes its least upper and greatest lower bounds. That is, there is a definite number Δ which is the least difference between k and the set of values of f (x) on the interval || a b . Theorem 51. If a function is continuous on an interval || a b , then the function takes on all values between its least upper and its greatest lower bound. Proof. If there is a value k between these bounds which is not assumed by a continuous function f (x), then by the corollary of the preceding theorem there is a value Δ such that no values of f (x) are between k−Δ and k + Δ. With ε less than Δ divide the interval || a b into subintervals according to Theorem 47, such that the oscillation on every interval is less than ε. No interval of this set can contain values of f (x) both greater and less than k, and no two consecutive intervals can contain such values. Suppose the values of f (x) on the first interval of this set are all greater than k, then the same is true of the second interval of the set, and so on. Hence it follows that all values of f (x) on || a b are either greater than or less than k, which is contrary to the hypothesis that k lies between the least upper and the greatest lower bounds of the function on || a b . Hence the hypothesis that f (x) does not assume the value k is untenable. By the aid of Theorem 51 we are enabled to prove the following: Theorem 51a. If f1(x) is continuous at every point of an interval || a′b′except at a certain point a, and if L x .=a f1(x) = +∞and L x .=a f2(x) =−∞, then for every b, finite or +∞or−∞, there exist two sequences of points, [xi] and [x′ i] (i = 0, 1, 2, . . .), each sequence having a as a limit point, such that L i .=∞ {f1(xi) + f2(x′ i)}= b.

CONTINUOUS FUNCTIONS. 73 Proof. Let [x′ i] be any sequence whatever on || a′b′having a as a limit point, and let x0 be an arbitrary point of || a′b′. Since f1(x) assumes all values between f1(x0) and +∞, and since L x .=a f2(x) =−∞, it follows, in case b is finite, that for every i greater than some fixed value there exists an xi such that f1(xi) + f2(x′ i) = b. In case b = +∞, xi is chosen so that f1(xi) + f2(x′ i) > i. Corollary.— Whether f1(x) and f2(x) are continuous or not, if L x .=a f1(x) = + ∞and L x .=a f2(x) =−∞, there exists a pair of sequences [xi] and [x′ i] such that L i .=∞ {f1(xi) + f2(xi)} is +∞or−∞. Theorem 52. If y is a function, f (x), of x, monotonic and continuous on an interval || a b , then x = f−1(y) is a function of y which is monotonic and continuous on the interval | | f (a) f (b). Proof. By Theorem 20 the function f−1(y) is monotonic and has as upper and lower bounds a and b. By Theorems 50 and 51 the function is defined for every value of y between and including f (a) and f (b) and for no

other values. We prove the function continuous on the interval | | f (a) f (b) by showing that it is continuous at any point y = y1 on this interval. As y approaches y1 on the interval | | f (a) y1 , f−1(y) approaches a definite limit g by Theorem 25, and by Theorem 40 a < g □ f−1(y1) □ b. If g < f−1(y1), then for values of x on the interval | | g f (y1) there is no corresponding value of y, contrary to the hypothesis that f (x) is defined at every point of the interval || a b . Hence g = f−1(y1), and by similar reasoning we show that f−1(y) approaches f−1(y1) as y approaches y1 on the interval, | | y1 f−1(b). Theorem 53. If f (x) is single-valued and continuous with A, B as lower and upper bounds, on an interval || a b and has a single-valued inverse on the interval, | | A B then f (x) is monotonic on || a b .

74 INFINITESIMAL ANALYSIS. Proof. If f (x) is not monotonic, then there must be three values of x, x1 < x 2 < x 3, such that either f (x1) □ f (x2) □ f (x3) or f (x1) □ f (x2) □ f (x3). In either case, if one of the equality signs holds, the hypothesis that f (x) has a single-valued inverse is contradicted. If there are no equality signs, it follows by Theorem 51 that there are two values of x, x4 and x5, such that x1 < x 4 < x 2 < x 5 < x 3, and f (x4) = f (x5), in contradiction with the hypothesis that f (x) has a single-valued inverse. Corollary.— If f (x) is single-valued, continuous, and has a single-valued inverse on an interval || a b , then the inverse function is monotonic on | | A B . § 3 Functions Continuous on an Everywhere Dense Set. Theorem 54. If the functions f1(x) and f2(x) are continuous on the interval || a b , and if f1(x) = f2(x) on a set everywhere dense, then f1(x) = f2(x) on the whole interval. 6 Proof. Let [x′] be the set everywhere dense on || a b for which, by hypothesis, f1(x) = f2(x). Let x″be any point of the interval not of the set [x′]. By hypothesis x″is a limit point of the set [x′], and further f1(x) and f2(x) are continuous at x = x″. Hence L x .=x″ f1(x) = f1(x″) and L x .=x″ f2(x) = f2(x″). But by Theorem 41 L x′.=x″ f1(x′) = L x .=x″ f1(x), 6I.e., if a function f (x), continuous on an interval | | a b , is known on an everywhere dense set on that interval, it is known for every point on that interval.

CONTINUOUS FUNCTIONS. 75 and by Theorem 41 L x′.=x″ f2(x′) = L x .=x″ f2(x). Therefore f1(x″) = f2(x″). Definition.— On an interval || a b a function f (x′) is uniformly continuous over a set [x′] if for every ε > 0 there exists a δε > 0 such that for any two values of x′, x′ 1, and x′ 2 an || a b , for which|x′ 1−x′ 2|< δε,|f (x′ 1)−f (x′ 2)|< ε. Theorem 55. If a function f (x′) is defined on a set everywhere dense on the interval || a b and is uniformly continuous over that set, then there exists one and only one function f (x) defined on the full interval || a b such that: (1) f (x) is identical with f (x′) where f (x′) is defined. (2) f (x) is continuous on the interval || a b . Proof. Let x″be any point on the interval || a b , but not of the set [x′]. We first prove that L x′.=x″ f (x′) exists and is finite. By the definition of uniform continuity, for every ε there exists a δε such that for any two values of x′, x′ 1, and x′ 2, where |x′ 1−x′ 2|< δε,|f (x′ 1)−f (x2)|< ε. Hence we have for every pair of values x′ 1 and x′ 2 where|x′ 1−x″|< δε 2 and|x′ 2−x″|< δε 2 that|f (x′ 1)−f (x′ 2)|< ε. By Theorem 23 this is a sufficient condition that L x′.=x″ f (x′) shall exist and be finite. Let f (x) denote a function identical with f (x′) on the set [x′] and equal to L x′.=x″ f (x′) at all points x″. This function is defined upon the continuum, since all points x″on || a b are limit points of the set [x′]. Hence the function has the property that L x1 .=x f (x′) = f (x) for every x of || a b .

76 INFINITESIMAL ANALYSIS. We next prove that f (x) is continuous at every point on the interval || a b , in other words that f (x) cannot approach a value b different from f (x1) as x approaches x1. We already know that f (x) approaches f (x1) on the set [x′]. If b is another value approached, then for every positive ε and δ there is an xεδ such that |xεδ−x1|< δ, |f (xεδ)−b|< ε. (1) Since f (xεδ) = L x′.=xεδ f (x′) we have that for every ε > 0 there exists a δε > 0 such that for every x′for which|x′−xεδ|< δε, |f (x′)−f (xεδ)|< ε. (2) From (1) and (2) we have |f (x′)−b|< 2ε. (3) Since the δ of (1) is any positive number, there is an xεδ on every neighborhood of x1 and hence by (2) and (3) an x′on every neighborhood of x1 such that |f (x′)−b|< 2ε, ε being arbitrary and b a constant different from f (x′′ 1). But this is contrary to the fact proved above, that L x′.=x1 f (x′) exists and is equal to f (x1). Hence the function is continuous at every point of the interval || a b . The uniqueness of the function follows directly from Theorem 54. This theorem can be applied, for example, to give an elegant definition of the exponen- tial function (see Chap. III). We first show that the function a m n is uniformly continuous on the set of all rational values between x1 and x2, and then define ax on the continuum as that continuous function which coincides with a m n for the rational values m n . The properties of the function then follow very easily. It will be an excellent exercise for the reader to carry out this development in detail. § 4 The Exponential Function. Consider the function defined by the infinite series 1 + x + x2 2! + x3 3! + . . . + xn n! + ... (1) Applying the ratio test for the convergence of infinite series we have xn n!÷xn−1 (n−1)! = x n . If n′is a fixed integer larger than x, this ratio is always less than x n′< 1. The series (1) therefore converges absolutely for every value of x, and we may denote its sum by e(x).

CONTINUOUS FUNCTIONS. 77 From Chap. I, page 12, we have that e(1) = L n .=∞ (1 + 1 n)n = e. Theorem 56. L n .=∞ (1 + x n)n , where [n] is the set of all positive integers, exists and is equal to e(x) for all values of x. Proof. Let En(x) = n∑ k=0 xk k! (where 0! = 1). Then, since (1 + x n)n = 1 + n! (n−1)!·x n + n! (n−2)!·2! (x n)2 + . . . + n! n! (x n)n , it follows that □□□En(x)− (1 + x n)n□□□= □□□□□ n∑ k=2 (1 k!− n! (n−k)!·k! nk) xk □□□□□ □ n∑ k=2 (1 k!−n(n−1) . . . (n−k + 1) k! nk) ·|xk| < n∑ k=2 nk−(n−k + 1)k k! nk ·|xk|. Now, since nk−(n−k + 1)k = (k−1){nk−1 + nk−2·(n−k + 1) + . . . + (n−k + 1)k−1}< (k−1)k·nk−1, it follows that □□□En(x)− (1 + x n)n□□□< n∑ k=2 |x|k (k−2)!·n < x2·e(|x|) n . For a fixed value of x, therefore, we have (1 + x n)n = En(x) + ε1(n), where ε1(n) is an infinitesimal as n .=∞. At the same time e(x) = En(x) + ε2(n), where ε2(n) is an infinitesimal as n .=∞. Hence L n .=∞ (1 + x n)n = e(x).

78 INFINITESIMAL ANALYSIS. Theorem 57. L z .=∞ (1 + x z) , where [z] is the set of all real numbers, exists and is equal to e(x). Proof. If z is any number greater than 1, let nz be the integer such that nz □ z < n z + 1. Hence, if x > 0, 1 + x nz □ 1 + x z > 1 + x nz + 1 . (1) Hence (1 + x nz)nz+1 □ (1 + x z)z > (1 + x nz + 1)nz , (2) or (1 + x nz)(1 + x nz)nz □ (1 + x z)z > (1 + x nz + 1)nz+1 · 1 1 + x nz+1 . (3) Since L z .=∞ (1 + x nz) = 1, and L z .=∞ (1 + x nz + 1) = 1, and L z .=∞ (1 + x nz)nz = e(x), and L z .=∞ (1 + x nz + 1)nz+1 = e(x), the inequality (3), together with Corollary 3, Theorem 40, leads to the result: L z .=∞ (1 + x z)z = e(x). The argument is similar if x < 0. Corollary.— L z .=∞ (1 + x z)z = e(x), where [z] is any set of numbers with limit point + ∞. Theorem 58. The function e(x) is the same as ex where

e = 1 + 1 + 1 2! + 1 3! + . . .

CONTINUOUS FUNCTIONS. 79 Proof. By the continuity of zx as a function of z (see Corollary 2 of Theorem 39), it follows that, since L n .=∞ (1 + 1 n)n = e, L n .=∞ (1 + 1 n)nx = ex. But (1 + 1 n)nx = (1 + x nx)nx = (1 + x z)z , where z = nx. Hence by Theorem 39 ex = L z .=∞ (1 + x z)z , and by the corollary of Theorem 57 the latter expression is equal to e(x). Hence we have ex = 1 + x + x2 2! + x3 3! + ... (1) (1) is frequently used as the definition of ex, ax being defined as ex·loge a.

80 INFINITESIMAL ANALYSIS.

Chapter 6 INFINITESIMALS AND INFINITES. § 1 The Order of a Function at a Point. An infinitesimal has been defined (page 59) as a function f (x) such that L x .=a f (x) = 0 . A function which is unbounded in every vicinity of x = a is said to have an infinity at a, to be or become infinite at x = a, or to have an infinite singularity at x = a.1 The reciprocal of an infinitesimal at x = a is infinite at this point. A function may be infinite at a point in a variety of ways: (a) It may be monotonic and approach + ∞or−∞as x .= a; for example, 1 x as x approaches zero from the positive side. (b) It may oscillate on every neighborhood of x = a and still approach +∞or−∞as a unique limit; for example, sin 1 x + 2 x as x approaches zero. (c) It may approach any set of real numbers or the set of all real numbers; an example of the latter is sin 1 x x as x approaches zero. See Fig. 13, page 50. 1It is perfectly compatible with these statements to say that while f (x) has an infinite singularity at x = a, f (a) = 0 or any other finite number. For example, a function which is 1 x for all values of x except x = 0 is left undefined for x = 0 and hence at this point the function may be defined as zero or any other number. This function illustrates very well how a function which has a finite value at every point may nevertheless have infinite singularities. 81

82 INFINITESIMAL ANALYSIS. (d) +∞and−∞may both be approached while no other number is approached; for example, 1 x as x approaches zero from both sides. Definition of Order.— If f (x) and φ(x) are two functions such that in some neighborhood V∗(a) neither of them changes sign or is zero, and if L x .=a f (x) φ(x) = k, where k is finite and not zero, then f (x) and φ(x) are said to be of the same order at x = a. If L x .=a f (x) φ(x) = 0, then f (x) is said to be infinitesimal with respect to φ(x), and φ(x) is said to be infinite with respect to f (x). If L x .=a f (x) φ(x) = +∞or −∞, then, by Theorem 37, φ(x) is infinitesimal with respect to f (x), and f (x) infinite with respect to φ(x). If f (x) and φ(x) are both infinitesimal at x = a, and f (x) is infinitesimal with respect to φ(x), then f (x) is infinitesimal of a higher order than φ(x), and φ(x) of lower order than f (x). If φ(x) and f (x) are both infinite at x = a, and f (x) is infinite with respect to φ(x), then f (x) is infinite of higher order than φ(x), and φ(x) is infinite of lower order than f (x).2 The independent variable x is usually said to be an infinitesimal of the first order as x approaches zero, x2 of the second order, etc. Any constant≠ 0 is said to be infinite of zero order, 1 x is of the first order, 1 x2 of the second order, etc. This usage, however, is best confined to analytic functions. In the general case there are no two infinitesimals of consecutive order. Evidently there are as many different orders of infinitesimals between x and x2 as there are numbers between 1 and 2; i.e., x1+k is of higher order than x for every positive value of k. Since L x .=a f1(x) f2(x) = 1 k whenever L x .=a f2(x) f1(x) = k, we have Theorem 59.

If f1(x) is of the same order as f2(x), then f2(x) is of the same order as f1(x). 2This definition of order is by no means as general as it might possibly be made. The restriction to functions which are not zero and do not change sign may be partly removed. The existence of L x .=a f (x) φ(x) is dispensed with for some cases in §4 on Rank of Infinitesimals and Infinites. For an account of still further generalizations (due mainly to Cauchy) see E. Borel , S´ eries` aTermes Positifs , Chapters III and IV, Paris, 1902. An excellent treatment of the material of this section together with extensions of the concept of order of infinity is due to E. Borlotti , Calcolo degli Infinitesimi , Modena, 1905 (62 pages).

INFINITESIMALS AND INFINITES. 83 Theorem 60. The function cf (x) is of the same order as f (x), c being any constant not zero. Proof. By Theorem 34, L x .=a cf (x) f (x) = c. Theorem 61. If f1(x) is of the same order as f2(x), and f2(x) is of the same order as f3(x), then f1(x) and f3(x) are of the same order. Proof. By hypothesis L x .=a f1(x) f2(x) = k1 and L x .=a f2(x) f3(x) = k2. By Theorem 34, L x .=a f1(x) f2(x)·L x .=a f2(x) f3(x) = L x .=a f1(x) f3(x). (By definition, f2(x) ≠ 0 and f3(x) ≠ 0 for some neighborhood of x = a.) Hence L x .=a f1(x) f3(x) = k1·k2. Theorem 62. If f1(x) and f2(x) are infinitesimal (infinite) and neither is zero or changes sign on some V∗(a), then f1(x)·f2(x) is infinitesimal (infinite) of a higher order than either. Proof. L x .=a f1(x)·f2(x) f2(x) = L x .=a f1(x) = 0 . (±∞.) Theorem 63. If f1(x), . . . , fn(x) have the same sign on some V∗(a) and if f2(x), . . . , fn(x) are infinitesimal (infinite) of the same or higher (lower) order than f1(x), then f1(x) + f2(x) + f3(x) + . . . + fn(x) is of the same order as f1(x), and if f2(x), f3(x), . . . , fn(x) are of higher (lower) order than f1(x), then f1(x)±f2(x)±f3(x)±. . . ±fn(x) is of the same order as f1(x). Proof. We are to show that L x .=a f1(x) + f2(x) + . . . + fn(x) f1(x) = k ≠ 0. By hypothesis, L x .=a f2(x) f1(x) = k2, L x .=a f3(x) f1(x) = k3, . . . , L x .=a fn(x) f1(x) = kn, and L x .=a f1(x) f1(x) = 1.

84 INFINITESIMAL ANALYSIS. Hence, by Theorem 30, L x .=a {f1(x) f1(x) + f2(x) f1(x) + f3(x) f1(x) + . . . + fn(x) f1(x) } = 1 + k2 + . . . +kn = k ≠ 0, since all the k's are positive or zero. Similarly, under the second hypothesis, L x .=a f1(x)±f2(x)±. . . ±fn(x) f1(x) = L x .=a {f1(x) f1(x)±f2(x) f1(x)±. . . ±fn(x) f1(x) } = 1 + 0 + . . . + 0 = 1 . Theorem 64. If f3(x) and f4(x) are infinitesimals with respect to f1(x) and f2(x), then L x .=a {f1(x) + f3(x)}·{f2(x) + f4(x)} f1(x)·f2(x) = 1. Proof. L x .=a {f1(x) + f3(x)}·{f2(x) + f4(x)} f1(x)·f2(x) = L x .=a f1(x)·f2(x) + f1(x)·f4(x) + f3(x)·f2(x) + f3(x)·f4(x) f1(x)·f2(x) = L x .=a f1(x)·f2(x) f1(x)·f2(x) + L x .=a f1(x)·f4(x) f1(x)·f2(x) + L x .=a f3(x)·f2(x) f1(x)·f2(x) + L x .=a f3(x)·f4(x) f1(x)·f2(x) = 1. § 2 The Limit of a Quotient. Theorem 65. If as x .= a, ε1(x) is an infinitesimal with respect to f1(x) and ε2(x) with respect to f2(x), then the values approached by f1(x) + ε1(x) f2(x) + ε2(x) and f1(x) f2(x) as x approaches a are identical. Proof. This follows from the identity f1(x) + ε1(x) f2(x) + ε2(x) = f1(x) f2(x)· (1 + ε1(x) f1(x)) (1 + ε2(x) f2(x)), since ε1(x) f1(x) and ε2(x) f2(x) are infinitesimal.

INFINITESIMALS AND INFINITES. 85 Corollary.— If f1(x) and f2(x) are infinite at x = a, then f1(x) + c f2(x) + d and f1(x) f2(x) approach the same values. Theorem 66. If L x .=a f1(x) φ1(x) = L x .=a f2(x) φ2(x) = k, and if L x .=a φ1(x) φ2(x) = l is finite, then k = L x .=a f1(x) + f2(x) φ1(x) + φ2(x) = L x .=a1 f1(x) φ1(x), provided l ≠ −1 if k

is finite, and provided l > 0 if k is infinite. Proof. f1(x) + f2(x) φ1(x) + φ2(x)−f2(x) φ2(x) = f1(x)φ2(x)−f2(x)φ1(x) φ2(x)(φ1(x) + φ2(x)) , f1(x) + f2(x) φ1(x) + φ2(x) = f2(x) φ2(x) + (f1(x) φ1(x)−f2(x) φ2(x)) · (1 1 + φ2(x) φ1(x)) . In case k is finite, the second term of the right-hand member is evidently infinitesimal if $\lambda = -1$ and the theorem is proved. In the case where k is infinite we write the above identity in the following form: f1(x) + f2(x) φ1(x) + φ2(x) = f1(x) φ1(x)· 1 1 + φ2(x) φ1(x) + f2(x) φ2(x)· 1 1 + φ1(x) φ2(x) . Both terms of the second member approach + ∞or both−∞if l > 0. Corollary.— If φ1(x) and φ2(x) are both positive for some V∗(a), and if k = L x .=a f1(x) φ1(x) = L x .=a f2(x) φ2(x), then L x .= a f1(x) + f2(x) φ1(x) + φ2(x) = k whenever k is finite. If k is infinite, the condition must be added that φ1(x) φ2(x) has a finite upper and a non-zero lower bound. Theorem 67. If f1(x) and f2(x) are both infinitesimals as x .= a, then a necessary and sufficient condition that L x .=a f1(x) f2(x) = k (k finite and not zero) is that in the equation f1(x) = k·f2(x) +ε(x), ε(x) is an infinitesimal of higher order than f1(x) or f2(x).

86 INFINITESIMAL ANALYSIS. Proof. (1) The condition is necessary. —Since L x .=a f1(x) f2(x) = k, f1(x) f2(x) = k + ε′(x), or f1(x) = f2(x)·k + f2(x)·ε′(x), where L x .=a ε′(x) = 0 (Theorem 31). By Theorems 60 and 61, f1(x) and f2(x)·k are of the same order, since $k \neq 0$, while by Theorem 62 ε′(x)·f2(x) is of higher order than either f1(x) or f2(x). Hence the function ε(x) = ε′(x)·f2(x) is infinitesimal. (2) The condition is sufficient. —By hypothesis f1(x) = f2(x)·k + ε(x), where f1(x) and f2(x) are of the same order as x .= a, while ε(x) is of higher order than these. Let ε′(x) = ε(x) f2(x) , which by hypothesis is an infinitesimal. We then have f1(x) f2(x) = k + ε′(x). Hence, by Theorem 31, L x .=a f1(x) f2(x) = k. § 3 Indeterminate Forms. 3 Lemma.— If a b and c d are any two fractions such, that b and d are both positive or both negative, then the value of a + c b + d lies on the interval || a b c d . Proof. Suppose b and d both positive and a b □ a + c b + d , then ab + ad □ ab + bc. □ ad □ bc; □ cd + ad □ cd + bc; □ a + c b + d □ c d. The other cases follow similarly. Theorem 68. If f (x) and φ(x), defined on some V (+∞), are both infinitesimal as x approaches +∞, and if for some positive number h, φ(x + h) is always less than φ(x) and L x .=∞ f (x + h)−f (x) φ(x + h)−φ(x) = k, 3The theorems of this section are to be used in §6 of Chap. VII.

INFINITESIMALS AND INFINITES. 87 then L x .=∞ f (x) φ(x) exists and is equal to k.4 Proof. Let V1(k) and V2(k) be a pair of vicinities of k such that V2(k) is entirely within V1(k). By hypothesis there exists an h and an X2 such that if x > X 2, f (x + h)−f (x) φ(x + h)−φ(x) (1) is in V2(k). Since this is true for every x > X 2, f (x + 2h)−f (x + h) φ(x + 2h)−φ(x + h) (2) is also in V2(k). From this it follows by means of the lemma that f (x + 2h)−f (x) φ(x + 2h)−φ(x) , (3) whose value is between the values of (1) and (2), is also in V2(k). By repeating this argument we have that for every integral value of n, and for every x > X 2, f (x + nh)−f (x) φ(x + nh)−φ(x) is in V2(k). By Theorem 65, for any x L n .=∞ f (x + nh)−f (x) φ(x + nh)−φ(x) = f (x) φ(x) . Hence for every x and for every ε there exists a value of n, Nxε, such that if n > N xε, □□□□ f (x + nh)−f (x) φ(x + nh)−φ(x)−f (x) φ(x) □□□□< ε. Taking ε less than the distance between the nearest end-points of V1(k) and V2(k) it is plain that for every x > X 2, f (x) φ(x) is on V1(k), which, according to Theorem 26, proves that L x .=∞ f (x) φ(x) = k. 4This and the following theorem are due to O. Stolz , who generalized them from the special cases (stated in our corollaries) due to Cauchy. See Stolz und Gmeiner,

Functionentheorie, Vol. 1, p. 31. See also the reference to Bortolotti given on page 82.

88 INFINITESIMAL ANALYSIS. Corollary.— If [n] is the set of all positive integers and φ(n + 1) < φ(n) and f (n) and φ(n) are both infinitesimal as n .=∞, then if L n .=∞ f (n + 1)−f (n) φ(n + 1)−φ(n) = k, it follows that L n .=∞ f (n) φ(n) exists and is equal to k. Theorem 69. If f (x) is bounded on every finite interval of a certain V (+∞), and if φ(x) is monotonic on the same V (+∞) and L x .=∞ φ(x) = +∞, and if for some positive number h L x .=∞ f (x + h)−f (x) φ(x + h)−φ(x) = k, then L x .=∞ f (x) φ(x) exists and is equal to k. Proof. By hypothesis, for every pair of vicinities V1(k) and V2(k), V2(k) entirely within V1(k), there exists an X2 such that if x > X 2, then f (x + h)−f (x) φ(x + h)−φ(x) is in V2(k). From this it follows as in the last theorem that f (x + nh)−f (x) φ(x + nh)−φ(x) is in V2(k). Now make use of the identity f (x + nh) φ(x + nh) = f (x + nh)−f (x) φ(x + nh) + f (x) φ(x + nh) = f (x + nh)−f (x) φ(x + nh)−φ(x) (1− φ(x) φ(x + nh)) + f (x) φ(x + nh). (1) Let [x′] be the set of all points on the interval | | X2 X2 + h , and for this interval let A2 be an upper bound of |f (x′)|and B2 an upper bound of φ(x′). Then φ(x′) φ(x′+ nh) = ε1(x′, n) < B2 φ(X2 + nh) and |f (x′)| φ(x′+ nh) = ε2(x′, n) < A2 φ(X2 + nh).

INFINITESIMALS AND INFINITES. 89 Hence for every ε there exists a value of n, NεV , such that if n > N εV ε1(x′, n) < ε and ε2(x′, n) < ε (2) independently of x′so long as x′is on | | X2 X2 + h . There are then three cases to discuss: (1) k finite. (2) k = +∞. (3) k =−∞. (1) k finite. By the preceding argument, for x > X 2, f (x + nh)−f (x) φ(x + nh)−φ(x) is in V2(k), and hence |f (x′+ nh)−f (x′)| φ(x′+ nh)−φ(x′) < K + εV2, where εV2, is the length of the interval V2(k) and K the absolute value of k. Then, in view of (1), □□□□ f (x′+ nh) φ(x′+ nh)−f (x′+ nh)−f (x′) φ(x′+ nh)−φ(x′) □□□□< (K + εV2)ε1(x′, n) + ε2(x′, n). Now take εV smaller in absolute value than the length of the interval between the closer end-points of V1(k) and V2(k). By (2) there exists a value of n, NεV , such that if n > N εV , ε1(x′, n) < εV 2(K + εV2) and ε2(x′, n) < εV 2 for all values of x′on | | X2 X2 + h . Hence for n > N εV □□□□ f (x′+ nh) φ(x′+ nh)−f (x′+ nh)−f (x′) φ(x′+ nh)−φ(x′) □□□□< (K + εV2) εV 2(K + εV2) + εV 2 = εV , and since for x > X 2 + NεV h there is an n > N εV and an x′between X2 and X2 + h such that x′+ nh = x, it follows that if x > X 2 + NεV , □□□□ f (x) φ(x)−f (x′+ nh)−f (x′) φ(x′+ nh)−φ(x) □□□□< εV ,

90 INFINITESIMAL ANALYSIS. and therefore, f (x) φ(x) is on V1(k). This means, according to Theorem 26, that L x .=∞ f (x) φ(x) = k. (2) k = +∞. If the numbers m1 and m2 are the lower end points of V1(k) and V2(k), then f (x′+ nh)−f (x′) φ(x′+ nh)−φ(x′) > m 2 for x′> X 2. If εV is then chosen less than m2−m1, there will exist a value of NεV such that ε1(x′, n) < εV 2m2 and ε2(x′, n) < εV 2m1 for all values of n > N εV independently of x′so long as x′is in | X2 X2 + h . Then, in view of (1), f (x′+ nh) φ(x′+ nh) > m 2 (1−εV 2m2) −εV 2m2 > m 2−εV 2 (1 + 1 m2) . Since there is no loss of generality if m2 > +1, this proves that for x > X 2 + NεV n, f (x) φ(x) > m 2−εV > m 1, and hence f (x) φ(x) is on V1(k). (3) k =−∞is treated in an analogous manner. Corollary 1. If [n] is the set of all positive integers and if φ(n + 1) > φ(n) and L n=∞ φ(n) =∞, then if L n=∞ L f (n + 1)−f (n) φ(n + 1)−φ(n) = k, it follows that L n=∞ f (n) φ(n) exists and is equal to k . Corollary 2. If f (x) is bounded on every interval, | | x (x

+ 1) , and if L x=∞ f (x + 1)−f (x) = k, then L x=∞ f (x) x exists and is equal to k.

INFINITESIMALS AND INFINITES. 91 § 4 Rank of Infinitesimals and Infinites. Definition.— If on some V∗(a) neither f1(x) nor f2(x) vanishes, and □□□□ f1(x) f2(x) □□□□and □□□□ f2(x) f1(x) □□□□ are both bounded as x approaches a, then f1(x) and f2(x) are of the same rank whether L x .=a f1(x) f2(x) exists or not. 5 The following theorem is obvious. Theorem 70. If f1(x) and f2(x) are of the same order, they are of the same rank, and if f1(x) and f2(x) are of different orders, they are not of the same rank. If f1(x) and f2(x) are of the same rank, they may or may not be of the same order. Theorem 71. If f1(x) and f2(x) are of the same rank as x approaches a, then c·f1(x) and f2(x) are of the same rank, c being any constant not zero. Proof. By hypothesis for some positive number M , □□□□ f1(x) f2(x) □□□□< M and □□□□ f2(x) f1(x) □□□□< M, hence □□□□ c·f1(x) f2(x) □□□□< M·|c|and □□□□ f2(x) c·f1(x) □□□□< M |c|. Theorem 72. If f1(x) and f2(x) are of the same rank and f2(x) and f3(x) are of the same rank as x approaches a, then f1(x) and f3(x) are of the same rank as x approaches a. Proof. By hypothesis, □□□□ f1(x) f2(x) □□□□< M 1 and □□□□ f2(x) f3(x) □□□□< M 2 in some neighborhood of x = a. Therefore □□□□ f1(x) f2(x) □□□□· □□□□ f2(x) f3(x) □□□□< M 1·M2 or □□□□ f1(x) f3(x) □□□□< M 1·M2. In the same manner □□□□ f2(x) f1(x) □□□□< M 1 and □□□□ f3(x) f2(x) □□□□< M 2, whence □□□□ f3(x) f1(x) □□□□< M 1·M2. 5x and x·(sin 1 x + 2) are of the same rank but not of the same order as x approaches zero.

92 INFINITESIMAL ANALYSIS. Theorem 73. If f1(x) is infinitesimal (infinite) and does not vanish on some V∗(a), and if f2(x) and f3(x) are infinitesimal (infinite) of the same rank as x approaches a, then f1(x)·f2(x) is of higher order than f3(x), and f1(x)·f3(x) is of higher order than f2(x). Conversely, if for every function, f1(x), infinitesimal (infinite) at a, f1(x)·f2(x) is of higher order than f3(x), and f1(x)·f3(x) is of higher order than f2(x), then f2(x) and f3(x) are of the same rank. Proof. Since □□□□ f1(x) f3(x) □□□□is bounded as x approaches a, it follows by Theorem 33 that L x .=a f1(x)·f2(x) f3(x) = 0, which proves the first part of the theorem. Since likewise □□□□ f3(x) f2(x) □□□□is bounded, we have that L x .=a f1(x)·f3(x) f2(x) = 0. Suppose that for every f1(x) L x .=a f1(x)·f2(x) f3(x) = 0 and L x .=a f1(x)·f3(x) f2(x) = 0, and that f2(x) and f3(x) are not of the same rank. Then, on a certain subset [x′], L x .=a f2(x′) f3(x′) = 0, or on some other subset [x″], L x .=a f3(x″) f2(x″) = 0. Let f1(x) = f2(x) f3(x) on the set [x′] for which L x .=a f2(x) f3(x) = 0, and x−a on the other points of the continuum; then f1(x) is an infinitesimal as x approaches a, while for the set [x′] L x .=a f1(x′)·f3(x′) f3(x′) = L x .=a f2(x′) f3(x′)·f3(x′) f2(x′) = 1, which contradicts the hypothesis that L x .=a f1(x)·f3(x) f2(x) = 0. Similarly if on a certain subset L x .=a f3(x) f2(x) = 0, we obtain a contradiction by putting f1(x) = f3(x) f2(x) .

Chapter 7 DERIV ATIVES AND DIFFERENTIALS. § 1 Definition and Illustration of Derivatives. Definition.— If the ratio f (x)−f (x1) x−x1 approaches a definite limit, finite or infinite, as x approaches x1, the derivative of f (x) at the point x1 is the limit L x .=x1 f (x)−f (x1) x−x1 . It is implied that the function f (x) is a single-valued function of x. x−x1 is sometimes denoted by Δ x1, and f (x)−f (x1) by Δ f (x1), or, if y = f (x), by Δ y1. An obvious illustration of a derivative occurs in Cartesian geometry when the function is represented by a graph (Fig. 14). f (x)−f (x1) x−x1 is the slope of the line AB. If we suppose that the line AB approaches a fixed direction (which in this

figure would obviously be the case) as x approaches x1, then L x .=x1 f (x)−f (x1) x−x1 will exist and will be equal to the slope of the limiting position of AB. If the point x were taken only on one side of x1, we should have two similar limiting processes. It is quite conceivable, however, that limits should exist on each side, but that they should differ. That case occurs if the graph has a cusp as in Fig. 15. These two cases are distinguished by the terms progressive and regressive derivatives. When the independent variable approaches its limit from below we speak of the progressive derivative, and when from above we speak of the regressive derivative. It follows from the definition of derivative that, except in one singular case, it exists only when both these limits exist and are equal. The exception is the case of a derivative of a function at an end-point of an interval upon which the function is defined. Obviously both the progressive and the regressive derivative cannot exist at such a point. In this case we say the derivative exists if either the progressive or the regressive derivative exists. Whether the progressive and regressive derivatives exist or not, there exist always four so-called derived numbers (which may be ±∞), namely, the upper and lower bounds of 93

94 INFINITESIMAL ANALYSIS. □□□□□□□ x1 x A B f (x) f (x1) Fig. 14. x x 1 x B A B f (x) f (x) f (x1) Fig. 15.

DERIV ATIVES AND DIFFERENTIALS. 95 indetermination of f (x)−f (x1) x−x1 , as x .= x1 from the right or from the left. (Compare page 65, Chapter IV.) The derived numbers are denoted by the symbols. − →D , D− →, ← −D , D← −, analogous to the symbols on page 65. Of course, in every case, − →D □ D− →and← −D □ D← −. If we consider the curve representing the function y = x·sin 1 x at the point x = 0, it is apparent that the limiting position of AB does not exist, although the function is continuous at the point x = 0 if defined as zero for x = 0. For at every maximum and minimum of the curve sin 1 x , x·sin 1 x =±x, and the curve touches the lines x = y and x =−y. That is, f (x)−f (x1) x−x1 approaches every value between 1 and −1 inclusive, as x approaches zero. The notion derivative is fundamental in physics as well as in geometry. If, for instance, we consider the motion of a body, we may represent its distance from a fixed point as a function of time, f (t). At a certain instant of time t1 its distance from the fixed point is f (t1), and at another instant t2 it is f (t2); then f (t1)−f (t2) t1−t2 is the average velocity of the body during the interval of time t1−t2 in a direction from or toward the assumed fixed point. Whether the motion be from or toward the fixed point is of course indicated by the sign of the expression f (t1)−f (t2) t1−t2 . If we consider this ratio as the time interval is taken shorter and shorter, that is, as t2 approaches t1, it will in ordinary physical motion approach a perfectly definite limit. This limit is spoken of as the velocity of the body at the instant t1. Definition.— The derivative of a function y = f (x) is denoted by f′(x) or by Dxf (x) or d f(x) dx or dy dx. f′(x) is also referred to as the derived function of f (x). § 2 Formulas of Differentiation. Theorem 74. The derivative of a constant is zero. More precisely: If there exists a neighborhood of x1 such that for every value of x on this neighborhood f (x) = f (x1), then f′(x1) = 0 .

96 INFINITESIMAL ANALYSIS. Proof. In the neighborhood specified f (x)−f (x1) x−x1 = 0 for every value of x. Corollary.— If f′(x1) exists and if in every V∗(x1) there

is a value of x such that f (x) = f (x1), then f′(x1) = 0. Theorem 75. When for two functions f1(x) and f2(x) the derived functions f′ 1(x) and f′ 2(x) exist at x1 it follows that, except in the indeterminate case ∞−∞, (a) If f3(x) = f1(x) + f2(x), then f3(x) has a derivative at x1 and f′ 3(x1) = f′ 1(x1) + f′ 2(x1). (b) If f3(x) = f1(x)·f2(x), then f3(x) has a derivative at x1 and f′ 3(x1) = f′ 1(x1)·f2(x1) + f1(x1)·f′ 2(x1). (c) If f3(x) = f1(x) f2(x) , then, provided there is a V (x1) upon which f2(x) ≠ 0 , f3(x) has a derivative and f′ 3(x1) = f′ 1(x1)·f2(x1)−f1(x1)·f′ 2(x1) {f2(x1)}2 . Proof. By definition and the theorems of Chapter IV (which exclude the case ∞−∞), (a) f′ 1(x1) + f′ 2(x1) = L x .=x1 f1(x)−f1(x1) x−x1 + L x .=x1 f2(x)−f2(x1) x−x1 (1) = L x .=x1 {f1(x)−f1(x1) x−x1 + f2(x)−f2(x1) x−x1 } (2) = L x .=x1 f1(x) + f2(x)−f1(x1)−f2(x1) x−x1 (3) = L x .=x1 f3(x)−f3(x1) x−x1 . But by definition, f′ 3(x1) = L x .=x1 f3(x)−f3(x1) x−x1 . (4) Hence f′ 3(x1) exists, and f′ 3(x1) = f′ 1(x1) + f′ 2(x1). (b) f3(x) = f1(x)·f2(x). Whenever x ≠ x1 we have the identity f3(x)−f3(x1) x−x1 = f1(x)·f2(x)−f1(x1)·f2(x1) x−x1 = f1(x)·f2(x)−f1(x1)·f2(x) + f1(x1)·f2(x)−f1(x1)·f2(x1) x−x1 = f2(x) {f1(x)−f1(x1) x−x1 } + f1(x) {f2(x)−f2(x1) x−x1 } .

DERIV ATIVES AND DIFFERENTIALS. 97 But the limit of the last expression exists as x .= x1 (except perhaps in the case ∞−∞) and is equal to f2(x1)·f′ 1(x1) + f1(x1)·f′ 2(x1). Hence L x .=x1 f3(x)−f3(x1) x−x1 exists and f′ 3(x1) = f2(x1)·f′ 1(x1) + f′ 2(x1)·f1(x1). (c) f3(x) = f1(x) f2(x). The argument is based on the identity f1(x) f2(x)−f1(x1) f2(x1) x−x1 = f1(x)·f2(x1)−f2(x)·f1(x1) f2(x)·f2(x1)·(x−x1) , which holds when x ≠ x1 and when f2(x) ≠ 0. But f1(x)·f2(x1)−f2(x)·f1(x1) f2(x)·f2(x1)(x−x1) = f1(x)·f2(x1)−f1(x1)·f2(x1) + f1(x1)·f2(x1)−f2(x)·f1(x1) f2(x)·f2(x1)(x−x1) = f2(x1){f1(x)−f1(x1)}−f1(x1){f2(x)−f2(x1)} f2(x)·f2(x1)(x−x1) . As before (excluding the case ∞−∞) we have f′ 3(x1) = f2(x1)·f′ 1(x1)−f′ 2(x1)·f1(x1) {f2(x1)}2 . Corollary.— It follows from Theorems 74 and 75 of this chapter that if f2(x) = a·f1(x) where f′ 1(x) exists, then f′ 2(x) = a·f′ 1(x). Theorem 76. If x > 0, then d dxxk = k·xk−1. (a) If k is a positive integer, we have L x .=x1 xk−xk 1 x−x1 = L x .=x1 { xk−1 + xk−2·x1 + . . . + xk·xk−2 1 + xk−1 1 } = k·xk−1 1 .

98 INFINITESIMAL ANALYSIS. (b) If k is a positive rational fraction m n , we have L x .=x1 x m n−x1 m n x−x1 = L x .=x1 (x 1 n)m − (x1 1 n)m (x 1 n)n − (x1 1 n)n = L x .=x1 1(x 1 n)n−1 + (x 1 n)n−2 · (x1 1 n) + . . . + (x1 1 n)n−1· (x 1 n)m − (x1 1 n)m x 1 n−x1 1 n = 1 n· (x1 1 n)n−1·m (x1 1 n)m−1 , by the preceding case. But 1 n· (x1 1 n)n−1·m (x1 1 n)m−1 = m n x1 m n−1 = k·x1 k−1. (c) If k is a negative rational number and equal to −m, then, by the two preceding cases, L x .=x1 x−m−x1−m x−x1 =−L x .=x1 ·1 xm·xm 1 ·xm−xm 1 x−x1 =−1 x2m 1 ·mxm−1 1 =−mx1 −m−1 . But −mx1 −m−1 = k·xk−1. (d) If k is a positive irrational number, we proceed as follows: Consider values of x greater than or equal to unity. Let x approach x1 so that x > x 1. Since, by Theorem 23, xk is a monotonic increasing function of k for x > 1, it follows that xk−xk 1 x−x1 = xk 1· (x x1)k −1 x−x1 > x k′ 1 · (x x1)k′ −1 x−x1 for all values of k′less than k, and all values of x greater than x1. If k′is a rational number, we have by the preceding cases that L x .=x1 xk′ 1 · (x x1)k′ −1 x−x1 = k′xk′−1 1 . Since xk−1 1 is a continuous function of k, it follows that for every number N less than kxk−1 1 there exists a rational number k′ 1 less than k such that N < k′ 1·xk′−1 1 < k·xk−1 1 .

DERIV ATIVES AND DIFFERENTIALS. 99 Hence, by Theorem 40, xk 1· (x x1)k −1 x−x1 cannot approach a value N less than kxk−1 1 as x approaches x1. By a precisely similar argument we show that a number greater than kxk−1 1 cannot be a value approached. Since there is always at least one value approached, we have that L x .=x1 xk−xk 1 x−x1 = k·xk−1 1 . If x < x 1 as x approaches x1, we write xk−xk 1 x−x1 = xk· (x1 x)k −1 x1−x and proceed as before. If k is a negative number we proceed as under (c). The case in which x1 < 1 is treated similarly. For another proof see page 102. Theorem 77. d dx loga x = 1 x·loga e. Proof. loga(x + Δ x)−loga x Δ x = 1 Δ x loga x + Δ x x = 1 x·loga (1 + Δ x x) x Δ x . But, by Theorem 57, L Δ x .=0 (1 + Δ x x) x Δ x = e. Therefore L Δ x .= 0 loga(x + Δ x)−loga x Δ x = 1 x·loga e. Corollary.— d dx loga x = 1 x . Theorem 78. If f′ 1(x) exists and if there is a V (x1) upon which f1(x) is continuous and possesses a single-valued inverse x = f2(y), then f2(y) is differentiable and f′ 1(x1) = 1 f′ 2(y1), where y1 = f1(x1). 1 1Theorem 78 gives a sufficient condition for the equality dy dx = 1 dx dy .

100 INFINITESIMAL ANALYSIS. If f′(x) is 0 or +∞or−∞the convention 1 +∞= 1 −∞= 0 is understood. Cf. Theo- rem 37. Proof. To prove this theorem we observe that f′ 1(x1) = L x .=x1 f1(x)−f1(x1) x−x1 = L x .=x1 1 x−x1 f1(x)−f1(x1) . By the definition of single-valued inverse (p. 34), x−x1 f1(x)−f (x1) = f2(y)−f2(y1) y−y1 . Hence, by Theorems 38 and 34 and 37, L x .=x1 1 x−x1 f (x)−f (x1) = L y .=y1 1 f2(y)−f2(y1) y−y1 = 1 f′ 2(y). Theorem 79. If (1) f′ 1(x) exists and is finite for x = x1, and f1(x) is continuous at x = x1, (2) f′ 2(y) exists and is finite for y1 = f1(x1), then d dx1 f2{f1(x1)}= f′ 2(y1)·f′ 1(x1). 2 Proof. We prove this theorem first for the case when there is a V∗(x1) upon which f1(x)≠ f1(x1). In this case the following is an identity in x: f2{f1(x)}−f2{f1(x1)} x−x1 = f2{f1(x)}−f2{f1(x1)} f1(x)−f1(x1) ·f1(x)−f1(x1) x−x1 . (1) By hypothesis (2) and Theorem 38, f′ 2(y1) = L y .=y1 f2(y)−f2(y1) y−y1 = L x .=x1 f2{f1(x)}−f2{f1(x1)} f1(x)−f1(x1) . By hypothesis (1), f′ 1(x) = L x .=x1 f1(x)−f1(x1) x−x1 . 2Theorem 79 gives a sufficient condition for the equality dz dx = dz dy dy dz .

DERIV ATIVES AND DIFFERENTIALS. 101 Hence, by equation (1) and Theorem 34, we have the existence of d dx f2{f1(x)}= L x .=x1 f2{f1(x)}−f2{f1(x1)} x−x1 = f′ 2(y1)·f′ 1(x1). If f1(x) = f1(x1) for values of x on every neighborhood of x = x1, then, by hypothesis (1) and the corollary of Theorem 74, f′(x1) = 0 . Let [x′] be the set of points upon which f1(x)≠ f1(x1). (There is such a set unless f (x) is constant in the neighborhood of x = x1.) Then, by the same argument as in the first case, we have d dx′f2{f1(x1)}= f′ 2(y1)·f′ 1(x1) = 0 for x on the set [x′]. Let [x″] be the set of values of x not included in [x′]. Then d dx″f2{f1(x1)}= L x″.=x′ f2{f1(x″)}−f2{f1(x1)} x″−x1 = 0, since the limitand function is zero. Hence both for the set [x′] and for the set [x″] the conclusion of our theorem is that the derivative required is zero. Theorem 80. d dx ax = ax log a. Proof. Let y = ax, therefore log y = x·log a and, by Theorem 77, dy dx y = log a, whence dy dx = y·log a = ax log a.

102 INFINITESIMAL ANALYSIS. This method also affords an elegant proof of Theorem 76, viz., d dx xn = nxn−1. Let y = xn, log y = n log x, dy dx y = n x , dy dx = n·y x = n·xn−1. § 3 Differential Notations. If y = f (x) and L x .=a f (x)−f (x1) x−x1 = K, we denote f (x)−f (x1) by Δ y, and x−x1 by Δ x. Then, by Theorem 31, Δ y = Δ x·K +

Δ x·ε(x), where Δ x·ε(x) is an infinitesimal with respect to Δ y and Δ x for x .= a. This fact is expressed by the equation dy = K·dx, where K = f′(x). Here dy and dx are any numbers that satisfy this equation. There is no condition as to their being small, either expressed or implied, and dx and dy may be regarded as variable or constant, large or small, as may be found convenient. When either dx or dy is once chosen, the other is, of course, determined. The numbers dx and dy are called the differentials of x and y respectively. In Fig. 16, f′(x1) is the tangent of the angle CAB , dx is the length of any segment AB with one extremity at A and parallel to the x-axis, and dy is the length of the segment BC . If x is regarded as approaching x1, then AB′is the infinitesimal Δ x, B′D′is Δ y, while D′C′is ε(x)·Δ x. Hence, by Theorem 73, D′C′is an infinitesimal of higher order than Δ x or Δ y. We thus obtain a complete correspondence between derivatives and the ratios of dif- ferentials. Accordingly, for any formula in derivatives there is a corresponding formula in differentials. Thus corresponding to Theorem 75 we have: Theorem 81. When for two functions f1(x) and f2(x) d f1(x) = f′ 1(x)·dx and d f2(x) = f2(x)·dx at x1, it follows that

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a b Fig. 17. Theorem 83. If f′(x1) exists and if f (x) has a maximum or a minimum at x = x1, then f′(x1) = 0 . Proof. In case of a maximum at x1, it follows directly from the hypothesis that L x.=x1 x>x1 f (x)−f (x1) x−x1 <= 0, and also L x.=x1 x<x1 f (x)−f (x1) x−x1 >= 0, Since f′(x1) exists these limits are equal, that is, the derivative is equal to zero. Similarly in case of a minimum. Theorem 84. If f (x1) = f (x2), f (x) being continuous on the interval | | x1 x2 , and if the derivative exists 3 at every point between x1 and x2, then there is a value ξ between x1 and x2 such that f′(ξ) = 0 . The derivative need not exist at x1 and x2. Proof. (a) The function may be a constant between x1 and x2, in which case f′(x) = 0 for all values of x between x1 and x2 by Theorem 74. (b) There may be values of the function between x1 and x2 which are greater than f (x1) and f (x2). Since the function is continuous on the interval | | x1 x2 , it reaches a least upper bound on this interval at some point x3 (different from x1 and x2). By Theorem 83, f′(x3) = 0 . 3Not necessarily finite.

 (c) In case there are values of the function on the interval | | x1 x2 less than f (x1), the derivative is zero at the minimum point in precisely the same manner as under case (b). ` ` ` ` ` ` ` ` ` ` ` ` ` ` ` ` ` ` ` ` x1 A ξ x2 B Fig. 18. This theorem is called Rolle’sTheorem. The restriction that f (x) shall be continuous is unnecessary if the derivative exists, but simplifies the argument. The proof without this restriction is suggested as an exercise for the reader. The geometric interpretation is that any curve representing a continuous function, f (x), such that f (x1) = f (x2), and having a tangent at every point between x1 and x2 has a horizontal tangent at some point between them. An immediate generalization of this is that between any two points A and B on a curve which satisfies the hypothesis of this theorem there is a tangent to the curve which is parallel to the line AB. The following theorem is a corresponding analytical generalization: Theorem 85. If f (x) is continuous on the interval | | x1 x2 , and if the derivative exists at every point between x1 and x2, then there is a value of x, x = ξ, between x1 and x2 such that f′(ξ) = f (x1)−f (x2) x1−x2 . Proof. Consider a function f1(x) such that f1(x) = f (x)−(x−x2)·f (x1)−f (x2) x1−x2 ; then f1(x1) = f (x2) and f1(x2) = f (x2). Therefore f1(x1) = f1(x2). Hence, by Theorem 84, there is an x, x = ξ on the segment x1 x2 such that f′ 1(ξ) = 0. That is, f′ 1(ξ) = f′(ξ)−f (x1)−f (x2) x1−x2 = 0.

 Therefore f′(ξ) = f (x1)−f (x2) x1−x2 . This is the “mean-value theorem.” Its content may also be expressed by the equation f (x2) = f (x1) + (x2−x1)f′(ξ). Denoting x1−x by dx and ξ by x + θdx, where 0 < θ < 1, it takes the form f (x1 + dx) = f (x1) + f′(x1 + θdx)dx. Theorem 86. If f1(x) and f2(x) are continuous on an interval || a b , and if f′ 1(x) and f′ 2(x) exist between a and b, f′ 2(x)≠±∞, and f′ 2(x)≠ 0, f2(a)≠ f2(b), then there is a value of x, x = ξ between a and b such that f1(a)−f1(b) f2(a)−f2(b) = f′ 1(ξ) f′ 2(ξ). Proof. Consider a function f3(x) = f1(a)−f1(b) f2(a)−f2(b){f2(x)−f2(b)}−{f1(x)−f1(b)}. Since f3(a) = 0 and f3(b) = 0, we have as before f′ 3(ξ) = 0. But f′ 3(ξ) = f1(a)−f1(b) f2(a)−f2(b)·f′ 2(ξ)−f′ 1(ξ). Therefore f1(a)−f1(b) f2(a)−f2(b) = f′ 1(ξ) f′ 2(ξ). This is called the second mean-value theorem. The first mean-value theorem has a very important extension to “Taylor’s series with a remainder,” which follows as Theorem 87. § 5 Taylor’s Series. The derivative of f′(x) is denoted by f″(x) and is called the second derivative of f (x). In general the nth derivative is the derivative of the n−1st derivative and is denoted by f

(n)(x).

108 INFINITESIMAL ANALYSIS. Theorem 87. If the first n derivatives of the function f (x) exist and are finite upon the interval || a b , there is a value of x, xn on the interval || a b such that f (b) = f (a) + (b−a) 1! f′(a) + (b−a)2 2! f″(a) + . . . + (b−a)n−1 (n−1)! ·f (n−1)(a) + (b−a)n n! f (n)(xn). Proof. Let Rn be a constant such that F (x) = f (x)−f (a)−(x−a)f′(a)−(x−a)2 2! f″(a)−. . . −(x−a)n−1 (n−1)! f (n−1)(a)−(x−a)n n! Rn is equal to zero for x = b. Since F (x) = 0 for x = a, there is, by Theorem 84, some value of x, x1, a < x 1 < b such that F′(x1) = 0. That is, F′(x) = f′(x)−f′(a)−(x−a)f″(a)−. . . −(x−a)n−2 (n−2)! f (n−1)(a)−(x−a)n−1 (n−1)! Rn is equal to zero for x = x1. Since also F′(a) = 0, there is a value of x, x2, a < x 2 < x 1 such that F″(x2) = 0. Proceeding in this manner we obtain a value of x, xn, a < x n < x n−1 such that F (n)(xn) = 0 . But F (n)(xn) = f (n)(xn)−Rn = 0. Therefore Rn = f (n)(xn), whence the theorem. Corollary.— In Theorem 87, f (n)(x) need be supposed to exist only on a b . Definition.— The expression (b−a)n n! Rn = (b−a)n n! f n(xn) = f (b)− n−1∑ k=0 (b−a) k! f (k)(a) is called the remainder, and the infinite series ∞∑ k=0 (b−a)n n! f (n)(a) is called Taylor’s Series.

DERIV ATIVES AND DIFFERENTIALS. 109 If L n=∞ f (n)(xn)(b−a)n n! = c, a constant different from zero, then ∞∑ n=0 f (n)(a)(b−a)n n! is convergent but not equal to f (b), i.e., ∞∑ n=0 f (n)(a) n! ·(b−a)n = f (b)−c. If L n=∞ f (n)(xn) n! ·(b−a)n fails to exist and be finite, then ∞∑ n=0 f (n)(a) n! ·(b−a)n is a divergent series. Hence an obvious necessary and sufficient condition that for a function f (x) all of whose derivatives exist for the values of x, a <= x <= b, f (b) = ∞∑ n=0 f (n)(a) n! ·(b−a)n, is that L n .=∞ f (n)(xn) n! (b−a)n = 0. 4 This leads at once, by Theorem 33, to the following sufficient condition: Theorem 88. If f (n)(x) exists and □□f (n)(x) □□is less than a fixed quantity M for every x on the interval || a b and for every n (n = 1, 2, . . .), then f (b) = f (a) + (b−1) 1! f′(a) + . . . + (b−a)n n! f (n)(a) + ... 4 L n .= ∞ f (n)(xn) n! (b−a)n = 0. for every value of x on | | a b is not sufficient, since xn depends upon n.

110 INFINITESIMAL ANALYSIS. Functions are well known all of whose derivatives exist at every point on an interval || a b , but such that for some point on this interval ∞∑ n=0 f (n)(a) n! (x−a)n = f (x) + R(x), where R is a function of x not identically zero. Other functions are known for which the series is divergent. The classical example of the former is that given by Cauchy, 5 e−1 x2 at the point x = 0. If this function is defined to be zero for x = 0, all its derivatives are zero for x = 0, whence Taylor’s development gives a function which is zero for all values of x. Pringsheim6 has given a set of necessary and sufficient conditions that a function shall be representable for the values of h, 0 < h < R , by means of the series ∞∑ n=0 1 n!·f (n)(0)·hn. It was remarked above, p. 105, that a necessary condition for f (x) to be a maximum at x = a is f′(a) = 0 if the derivative exists. Taylor’s series permits us to extend this as follows: Theorem 89. If on some V (a) the first n derivatives of f (x) exist and are finite and on V∗(a)f (n+1)(x) exists and is bounded, 7 and if 0 = f′(a) = f″(a) = . . . = f (n−1)(a), f (n)(a)≠ 0, then: (1) If n is odd, f (x) has neither a maximum nor a minimum at a; (2) If n is even, f (x) has a maximum or a minimum according as f (n)(a) < 0 or f (n)(a) > 0. Proof. By Taylor’s theorem, for every x in the vicinity of a f (x) = f (a) + (x−a)nf (n)(a) + (x−a)n+1·f (n+1)(ξx), where ξx is between x and a. Hence f (x)−f (a) = (x−a)n{f

(n)(a) + (x−a)f (n+1)(ξx)}. 5Cauchy, Collected Works , 2d series, Vol. 4, p. 250. 6A. Pringsheim , Mathematische Annalen, Vol. 44 (1893), p. 52, 53. See also K¨onig, Mathematische Annalen, Vol. 23, p. 450. 7Instead of assuming the existence of f (n+1)(x) we might have assumed f (n)(x) continuous without essentially changing the proof.

DERIV ATIVES AND DIFFERENTIALS. 111 But since f (n+1)(ξx) is bounded and x−a is infinitesimal, there exists a V∗(a) such that if x is in V∗(a), f (x)−f (a) is positive or negative according as (x−a)n·f (n)(a) is positive or negative. (1) If n is odd, (x−a)n is of the same sign as x−a, and hence for f (n)(a) > 0 f (x)−f (a) > 0 if x > a, f (x)−f (a) < 0 if x < a ; while for f (n)(a) < 0 f (x)−f (a) > 0 if x < a, f (x)−f (a) < 0 if x > a. (2) If n is even, (x−a)n is always positive, and hence if f (n)(a) > 0, f (x)−f (a) > 0 if x > a, f (x)−f (a) > 0 if x < a ; } then f (a) is a maximum. If f (n)(a) < 0, f (x)−f (a) < 0 if x > a, f (x)−f (a) < 0 if x < a ; } then f (a) is a minimum. § 6 Indeterminate Forms. The mean-value theorems have an important application in the derivation of l'Hospital's rule for calculating "indeterminate forms." There are seven cases. (1) 0 0 , i.e., to compute L x .=a f (x) φ(x) if L x .=a f (x) = 0 and L x .=a φ(x) = 0. (2) ∞ ∞, i.e., to compute L x .=a f (x) φ(x) if L x .=a f (x) =±∞and L x .=a φ(x) =±∞. (3)∞−∞, i.e., to compute L x .=a {f (x)−φ(x)}if L x .=a f (x) =±∞and L x .=a φ(x) =±∞.

112 INFINITESIMAL ANALYSIS. (4) 0·∞, i.e., to compute L x .=a f (x)·φ(x) if L x .=a f (x) = 0 and L x .=a φ(x) =±∞. (5) 1∞, i.e., to compute L x .=a f (x)φ(x) if L x .=a f (x) = 1 and L x .=a φ(x) =±∞. (6) 00, i.e., to compute L x .=a f (x)φ(x) if L x .=a f (x) = 0 and L x .=a φ(x) = 0. (7)∞0, i.e., to compute L x .=a f (x)φ(x) if L x .=a f (x) =±∞and L x .=a φ(x) = 0. These problems may all be reduced to one or the other of the first two. The third may be written (since f (x)= 0 on some V∗(a)) f (x)−φ(x) = 1 1 f (x) −φ(x) = 1−φ(x) f (x) 1 f (x) , which is either determinate or of type (1). To the cases (5), (6), and (7) we may apply the corollaries of Theorem 39 of Chapter IV, from which it follows (provided f (x)= 0 on some V∗(a)), that L x .=a f (x)φ(x) exists if and only if log L x .=a f (x)φ(x) = L x .=a log f (x)φ(x) = L x .=a φ(x) log f (x) exists. The evaluation of L x .=a log f (x) 1 φ(x) comes under case (1) or case (2). The evaluation of cases (1) and (2) is effected by the following theorems: Theorem 90. If f (x) and φ(x) are continuous and differentiable and φ(x) is monotonic and φ′(x)= 0 and φ′(x)=∞and (1) if L x .=∞ f (x) = 0 and L x .=∞ φ(x) = 0 or (2) if L x .=∞ φ(x) =±∞,8 then if L x .=∞ f′(x) φ′(x) = K, L x .=∞ f (x) φ(x) exists and is equal to K. 8It is not necessary that Lf (x) =∞; cf. Theorem 69.

DERIV ATIVES AND DIFFERENTIALS. 113 Proof. For every positive h we have, by the second mean-value theorem, f (x + h)−f (x) φ(x + h)−φ(x) = f′(ξx) φ′(ξx) , where ξx lies between x and x + h. But since ξx takes on values which are a subset of the values of x, and since L x .=∞ ξx =∞, L x .=∞ f′(x) φ′(x) = K implies L x .=∞ f′(ξx) φ′(ξx) = K, which in turn implies L x .=∞ f (x + h)−f (x) φ(x + h)−φ(x) = K, and this, according to Theorems 68 and 69, gives L x .=∞ f (x) φ(x) = K. Corollary.— If f (x) is continuous and differentiable, then L x .=∞ f (x) x = L x .=∞ f′(x). The theorem above can be extended by the substitution z = 1 x−a to the case where x approaches a finite value a. The approach must of course be one-sided. Theorem 91. If f (x) and φ(x) are continuous and differentiable on some V∗(a) and f (x) is bounded on every finite

interval, while φ(x) is monotonic and (1) L x .=a f (x) = 0 , L x .= a φ(x) = 0 or (2) L x .= a φ(x) = +∞or−∞: then if L x .=a f′(x) φ′(x) = K, it follows that L x .=a f (x) φ(x) exists and is equal to K.

114 INFINITESIMAL ANALYSIS. Proof. If L x .=a f′(x) φ′(x) exists, the limit exists when the approach is only on values of x > a . Consider only such values of x. Then if z = 1 x−a , f (x) = f (a + 1 z) = F (z) and φ(x) = φ(a + 1 z) = Φ(z), by hypothesis and Theorem 79, F′(z) and Φ ′(z) exist and F′(z) = f′(x)dx dz , Φ′(z) = φ′(x)dx dz . Hence if L x .=a f′(x) φ′(x) = K, then, according to Theorem 38, L x .=∞ F′(z) Φ′(z) exists and is equal to K. Hence, by Theorem 90, L x .=∞ F (z) Φ(z) exists and is equal to K. Hence, by Theorem 38, L x .=a f (x) φ(x) exists and is equal to K. We have now derived conditions under which we can state a general rule for computing an indeterminate form. Provided f (x) is not zero on every V∗(a), any of the forms (3) to (7) can be reduced to F (x) Φ(x) (a) where this is of type (1) or (2). Provided F (x) and Φ(x) satisfy the conditions of Theo- rem 91, the existence of the limit of (a) depends on the existence of the limit of F′(x) Φ′(x) . (b)

DERIV ATIVES AND DIFFERENTIALS. 115 If (b) is indeterminate, and F′(x) and Φ′(x) satisfy the conditions of Theorem 91, the limit of (b) depends on the limit of F′′(x) Φ′′(x) , (c) and so on in general. If at each step the conditions of Theorem 91 are satisfied and the form is still indeterminate, the limit of F (n)(x) Φ (n)(x) (n) depends on the limit of F (n+1)(x) Φ (n+1)(x) . (n + 1) If (n) is indeterminate for all values of n, this rule leads to no result. If for some value of n L x .=a F (n)(x) Φ (n)(x) = K, then all the preceding limits exist and are equal to K, and so L x .=a F (x) Φ(x) = K. The original expression is equal to K or eK according to the case under consideration. § 7 General Theorems on Derivatives. Theorem 92. If f (x) is continuous and f′(x) exists for every x on an interval || a b , then f′(x) takes on every value between any two of its values. Proof. Consider any two values of f′(x), f′(x1), and f′(x2) on the interval || a b . Consider, further, the function f (x)−f (x1) x−x1 on the interval between x1 and x2. Since f (x)−f (x1) x−x1 is a continuous function of x on this interval, it takes on every value between f (x2)−f (x1) x2−x1 and f′(x1), which is its limiting value as x approaches x1. Hence, by Theorem 85, f′(x) takes on all values between and including f′(x1), and f (x2)−f (x1) x2−x1 for values of x on the interval | | x1 x2 . By considering in a similar manner the function f (x2)−f (x) x2−x on the interval | | x1 x2 , we show that f′(x) takes on all values between f (x2)−f (x1) x2−x1 and f′(x2). Hence f′(x) takes on all values between f′(x1) and f′(x2). Theorem 93. If the derivative exists at every point on an interval, including its end- points, it does not follow that the derivative is continuous or that it takes on its upper and lower bounds.

116 INFINITESIMAL ANALYSIS. Proof. This is shown by the following example. The curve shall lie between the x-axis and the parabola y = 1 2 x2. The straight lines of slopes 1 , 1 1 2 , 1 3 4 , . . . , 1 + 2n−1 2n . . . through the points (1 2 , 0), (1 4 , 0), . . . , (1 2n+1 , 0) , . . . , re- spectively, meet the parabola in points A1, A 2, A 3, . . . , A n, . . . The broken line A1 (1 2 , 0) A2 (1 4 , 0) A3 . . .An (1 2n , 0) . . . ∞, has an infinitude of vertices. In each angle of the broken line consider an arc of circle tangent to and terminated by the sides of the angle, the points of tangency being one fourth of the distance to the nearest vertex. The function whose graph consists of

these circular arcs and the portions of the broken line between them is continuous and differentiable on the interval || 0 1. Its derivative is discontinuous at x = 0 and has the least upper bound 2, which is never reached. Theorem 94. If f′(x) exists and is equal to zero for every value of x on the interval || a b , then f (x) is a constant on that interval. Proof. By Theorem 82, f (x) is continuous. Suppose f (x) not a constant, so that for two values of x, x1, and x2, f (x1)≠ f (x2), then, by Theorem 85, there is a value of x, x = ξ between x1 and x2 such that f′(ξ) = f (x2)−f (x1) x2−x1 , which is different from zero, whence f′(x) is not zero for every value of x on the interval || a b . Hence f (x) is a constant on || a b .

DERIV ATIVES AND DIFFERENTIALS. 117 Corollary.— If f′ 1(x) = f′ 2(x) and is finite for every value of x on an interval || a b , then f1(x)−f2(x) is a constant on this interval. Theorem 95. If f′(x) exists and is positive for every value of x on the interval || a b , then f (x) is monotonic increasing on this interval. If f′(x) is negative for every value of x on this interval, then f (x) is monotonic decreasing. Proof. If f′(x) is positive for every value of x, then it follows from Theorem 85, provided that f (x) is continuous, that the function is monotonic increasing, for if there were two values of x, x1 and x2, such that f (x1) □ f (x2) while x1 < x 2, then there would be a value of x, x = ξ, between x1 and x2 such that f′(ξ) = f (x2)−f (x1) x2−x1 □ 0. In case f (x) is not supposed continuous, the argument can be made as follows: If f′(x1) > 0, then, by Theorem 23, there exists about the point x1 a segment (x1−δ) (x1 + δ), upon which f (x)−f (x1) x−x1 > 0, and hence, if x > x 1, f (x) > f (x1) and if x < x 1, f (x) < f (x1). Now about every point of the segment a b there is such a segment. Let x′and x′′be any two points of || a b such that x′< x′′. By Theorem 10, there is a finite set of these segments of lengths δ1 . . . δ n which include within them every point of the interval | | x′x′′. We thus have a finite set of points, namely, the mid-point and points on the overlapping parts of the segments, x′< x 1 < x 2 < . . . < x k < x′′, such that f (x′) < f (x1) < f (x2) < . . . f (xk) < f (x′′). Hence f (x′) < f (x′′). In a similar manner we prove that the function is monotonic de- creasing in case f′(x) is negative. Theorem 96. If a function f (x) is monotonic increasing on an interval || a b , and if f′(x) exists for every value of x on this interval, then there is no point on the interval for which f′(x) is negative. That is, f′(x) is either positive or zero for every point of || a b . Proof. If f′(x) is negative for some value of x, say x1, then L x .=x1 f (x)−f (x1) x−x1 = C, a negative number , whence there is a neighborhood of x1 on which f (x) > f (x1), while x < x 1, or f (x1) > f (x), while x > x 1, which is contrary to the hypothesis that the function is monotonic increasing in the neighborhood of x = x1. In the same manner we prove that if the function is monotonic decreasing, and if the derivative exists, then f′(x) cannot be positive.

118 INFINITESIMAL ANALYSIS. The following theorem states necessary and sufficient conditions for the existence of the progressive and regressive derivatives. Conditions for the existence of a derivative proper are obtained by adding the condition that the progressive and regressive derivatives are equal. Theorem 97. If f (x), x < x 1, is continuous in some neighborhood of x = x1, then a necessary and sufficient condition that f′(x1) shall exist and be finite is that there exists not more than one linear function of x, ax + c, such that f (x) + ax + c vanishes on every neighborhood of x = x1. Proof. (1) The condition is necessary. We prove that if f′(x)

exists and is finite, then not more than one function of the form ax + c exists such that f (x) + ax + c vanishes on every neighborhood of x = x1. If no such function exists, the theorem is verified. If there is one such function, the following argument will show that there is only one. Since, by hypothesis, L x .=x1 f (x)−f (x1) x−x1 exists, we have, by Theorem 75, that L x .=x1 f (x) + ax + c−f (x1)−ax1−c x−x1 exists. Let [x′] be the subset of the set of values of x on any neighborhood of x = x1 such that f (x′) + ax′+ c vanishes on the set [x′]. By Theorem 41, L x′.=x1 f (x′) + ax′+ c−f (x1)−ax1−c x′−x1 = L x .=x1 f (x) + ax + c−f (x1)−ax1−c x−x1 = f′(x1) + a. Since f′(x1) and a are both finite, L x′.=x1 f (x′) + ax′+ c′−f (x1)−ax1−c x′−x1 is finite. But the numerator of this fraction is a constant, f (x) + ax + c being zero on the set [x′]. Hence L x .=x1 f (x) + ax + c−f (x1)−ax1−c x−x1 = 0, or f′(x1) + a = 0, and, being continuous, f (x1) + ax1 + c = 0. The numbers a and c are uniquely determined by the equations { f′(x1) + a = 0, f (x1) + ax1 + c = 0.

DERIV ATIVES AND DIFFERENTIALS. 119 (2) The condition is sufficient. We are to show that L x .=x1 f (x)−f (x1) x−x1 can fail to exist only when there are at least two functions of the form ax + c such that f (x) + ax + c vanishes on every neighborhood of x = x1. If L x .=x1 f (x)−f (x1) x−x1 does not exist, then f (x)−f (x1) x−x1 approaches at least two distinct values K1 and K2. Let K2 < K 1. Let A and B be two finite values such that K2 < A < B < K 1. On every neighborhood of x = x1 there are values of x for which f (x)−f (x1) x−x1 is greater than B, and also values of x for which f (x)−f (x1) x−x1 is less than A. Hence, since f (x)−f (x1) x−x1 is continuous at every point except possibly x1, in a certain neighborhood of x1 there are values of x in every neighborhood of x1 for which f (x)−f (x1) x−x1 = A, or f (x)−f (x1) = A(x−x1), which gives −f (x1)−A(x−x1) as one function of the form ax + c. In the same manner we show that −f (x1)−B(x−x1) is another function ax + c, which makes f (x) + ax + c vanish on every neighborhood of x = x1.

120 INFINITESIMAL ANALYSIS. The geometric meaning of this theorem is obvious. If P is a point on the curve repre- senting f (x), then a necessary and sufficient condition that this curve shall have a tangent at P is that there exists not more than one line through P which intersects the curve an infinite number of times on any neighborhood of P . Compare the functions x sin 1 x and x2 sin 1 x on page 38. The earlier mathematicians supposed that every continuous function must have a derivative except at particular points. The first example of a function which has no deriva- tive at any point is due to Weierstrass.9 The function is f (x) = ∞∑ n=0 bn cos(anπx), where a is an odd integer, 0 < b < 1 and ab > 1 + 3 2 π. 9For references and remarks see page 38.

Chapter 8 DEFINITE INTEGRALS. § 1 Definition of the Definite Integral. The area of a rectangle the lengths of whose sides are exact multiples of the length of the side of a unit square, is the number of squares equal to the unit square contained within the rectangle, and is easily seen to be equal to the product of the lengths of its base and altitude.1 In case the sides of the rectangle and the side of the unit square are commensurable, the sides of the rectangle not being exact multiples of the side of the square, the rectangle and the square are divided into a set of equal squares. The area of the rectangle is then defined as the ratio of the number of squares in the rectangle

to be measured to the number of squares in the unit square. Again, the area is equal to the product of the base and altitude. Any figure so related to the unit square that both figures can be divided into a finite set of equal squares is said to be commensurable with the unit. The area of a rectangle incommensurable with the unit is defined as the least upper bound of the areas of all commensurable rectangles contained within it. It follows directly from the definition of the product of irrational numbers that this process gives the area as the product of the base and altitude. 2 Turning to the figure bounded by the segment a b (which we take on the x axis in a system of rectangular coordinates) the graph of a function y = f (x) and the ordinates x = a and x = b, we obtain as follows an approximation to the common notion of the area of such figures. Let x0 = a, x1, x2, . . . , xn = b be a set of points lying in order from a to b. Such a set of points is called a partition of || a b , and is denoted by π. The intervals | | x0 x1 , | | x1 x2 , . . . , | | xn−1 xn are intervals of π. 1Of course the units are not necessarily squares; they may be triangles, parallelograms, etc. 2For the meaning of the length of a segment incommensurable with the unit segment, compare Chap- ter II, page 25. 121

122 INFINITESIMAL ANALYSIS. a ξ1 x1 ξ2 x2 xn−1 ξn b Fig. 20 Let x1−x0 = Δ 1x, x2−x1 = Δ 2x, . . . , xn−xn−1 = Δ nx, and let ξ1, ξ 2, . . . , ξ n be a set of points such that ξ1 is on the interval | | x0 x1 , ξ2 is on | | x1 x2 . . . , and ξn is on | | xn−1 xn . Then f (ξ1), f (ξ2), . . . , f (ξn) are the altitudes of a set of rectangles whose combined area is a more or less close approx- imation of the area of our figure. Denote this approximate area by S. Then S = f (ξ1)Δ 1x + f (ξ2)Δ 2x + . . . + f (ξn)Δ nx = n∑ k=1 f (ξk)Δ kx. As the greatest Δ kx is taken smaller and smaller, the figure composed of the rectangles comes nearer to the figure bounded by the curve. In consequence of these geometrical notions we define the area of the figure as the limit of S as the Δ kx's decrease indefinitely. The area S is the definite integral of f (x) from a to b. It has been tacitly assumed that the graph of y = f (x) is continuous, since we do not usually speak of an area being enclosed by a discontinuous curve. The definition of the definite integral when stated in its general form admits, however, of functions which are discontinuous in a great variety of ways. A more general definition of the definite integral is as follows: Let || a b (or || b a) be an interval upon which a function f (x) is defined, single-valued and bounded. Let πδ stand for any partition of || a b or || b a by the points a = x0, x 1, x 2, . . . , x n = b such that the numbers Δ 1x = x1−a, Δ 2x = x2−x1, . . . , Δ nx = b−xn−1 are each numerically less than or equal to δ. Let ξ1, ξ 2, . . . , ξ n

DEFINITE INTEGRALS. 123 be a set of points on the intervals | | x0 x1 , | | x1 x2 ,. . . , | | xn−1 xn (or if b < a , | | x1 x0 , | | x2 x1 , | | x3 x2 , . . . , | | xn xn−1) respectively, and let Sδ = f (ξ1)Δ 1x + f (ξ2)Δ 2x + . . . + f (ξn)Δ nx = n∑ k=1 f (ξk)Δ kx. If the many-valued function of δ, Sδ, approaches a single limiting value as δ approaches zero, then L δ.=0 Sδ = ∫b a f (x)dx. When we desire to indicate the interval of integration we write b aSδ and b aπδ instead of Sδ and πδ. a and b are called the limits of integration . The details of this definition should be carefully noted. For every δ there is an infinite number of different partitions πδ, and for every partition there is an infinite set of different sets of ξk, so that for every δ the function Sδ has an infinite set of values. The graph of the function Sδ is of the type shown in Fig. 21.

Every value of Sδ for one δ is assumed by S for every larger δ. For any particular value of δ the values of Sδ lie on a definite interval | | BSδ BSδ, whose length never increases as δ decreases. If this interval approaches 0 as δ approaches 0, the required limit exists. It is to be noticed that the set of π's, [πδ] includes every possible π whose largest Δ kx is less than δ. Thus we cannot obtain the set of all π's by sequential repartitioning of any

124 INFINITESIMAL ANALYSIS. given π, since there are partitions of the set [πδ] which have no partition points in common with any given partition. Inattention to this point is perhaps the greatest source of error in the development of the notion of a definite integral. § 2 Integrability of Functions. The class of integrable functions is very large, including nearly all the bounded functions studied in mathematics and physics. Even such an arbitrary function as { y = 0 if x irrational, y = 1/n 3 if x = m/n , is integrable. (See page 147, Theorem 127.) Examples of non-integrable functions are y = 1 /x on the interval || 0 1 (where it is not bounded, see page 153), and the function, { y = 0 if x is irrational and y = 1 if x is rational. To determine the conditions of integrability we introduce the concept of integral oscil- lation. On any interval || a b , f (x) has a least upper bound A and a greatest lower bound B, between which the function varies. If A−B = Δ y = b aOf (x) is multiplied by the length of the interval, Δ x =|b−a|, it gives the area of a rectangle, including the graph of f (x). If the interval is subdivided by a partition π, the sum of the products Δ x·Δ y on the intervals of the partition is called the integral oscillation of f (x) for the partition π and is denoted by Oπ. If we call Δ ky the difference between the upper and lower bounds of f (x) on the intervals | | xk−1 xk , we have Oπ =|Δ 1x|·Δ 1y +|Δ 2x|·Δ 2y + . . . +|Δ nx|Δ ny = n∑ k=1 |Δ kx|·Δ ky. Geometrically Oπ represents the areas of the rectangles F1, . . . , F n (Fig. 22), and so we expect to find that if the lower bound of Oπ is zero, f (x) is integrable. This proposition, which requires some rather delicate argument for its proof, will be taken up in §7. At present we shall show in a simple manner that every continuous and every monotonic function is integrable. Lemma 1.— If Sπ and S′ π are two sums (formed by using different ξk's) on the same partition, then |Sπ−S′ π|□ Oπ.

DEFINITE INTEGRALS. 125 Proof. Sπ = n∑ k=1 f (ξk)Δ kx, S′ π = n∑ k=1 f (ξ′ k)Δ kx, |Sπ−S′ π|= □□□□□ n∑ k=1 {f (ξk)−f (ξ′ k)}Δ kx □□□□□□ n∑ k=1 |f (ξk)−f (ξ′ k)|·|Δ kx|. But|f (ξk)−f (ξ′ k)|□ Δ ky by the definition of Δ ky. Therefore |Sπ−S′ π|□ n∑ k=1 |Δ kx|·Δ ky (4) A repartition of a partition π is formed by introducing new points in π. Lemma 2.— If π1 is a repartition of π, |Sπ−Sπ1|□ Oπ. Proof. Any interval Δ kx of π is composed of one or more intervals Δ ′ kx, Δ ″ kx, etc., of π1, and these contribute to Sπ the terms f (ξ′ k)Δ′ kx + f (ξ″ k)Δ″ kx + . . . (1) The corresponding term of Sπ is f (ξk)Δ kx = f (ξk)Δ′ kx + f (ξk)Δ″ kx + . . . (2) But since|f (ξk)−f (ξk′)|□ Δ ky, the difference between (1) and (2) is less than or equal to Δ ky·|Δ′ kx + Δ″ kx + . . . |= Δ ky·|Δ kx| and hence |Sπ−S π1|□ n∑ k=1 Δ ky·|Δ kx|= Oπ. Theorem 98. Every function continuous on || a b is integrable on || a b . Proof. We have to investigate the existence of the limit L δ.=0 Sδ of the many-valued function Sδ as δ .= 0. Since Sδ approaches at least one value as δ approaches zero (see Theorem 24), we need only to prove that it cannot have more than one value approached. Suppose

126 INFINITESIMAL ANALYSIS. a F1 F2 Δ 3y Δ 3x Fn b Fig. 22

DEFINITE INTEGRALS. 127 there were two such values, B and C, B > C . Let ε = B−C 4 . By the definition of value approached, for every δ there must exist an S (which we call SB) such that |SB−B|< ε (1) and such that the corresponding πB has its largest Δ kx < δ . Similarly there must be an SC such that |SC−C|< ε, (2) and such that the corresponding πC has its largest Δ kx < δ . Let π be a partition made up of the points both of πB and πC, and let S be one of the corresponding sums. π is a repartition both of πB and πC. Therefore |S−SC|□ OπC (3) and |S−SB|□ OπB . (4) But since f (x) is continuous, by the theorem of uniform continuity, δ can be so chosen that if any two values of x differ by less than δ, the corresponding values of f (x) differ by less than ε |b−a|and hence on the partitions πB and πC, whose Δ kx's are all less than δ, the corresponding Δ ky's are all less than ε |b−a|. So we have (since n∑ k=1 Δ kx = b−a) OπB = n∑ k=1 |Δ kx|·Δ ky < n∑ k=1 |Δ kx|·ε |b−a|= ε. Hence OπB < ε and OπC < ε. So we have, since ε = B−C 4 and δ is so chosen that whenever |x′−x″|< δ,|f (x′)−f (x″)|< ε |b−a|: |SB−B|< ε, |SC−C|< ε, |SB−S|< ε, |SC−S|< ε. From these inequalities it follows that |B−C|< 4ε, which contradicts the statement that ε = B−C 4 . Hence the hypothesis that f (x) is not integrable is untenable. Theorem 99. Every non-oscillating bounded function is integrable.

128 INFINITESIMAL ANALYSIS. Proof. The proof runs, as in the preceding theorem, to the paragraph following (4). Let D and d be the upper and lower bounds of f (x). δ, being arbitrary, can be so chosen that δ = ε D−d. Then OπB = n∑ k=1 Δ ky·|Δ kx|< n∑ k=1 Δ ky·δ, and since f (x) is non-oscillating, n∑ k=1 Δ ky = D−d. Therefore OπB < (D−d)δ = ε. Similarly OπC < ε. Hence again we have |SB−B|< ε, |SC−C|< ε, |SB−S|< ε, |SC−S|< ε, and therefore|B−C|< 4ε, whereas ε was assumed equal to B−C 4 . Thus the hypothesis of a non-integrable non-oscillating function is untenable. § 3 Computation of Definite Integrals. In computing definite integrals it is important to observe that when the integral is known to exist the limit can be calculated on any properly chosen subset of the Sδ's. (See The- orem 41.) So we have that if Sδ1, Sδ2, . . . is any sequence of sums such that L n .=∞ δn = 0, then L n .=∞ Sδn = ∫b a f (x)dx. One case of this kind occurs when ξk is taken as an end-point of the interval | | xk−1 xk and all the Δ kx's are equal. Then we have ∫b a f (x)dx = L n .=∞ n∑ k=1 f (a + kΔ x)Δ x, where Δ x = b−a n . A simple example of this principle is the proof of the following theorem. Theorem 100. If f (x) is a constant, C, then ∫b a Cdx = C(b−a).

DEFINITE INTEGRALS. 129 Proof. The function f (x) = C is integrable either according to Theorem 98 or Theorem 99. Hence ∫b a Cdx = L n .=∞ n∑ k=1 C b−a n = L n .=∞ n·C·b−a n = C(b−a). A few other examples follow. In each case the function is known to be integrable by the theorems of §2. Theorem 101. ∫b a exdx = eb−ea. Proof. Let SΔ x = eaΔ x + ea+Δ x·Δ x + ea+2Δ x·Δ x + . . . + ea+(n−1)Δ x·Δ x = ea·Δ x[1 + eΔ x + e2Δ x + . . . + e(n−1)Δ x] = ea·Δ x·enΔ x−1 eΔ x−1 = eb−a−1 eΔ x−1 ea·Δ x = (eb−ea)·Δ x eΔ x−1. Whence the result follows since L Δ x .=0 Δ x eΔ x−1 = 1. (Differentiate numerator and denomi- nator with respect to Δ x according to Theorem 90.) Instead of arranging the partition-points in an arithmetical progression as in the cases above, we may put them in a geometrical progression, that is, we let (b a) 1 n = q, b a = qn, Δ 1x = aq−a, Δ 2x = aq2−aq, . . . , Δ nx = aqn−aqn−1, ξ1 = a, ξ 2 = aq, . . . , ξ n = aqn−1, and obtain the formula ∫b a f (x)dx = L q .=1 a(q−1)[f (a)

+ qf (aq) + . . . + qn−1f (aqn−1)] = L q .=1 a(q−1) n−1∑ k=0 qkf (aqk). We apply this scheme to the following.

130 INFINITESIMAL ANALYSIS. Theorem 102. In all cases where m is a whole number ≠−1, and if a > 0, b > 0 for every value of m ≠−1, ∫b a xmdx = bm+1−am+1 m + 1 . Proof. ∫b a xmdx = L q .=1 a(q−1) n−1∑ k=0 qk(aqk)m = am+1 L q .=1 (q−1)[1 + (qm+1) + (qm+1)2 + . . . + (qm+1)n−1] (1) = am+1 L q .=1 (q−1)(qm+1)n−1 qm+1−1 = L q .=1 am+1{(qn)m+1−1}q−1 qm+1−1 = (bm+1−am+1) L q .=1 q−1 qm+1−1. Hence ∫b a xmdx = bm+1−am+1 m + 1 , since L q .=1 q−1 qm+1−1 = 1 m + 1 . Theorem 103. ∫b a 1 x dx = log b−log a, (0 < a < b). Proof. By equation (1) in the last theorem, since qm+1 = q0 = 1, ∫b a 1 x dx = L n .=∞ n(q−1); but n = log(b a) log q , hence ∫b a 1 x dx = L q .=1 q−1 log q·log (b a) = log (b a) = log b−log a, since (§6, Chapter VII) l'Hospital's rule gives L q .=1 q−1 log q = 1.

DEFINITE INTEGRALS. 131 The following theorem is of frequent use in computing both derivatives and integrals. Theorem 104. If on an interval || a b two functions f (x) and F (x) have the property that for every two values of x, x1 and x2, where a < x 1 < x 2 < b, f (x1)(x2−x1) □ F (x2)−F (x1) □ f (x2)(x2−x1); or if f (x1)(x2−x1) □ F (x2)−F (x1) □ f (x2)(x2−x1), then (1) if f (x) is continuous, dF (x) dx = f (x); and (2) whether f (x) is continuous or not, ∫b a f (x)dx exists and is equal to F (b)−F (a). Proof. We consider first the case f (x1)(x2−x1) □ F (x2)−F (x1) □ f (x2)(x2−x1). This gives f (x1) □ F (x2)−F (x1) x2−x1 □ f (x2). Since f (x) is continuous at x = x1, L x2 .=x1 f (x2) = f (x1). Hence, by Theorem 40 (Corol- lary 2), L x2 .=x1 F (x2)−F (x1) x2−x1 = f (x1), which proves (1). To prove (2) we observe that f (x) is non-oscillating and therefore integrable according to Theorem 99. On any partition π whose dividing points are x1, x2, . . . , x n−1 we have f (a)(x1−a) □ F (x1)−F (a) □ f (x1)(x1−a), f (x1)(x2−x1) □ F (x2)−F (x1) □ f (x2)(x2−x1), · · · · · · · · · · · · f (xn−1)(b−xn−1) □ F (b)−F (xn−1) □ f (b)(b−xn−1),

132 INFINITESIMAL ANALYSIS. Adding, we get f (a)(x1−a) + f (x1)(x2−x1) + . . . + f (xn−1)(b−xn−1) □ F (b)−F (a) □ f (x1)(x1−a) + f (x2)(x2−x1) + . . . + f (b)(b−xn−1). But f (a)(x1−a) + . . . + f (xn−1)(b−xn−1) □ BSπ and f (x1)(x1−a) + . . . + f (b)(b−xn−1) □ BSπ. Since this holds for every π, we have by Theorem 40 that as (Theorem 99) ∫b a f (x)dx exists, ∫b a f (x)dx = F (b)−F (a). The proof in case f (x1)(x2−x1) □ F (x2)−F (x1) □ f (x2)(x2−x1) is identical with the above when we write □ instead of □. § 4 Elementary Properties of Definite Integrals. Theorem 105. If a < b < c , and if a bounded function f (x) is integrable from a to c, then it is integrable from a to b and from b to c. Proof. Suppose f (x) not integrable from a to b, then by the definition of a limit (see Chap. II.) there must be a set of values of b aSδ, [b aS′ δ], such that L δ.=0 b aS′ δ = A, and another set [b aS′′ δ] such that L δ.=0 b aS′′ δ = B, while A and B are distinct. Whether ∫c b f (x)dx exists or not, there must be a set of values of c bSδ, [c bS′ δ], such that the limit L δ.=0 c bS′ δ = C. Now for every b aS′ δ and c bS′ δ there exists a c aS′ δ such that c aS′ δ = b aS′ δ + c bS′ δ. Therefore A + C is a value approached by c aSδ. By similar reasoning, B + C is a value approached by c aSδ. Hence c aSδ has two values approached, which is contrary to the hypothesis. Hence ∫b a f (x)dx must exist. By similar reasoning ∫c b f (x)dx must exist. Theorem

106. If a < b < c and if a bounded function f (x) is integrable from a to b and from b to c, then f (x) is integrable from a to c and ∫c a f (x)dx = ∫b a f (x)dx + ∫c b f (x)dx.

DEFINITE INTEGRALS. 133 Proof. Since ∫b a f (x)dx and ∫c b f (x)dx exist, we know by Theorem 26 that for every ε there exists a δ′ c such that for b aSδ where δ □ δε, □□□□ b aSδ− ∫b a f (x)dx □□□□< ε 3, (1) and also a δ″ ε such that for every value of c bSδ where δ □ δ″ ε, □□□□ c bSδ− ∫c b f (x)dx □□□□< ε 3. (2) Now if the upper bound of f (x) on || a c is M and its lower bound is m, let δ‴ ε = ε 3(M−m) , and let δε, be smaller than the smallest of δ′ δ, δ″ δ, δ‴ δ . Consider any value of c aSδ. If the point b is one of the points of the partition upon which c aSδ is computed, then c aSδ is the sum of one value of b aSδ and one value of c bSδ. If b is not a point of this partition, let Δ bx be the length of the interval of c aпδ that contains b. Then for properly chosen b aSδ and c bSδ |b aSδ + c bSδ−c aSδ|< Δ bx(M−m) < ε 3. (3) So in every case (whether or not b is a partition-point of c aпδ) by combining (1), (2), and (3) we obtain the result that for every ε there exists a δε such that for every c aSδε □□□□ c aSδε− ∫b a f (x)dx− ∫c b f (x)dx □□□□< ε. Therefore L δ.=0 c aSδ = ∫b a f (x)dx + ∫c b f (x)dx, which proves the theorem. Theorem 107. Provided both integrals exist, 3 and a < b , ∫b a |f (x)|dx □ □□□□ ∫b a f (x)dx □□□□. Proof. ∑ |f (ξk)|Δ kx □ □□□ ∑ f (ξk)Δ kx □□□. Hence for every Sδ|f (x)|there is a smaller or equal Sδf (x), the δ’s being the same. Hence by Corollary 2, Theorem 40, L δ.=0 Sδ|f (x)|□|L δ.=0 Sδf (x)|. 3That the first integral exists if the second exists is shown in Theorem 135.

134 INFINITESIMAL ANALYSIS. Theorem 108. If ∫b a f (x)dx exists, then ∫a b f (x)dx exists and ∫b a f (x)dx =− ∫a b f (x)dx. Proof. This is a consequence of the theorem (Corollary 1 Theorem 27) that L x .=a (−f (x)) =−L x .=a f (x), for to every S used in defining ∫b a f (x)dx corresponds a sum equal to −S which is used in defining ∫a b f (x)dx. Similarly to every S′used in defining ∫a b f (x)dx there corresponds a sum −S′used in defining ∫b a f (x)dx. Hence the function Sδ in the definition of ∫b a f (x)dx is the negative of the function Sδ used in the definition of ∫a b f (x)dx. Hence the theorem follows from the theorem quoted. We adjoin the following two theorems, the first of which is an immediate consequence of the definition of an integral, and the second a corollary of Theorems 105, 106, and 108. Theorem 109. ∫b+h a+h f (x−h)dx exists and is equal to ∫b a f (x)dx, provided the latter integral exists.4 Theorem 110. If any two of the following integrals exist, so does the third, and ∫b a f (x)dx + ∫c b f (x)dx = ∫c a f (x)dx. Theorem 111. If C is any constant and if f (x) is integrable on || a b , then Cf (x) is inte- grable on || a b and ∫b a Cf (x)dx = C ∫b a f (x)dx. Proof. Sδ = n∑ k=1 f (ξk)Δ kx 4First stated formally by H. Lebesgue , Le¸ cons sur l’Int´ egration, Chapter VII, page 98.

DEFINITE INTEGRALS. 135 is an Sδ of the set which defines ∫b a f (x)dx and S′ δ = n∑ k=1 Cf (ξk)Δ kx is the corresponding Sδ of the set which defines ∫b a Cf (x)dx. Hence our theorem follows immediately from Theorem 34, a special case of which is L x .=a Cf (x) = C L x .=a f (x). Theorem 112. If f1(x) and f2(x) are any two functions each integrable on the interval || a b , then f (x) = f1(x)±f2(x) is integrable on || a b and ∫b a f (x)dx = ∫b a f1(x)dx± ∫b a f2(x)dx. Proof. The proof depends directly upon the theorem that if L x .=a φ1(x) = b1, and L x .=a φ2(x) = b2, then L x .=a (φ1(x)±φ2(x)) = b1±b2 (Theorem 34). Theorem 113. If f1(x) and f2(x) are integrable

on || a b and such that for every value of x on || a b f 1(x) □ f2(x), then ∫b a f1(x)dx □ ∫b a f2(x)dx. Proof. Since S1 is always greater than or equal to S2, then, by Theorem 34, L δ.=0 S1 □ L δ.=0 S2, which proves the theorem. Theorem 114. (Maximum-Minimum Theorem.) If (1) the product f1(x)·f2(x) and the factor f1(x) are integrable on || a b , (2) f1(x) is always positive or always negative on || a b , (3) M and m are the least upper and the greatest lower bounds respectively of f2(x) on || a b , then m· ∫b a f1(x)dx □ ∫b a f1(x)·f2(x)dx □ M· ∫b a f1(x)dx, or m· ∫b a f1(x)dx □ ∫b a f1(x)·f2(x)dx □ M· ∫b a f1(x)dx.

 Proof. By Theorem 111, M· ∫b a f1(x)dx = ∫b a M·f1(x)dx and m· ∫b a f1(x)dx = ∫b a m·f1(x)dx. But in case f1(x) is always positive, m·f1(x) □ f1(x)·f2(x) □ M·f1(x). Hence, by the preceding theorem, ∫b a m·f1(x)dx □ ∫b a f1(x)·f2(x)dx □ ∫b a M·f1 (x)dx, and therefore m· ∫b a f1(x)dx □ ∫b a f1(x)·f2(x)dx □ M· ∫b a f1(x)dx. If f1(x) is always negative, it follows in the same manner that m· ∫b a f1(x)dx □ ∫b a f1(x)·f2(x)dx □ M· ∫b a f1(x)dx. As an obvious corollary of this theorem we have the Mean-value Theorem: Theorem 115. Under the hypothesis of Theorem 114 there exists a number K, m □ K □ M , such that ∫b a f1(x)·f2(x)dx = K ∫b a f1(x)dx. Corollary 1. In case f2(x) is continuous we have a value ξ of x on || a b such that ∫b a f1(x)·f2(x)dx = f2(ξ) ∫b a f1(x)dx. In case f1(x) = 1, ∫b a f1(x)dx = b−a, and the theorem reduces to this:

 Theorem 116. If f (x) is any integrable function on the interval || a b , there exists a number M lying between the upper and lower bounds of f (x) on || a b such that ∫b a f (x)dx = M (b−a), and if f (x) is continuous, there is a value ξ of x on || a b such that ∫b a f (x)dx = f (ξ)(b−a). In many applications of the integral calculus the expression ∫b a f (x)dx b−a represents the notion of an average value of the dependent variable y = f (x) as x varies from a to b. An average of an infinite set of values of f (x) is of course to be described only by means of a limiting process. Consider a set of points x1, x2, . . . , x n−1, xn = b on the interval || a b such that x1−a = x2−x1 = x3−x2 = . . . = xn−1−xn−2 = b−xn−1. Then Mn = 1 n n∑ k=1 f (xk), and we define the mean value of f (x), b aM f(x) = L n .=∞ Mn if this limit exists. But xk+1−xk = b−a n = Δ x. If the definite integral ∫b a f (x)dx exists, we may write ∫b a f (x)dx = L δ.=0 Sδ, where Sδ = n∑ k=1 f (xk)Δ x = n∑ k=1 f (xk)b−a n = b−a n n∑ k=1 f (xk) = (b−a)Mn. Therefore L δ.=0 Sδ = (b−a) L n .=∞ Mn. We therefore have the theorem:

 Theorem 117. In case the integral of f (x) exists on the interval || a b , b aM f(x) = ∫b a f (x)dx b−a . We note that b aM is the same as the K which occurs in the mean-value theorem, and that the last theorem suggests a simple method of approximating the value of a definite integral by multiplying the average of a finite number of ordinates by b−a. § 5 The Definite Integral as a Function of the Limits of Integration. Theorem 118. If f (x) is integrable on an interval || a b , and if x is any point of || a b ,∫x a f (x)dx is a continuous function of x. Proof. ∫x a f (x)dx exists, by Theorem 105, and by the definition of a continuous function we need only to show that L x′.=x (∫x′ a f (x)dx− ∫x a f (x)dx) = 0. By the theorems of the preceding section, ∫x′ a f (x)dx− ∫x a f (x)dx = ∫x′ x f (x)dx □|x′ x B·(x′−x)|□|B·(x′−x)|, where x′ x B stands for the least upper bound of f (x) on the interval || x x′, and B for

the least upper bound of f (x) on || a b . Since B is a constant, B(x′−x) approaches zero as x′ approaches x, and therefore by Theorem 40, Corollary 4, the conclusion of our theorem follows. Theorem 119. If f (x) is continuous on an interval || a b , ∫x a f (x)dx (a < x < b) possesses a derivative with respect to x such that d dx ∫x a f (x)dx = f (x).

DEFINITE INTEGRALS. 139 Proof. By the preceding theorem ∫x a f (x)dx is continuous. To form the derivative we investigate the expression ∫x′ a f (x)dx− ∫x a f (x)dx x′−x = ∫x′ x f (x)dx x′−x (1) as x′approaches x. By Theorem 115 (the mean-value theorem), ∫x′ x f (x)dx = f (ξ(x′)) (x′−x), where ξ(x′) is a value of x between x and x′and is a function of x′. Hence (1) is equal to f (ξ). (2) But as x′approaches x, ξ also approaches x and so, by Theorem 39, as x′approaches x, (2) approaches f (x). Therefore L x′.=x ∫x′ a f (x)dx− ∫x a f (x)dx x′−x = f (x) = d dx ∫x a f (x)dx. Following is a more general statement of Theorem 119. Corollary.— If f (x) is continuous at a point x1 of || a b and integrable on || a b , then at x = x1 d dx ∫x a f (x)dx = f (x). The proof is like that of Theorem 112 except that ∫x x1 f (x)dx = (x−x1)M (x), and M (x) is a value between the upper and lower bounds of f (x) on | | x1 x. But by the continuity of f (x) at x1 L x .=x1 M (x) = f (x1), and hence the conclusion follows as in the theorem.

140 INFINITESIMAL ANALYSIS. Theorem 120. If f (x) is any continuous function on the interval || a b , and F (x) any function on this interval such that d dx F (x) = f (x), then F (x) differs from ∫x a f (x)dx at most by an additive constant. Proof. Let F (x) = ∫x a f (x)dx + φ(x). Since F (x) and ∫x a f (x)dx are both differentiable, d dx F (x) = d dx (∫x a f (x)dx + φ(x)) = d dx (∫x a f (x)dx) + d dx φ(x). By the preceding theorem d dx ∫x a f (x)dx = f (x). Hence d dxφ(x) = 0, whence, by Theorem 94, φ(x) is a constant. Theorem 121. If f (x) is a continuous function on an interval || a b and F (x) is such that d dx F (x) = f (x), then ∫b a f (x)dx = F (b)−F (a). Proof. By the last theorem, ∫x a f (x)dx = F (x) + c. But 0 = ∫a a f (x)dx = F (a) + c. Therefore −F (a) = c. Whence ∫b a f (x)dx = F (b) + c = F (b)−F (a). The symbol [F (x)]b a or|b aF (x) is frequently used for F (b)−F (a). In these terms the above theorem is expressed by the equation ∫b a f (x)dx =|b aF (x).

DEFINITE INTEGRALS. 141 By this last theorem the theory of definite and indefinite integrals is united as far as continuous functions are concerned, and a table of derivatives gives a table of integrals. For discontinuous functions the correspondence does not in general hold. That is, there are on the one hand integrable functions f (x) such that ∫x a f (x)dx is not differentiable with respect to x, and on the other hand differentiable functions φ(x) such that φ′(x) is not integrable. 5 § 6 Integration by Parts and by Substitution. The formulas for integration by parts and by substitution are ordinarily written as follows: ∫ udv = uv− ∫ v du, ∫ f (y)dy = ∫ f (y)·dy dx·dx. The following theorems state sufficient conditions for their validity. Theorem 122. (Integration by parts.) ∫b a f1(x)·f′ 2(x)dx = [f1(x)·f2(x)]b a− ∫b a f2(x)·f′ 1(x)dx, provided f′ 1(x) and f′ 2(x) exist and are continuous on the interval || a b . Proof. By Theorem 75, d dx (f1(x)·f2(x)) = f1(x)·f′ 2(x) + f2(x)·f′ 1(x). Therefore ∫b a d dx (f1(x)·f2(x)) dx = ∫b a f1(x)·f′ 2(x)dx + ∫b a f2(x)·f′ 1(x)dx. (The integral exists since it follows from the existence and continuity of f′ 1(x) and f′ 2(x) that f1(x) and f2(x) are continuous). By Theorem 121, ∫b a d dx{f1(x)·f2(x)}dx = f1(b)·f2(b)−f1(a)·f2(a). Therefore ∫b a f1(x)·f′ 2(x)dx = [f1(x)·f2(x)]b a− ∫b a f2(x)·f′ 1(x)dx. 5For a good

discussion of this subject the reader is referred to H. Lebesgue , Le¸ cons sur l'Int´ egration.

142 INFINITESIMAL ANALYSIS. Theorem 123. (Integration by substitution.) If y = φ(x) has a continuous derivative at every point of || a b and f (y) is continuous for all values taken by y = φ(x) as x varies from a to b, ∫B A f (y)dy = ∫b a f (y) dy dx dx, where A = φ(a), B = φ(b). Proof. By Theorem 120 and by Theorem 79, ∫φ(x) A f (y)dy = ∫x a d dx (∫φ(x) A f (y)dy) dx + C = ∫x a f (y) dy dx·dx + C, C being an arbitrary constant. C is determined by letting x = a. Then if x = b we have ∫B A f (y)dy = ∫b a f (y) dy dx·dx. Theorem 124. ∫b a f (x)dx = ∫B A f (φ(y)) dx dy dy, where x = φ(y) and a = φ(A), b = φ(B); provided that both integrals exist, and that φ(y) is non-oscillating and has a finite derivative. Proof. ∫b a f (x)dx = L n .=∞ n∑ k=1 f (ξk)Δ kx (1) whenever the least upper bound of Δ kx for each n approaches zero as n approaches +∞. Now let Δ y = B−A n , yk = A + k·Δ y, φ(yk)−φ(yk−1) = Δ kx. Hence, by Theorem 85, Δ kx = φ′(ηk)Δ y, where ηk lies between yk and yk−1. Now if ξk = φ(ηk), it will lie between φ(yk) and φ(yk−1); moreover the Δ kx's are all of the same sign or zero; and since the hypothesis makes φ(y) uniformly continuous, their least upper bound approaches zero as n approaches +∞. Therefore ∫b a f (x)dx = L n .=∞ n∑ k=1 f (ξk)Δ kx = L n .=∞ n∑ k=1 f (φ(ηk)) ·φ′(ηk)·Δ y = ∫B A f (φ(y)) φ′(y)dy,

DEFINITE INTEGRALS. 143 provided the latter integral exists. Hence ∫b a f (x)dx = ∫B A f (φ(y))·dx dy dy. Corollary.— The validity of this theorem remains if φ(y) has a finite number of oscillations. Proof. Suppose the maximum and minimum values of φ(y) are a1, a 2, a 3, . . . , a n, corresponding to the values of y, A1, A 2, A 3, . . . , A n. Then we have ∫b a f (x)dx = ∫a1 a f (x)dx + ∫a2 a1 f (x)dx + . . . + ∫b an f (x)dx = ∫A1 A f (φ(x))dx dy dy + ∫A2 A1 f (φ(x))dx dy dy . . . + ∫B An f (φ(x))dx dy dy = ∫B A f (φ(x))dx dy dy. The form of this proposition given in Theorem 123 would permit an infinitude of oscillations of φ(y). § 7 General Conditions for Integrability. The following lemmas are to be associated with those on pages 124 and 125. Lemma 3.— If π1 is a repartition of π, then for any function bounded on || a b Oπ1 □ Oπ. Proof. Any interval Δ kx of π is composed of one or more intervals Δ ′ kx, Δ ′′ kx, etc., of π1, and these contribute to Oπ1 the terms |Δ′ kx|Δ′ ky +|Δ′′ kx|Δ′′ ky + ... (1) The corresponding term of Oπ is |Δ kx|Δ ky =|Δ′ kx|Δ ky +|Δ′′ kx|Δ ky + ... (2) Since each of Δ ′ ky, Δ ′′ ky, etc., is less than or equal to Δ ky, (1) is less than or equal to (2), and hence Oπ1 □ Oπ.

144 INFINITESIMAL ANALYSIS. Lemma 4.— If π0 is any partition of the interval || a b , and ε0 any positive number, then for any bounded function there exists a number δ0 such that for every partition π whose greatest Δ is less than δ0 Oπ0 + ε0 □ Oπ. Proof. We prove the lemma by showing that if π0 has N +1 partition points x0, x1, x2, . . . , x N , an effective choice of δ0 is δ0 = ε0 R·N , where R is the oscillation of the function on || a b . Of the intervals of π there are at most N−1 which contain as interior points, points of x0, x1, x2, . . . , xN . Denote the lengths of these intervals of π by Δ px, and denote by Δ qx the lengths of the intervals of π which contain as interior points no points of x0, x1, x2, . . . , xN . Then Oπ =∑ p |Δ px|·Δ py +∑ q |Δ qx|·Δ qy. If π′is a repartition of π0 obtained by introducing the points of π, then ∑ q |Δ qx|·Δ qy is a subset of the terms whose sum constitutes Oπ′. Hence, by Lemma 3, ∑ q

|Δ qx|·Δ qy □ Oπ′□ Oπ0. Since |Δ px|□ δ0 = ε0 R·N , it follows that ∑ p |Δ px|·Δ py □ ε0. Therefore Oπ0 + ε0 □ Oπ. Lemma 5.— If π is any partition, Oπ is the least upper bound of the expression S′ π−S″ π, where S′ π and S″ π may be any two values of Sπ corresponding to different choices of the ξ's. Proof. Without loss of generality we may assume every Δ kx positive. Then BSπ−BSπ = B|Sπ ′−Sπ ″|.

DEFINITE INTEGRALS. 145 But BSπ = B {n∑ k=1 f (ξk)·Δ kx } = n∑ k=1 { Bf (ξk) } Δ kx and BSπ = B {n∑ k=1 f (ξk)·Δ kx } = n∑ k=1 {Bf (ξk)}Δ kx. Therefore BSπ−BSπ = n∑ k=1 [Bf (ξk)−Bf (ξk)] Δ kx = n∑ k=1 Δ ky·Δ kx = Oπ. Therefore B(Sπ ′−Sπ ″) = Oπ. Theorem 125. A necessary and sufficient condition that a function f (x), defined, single- valued, and bounded on an interval || a b shall be integrable on || a b , is that the greatest lower bound of Oπ for this function shall be zero. Proof. We first show that if f (x) is integrable the lower bound of Oπ is zero. By hypoth- esis, ∫b a f (x)dx = L δ.=0 Sδ exists. By Theorem 27, Chapter IV, this implies that for every ε there exists a δε such that for every δ1 < δε and δ2 < δε |Sδ1−Sδ2|< ε. Hence, if π be a partition whose intervals Δ kx are all less than δε, we must have |S′ π−S″ π|< ε for every S′ π and S″ π. By Lemma 5 this implies that Oπ □ ε. But if for every ε there exists a π such that Oπ □ ε, then BOπ = 0. Secondly, we show that if the lower bound of Oπ is zero, Sδ converges to a single value, ∫b a f (x)dx,

146 INFINITESIMAL ANALYSIS. as δ approaches zero. Given any positive quantity ε there exists a partition πε, such that Oπε < ε 4 . By Lemma 4 there exists a δε such that for every π whose intervals are numerically less than δε Oπ □ Oπε + ε 4 < ε 2. Now let Sπ′ε and Sπ″ε be any two values of Sδε, and let π‴ ε be the partition composed of the points of both π′ ε and π″ ε. Then for any value of Sπ‴ε we have, by Lemma 2, □□Sπ′ε−Sπ‴ε □□□ Oπ′ε < ε 2, □□Sπ″ε−Sπ‴ε □□□ Oπ″ε < ε 2. Therefore □□Sπ′ε−Sπ″ε □□< ε. Hence for every ε we have a δε such that for every two values of Sδ, δ < δε, |Sπ′ε−Sπ″ε|< ε. By Theorem 27, this implies the existence of L δ.=0 Sδ. In case the definite integral does not exist it is sometimes desirable to use the upper and lower bounds of indeterminateness of Sδ as δ approaches zero. These are denoted respectively by the symbols ∫b a f (x)dx and ∫b a f (x)dx6 and are called the upper and lower definite integrals of f (x). They are both equal to ∫b a f (x)dx if and only if the latter integral exists. They are usually defined by the equations ∫b a f (x)dx = BSπ, where Sπ = n∑ k=1 {Bf (ξk)}Δ kx for all partitions of π, and ∫b a f (x)dx = BS π, 6For a more extended theory of these integrals, cf. Pierpont, page 337.

DEFINITE INTEGRALS. 147 where Sπ = n∑ k=1 {Bf (ξk)}Δ kx for all partitions of π. That ∫b a f (x)dx exists when the upper and lower integrals are equal is evident under this definition, because Oπ = Sπ−Sπ, and thus BOπ = 0 if and only if ∫b a f (x)dx = ∫b a f (x)dx. For every value of δ > 0 there is an infinite set of partitions π, for which the largest Δ kx is less than δ, and for each of these there is a value of Oπ. If Oδ stands for any such Oπ, then Oδ is a many-valued function of δ. Theorem 126. A necessary and sufficient condition that a function f (x), defined, single- valued, and bounded on an interval || a b , is integrable is that L δ.=0 Oδ = 0. Proof. The condition is necessary. By Theorem 125 the integrability of f (x) implies BOπ = 0. Hence for every ε there exists a partition π such that Oπ < ε. By Lemma 4 there exists a δε such that for every π′whose greatest Δ x is less than δε Oπ′< O π + ε < 2ε. Hence L δ.=0

Oδ = 0. The condition is sufficient. Since L δ.=0 Oδ = 0, and Oδ > 0, BOπ = 0. Hence the function is integrable by Theorem 125. Theorem 127. A necessary and sufficient condition that a function, defined, single-valued, and bounded on an interval || a b , shall be integrable on that interval is that for every pair of positive numbers σ and λ there exists a partition π such that the sum of the lengths of those intervals on which the oscillation of the function is greater than σ is less than λ.

148 INFINITESIMAL ANALYSIS. Proof. The condition is necessary. If for a given pair of positive numbers σ and λ there exists no π such as is required by the theorem, then Oπ > σ·λ for every π, which is contrary to the conclusion of Theorem 125 that BOπ = 0. The condition is sufficient. For a given positive ε choose σ and λ so that σ(b−a) < ε 2 and λ·R < ε 2, where R is the oscillation of the function on || a b . Let π be a partition such that the sum of the lengths of those intervals on which the oscillation of the function is greater than σ is less than λ. Then the sum of the terms of Oπ which occur on these intervals is less than λ·R, and the sum of the terms of Oπ on the remaining intervals is less than σ(b−a). Therefore Oπ < λ·R + σ(b−a) < ε. Hence BOπ = 0, whence by Theorem 125 the integral exists. Definition.— The content of a set of points [x] on an interval || a b is a number C[x] defined as follows: Let π be any partition of || a b , none of the partition points of which are points of [x], and Dπ the sum of the lengths of those intervals of π which contain points of [x] as interior points. Then BDπ = C[x]. An important special case is where C[x] = 0 . It is evident that if a set [x] has content zero, for every ε there exists a finite set of segments of lengths ε1, ε 2, ε 3, . . . , ε n which contain every point [x] and such that n∑ i=1 εi < ε.

DEFINITE INTEGRALS. 149 It is also evident that if the sets [x1] and [x2] are of content zero, then the set of all x1 and x2 is of content zero. 7 Theorem 128. A necessary and sufficient condition for the integrability of a function f (x) on an interval || a b is that for every σ > 0 the set of points [xσ] at which the oscillation of f (x) is greater than or equal to σ shall be of content zero. 8 Proof. If at every point of an interval || c d the oscillation of f (x) is less than σ, then about each point of || c d there is a segment upon which the oscillation is less than σ, and hence by Theorem 11, Chapter II, there is a partition of || c d upon each interval of which the oscillation of f (x) is less than σ. Now to prove the condition sufficient we observe that if the content of [xσ] is zero, there exists for every λ a partition πλ, such that the sum of the lengths of the intervals containing points of [xσ] is less than λ. Moreover we have just seen that the intervals which do not contain points on [xσ] can be repartitioned into intervals on which the oscillation is less than σ. Hence, by Theorem 127, the function is integrable. To prove the condition necessary we note that on every interval containing a point, xσ, the oscillation of f (x) is greater than or equal to σ. Hence, if C[xσ] > 0, the sum of the intervals upon which the oscillation is greater than or equal to σ is greater than C[xσ]. Definition.— A set of points is said to be numerable if it is capable of being set into one-to-one correspondence with the positive integral numbers. If a set [x] is numerable, it can always be indicated by the notation x1, x2, x3, . . . , xn, . . . , or {xn}, but if it is not numerable, the notation{xn}cannot be applied with the understanding that n is integral. Theorem 129. A perfect set of points is not numerably infinite. 9 Proof. Suppose the theorem not true. Then there exists a sequence of points {xn}con- taining every point of a

perfect set [x]. Let P1 be any point of [x], and a1 b1 a segment containing P1. Let xn1 be the first of {xn}within a1 b1. Since xn is a limit point of points of [x], there are points of the set other than P1 and xn1 on the segment a1 b1. Let P2 be such a point, and let a2 b2 be a segment within a1 b1 and containing P2 but neither P1 nor xn1. Let xn2 be the first point of {xn}within a2 b2. Proceeding in this manner we obtain a sequence of segments {ai bi}such that every segment lies within the preceding and such that every segment ai bi contains no point xni−k of the sequence {xn}. By the lemma on 7For further discussion of the notion content see Pierpont, Real Functions , Vol. I, p. 352, and Lebesgue, Le¸ cons sur l'Int´ egration. 8Compare the example on page 124. 9For definition of perfect set see page 31.

150 INFINITESIMAL ANALYSIS. page 31, Chapter II, there is a point P on every segment of this set. Since there are points of [x] on every segment ai bi, P is a limit point of the set [x]. Since [x] is a perfect set, P is a point of [x]. But if P were in the sequence {xn}, there would be only a finite number of points of [x] preceding P , whereas by the construction there is an infinitude of such points. Theorem 130. A numerably infinite set of sets of points each of content zero cannot contain every point of any interval. Proof. Let the set of sets be ordered into a sequence {[x]n}. We show that on every segment a b there is at least one point not of {[x]n}. Since [x]1 is of content zero, there is a segment a1 b1 contained in a b which contains no point of [x]1. Let [x]n1 be the first set of the sequence which contains a point of a1 b1. Since [x]n1 is of content zero, there is a segment a2 b2 contained in a1 b1 which contains no point of [x]n1. Continuing in this manner we obtain a sequence of segments a b , a1 b1, . . . , an bn . . . such that every segment lies within the preceding, and such that an bn contains no point of [x]1, . . . , [x]n. By the lemma on page 31 there is at least one point P on all these segments. Hence P is a point of a b and is not a point of any set of {[x]n}. Theorem 131. The points of discontinuity of an integrable function form at most a set consisting of a numerable set of sets, each of content zero. Proof. Let σ1, σ2, σ3, . . . be any set of numbers such that σn > σn+1, and L n .=∞ σn = 0. By Theorem 128 the set of points [xσn] at which the oscillation of f (x) is greater than or equal to σn+1 and less than σn is of content zero. Since the set of sets {[xσn]}includes all the points of discontinuity of f (x), this proves the theorem. Theorem 132. If a function f (x) is integrable on an interval || a b , then it is continuous at a set of points which is everywhere dense on || a b . Proof. If the theorem fails to hold, then there exists an interval || a b on which the function is discontinuous at every point. By Theorem 131 an integrable function is discontinuous at most on a numerably infinite set of sets each of content zero, and by Theorem 130 such sets of sets fail to contain every point of any interval. Theorem 133. If ∫X a f (x)dx = 0 for every X of || a b , then f (x) = 0 on a set of points everywhere dense on || a b , and for every σ > 0 the points where |f (x)|> σ form a set of content zero.

DEFINITE INTEGRALS. 151 Proof. At every point X where f (x) is continuous, according to the corollary of Theo- rem 119, d dX ∫X a f (x)dx = f (X) = 0 , since ∫X a f (x)dx is a constant. The points of continuity of f (x) are everywhere dense, according to Theorem 132. Hence the zero points of f (x) are everywhere dense. At a point of discontinuity the oscillation of f (x) is greater than or equal to |f (x)|. Hence the points where|f (x)|> σ form a set of content zero. Theorem 134. If ∫X a f (x)dx = ∫X a φ(x)dx

for every X of || a b , then f (x) = φ(x) on a set of points everywhere dense on || a b , and for every σ > 0 the points where |f (x)−φ(x)|> σ forms a set of content zero. Proof. Apply the theorem above to f (x)−φ(x). Theorem 135. If f (x) is integrable from a to b, then |f (x)|is integrable from a to b.10 Proof. Since 0 □ Oπ|f (x)|□ Oπf (x), it follows that B Oπf (x) = 0 implies B Oπ|f (x)|= 0, and hence the integrability of f (x) implies the integrability of |f (x)|. Theorem 136. If f (x) and φ(x) are both integrable on an interval || a b , then f (x)·φ(x) (1) is integrable on || a b ; and, provided there is a constant m > 0 such that |φ(x)|−m > 0 for x on || a b , then f (x)÷φ(x) (2) is integrable on || a b . Proof. Since f (x) and φ(x) are both integrable on || a b , it follows that for every pair of positive numbers σ and λ there is a partition π1 for f (x) and a partition π2 for φ(x) such that the sums of the lengths of the intervals on which the oscillations of f (x) and φ(x) respectively are greater than σ are less than λ. Let π be the partition consisting of the points of both π1 and π2. Then the sum of the intervals of π on which the oscillation of 10The converse theorem is not true; cf. example given on page 154.

152 INFINITESIMAL ANALYSIS. either f (x) or φ(x) is greater than σ is less than 2 λ. Let M be the greater of B|f (x)| and B|φ(x)|on || a b . Then on any interval of π on which the oscillations of f (x) and φ(x) are both less than σ the oscillation of f (x)·φ(x) is less than σM . Hence the sum of the intervals on which the oscillation of f (x)·φ(x) is greater than σM is less than 2 λ. Since σ and λ may be chosen so that 2 λ and σM shall be any pair of preassigned numbers, it follows by Theorem 127 that f (x)·φ(x) is integrable on || a b . In view of the argument above it is sufficient for the second theorem to prove that 1 φ(x) is integrable on || a b if φ(x) is integrable and |φ(x)|> m. Consider a partition π such that the sum of the intervals on which the oscillation of φ(x) is greater than σ is less than λ. Since □□□□ 1 φ(x1)−1 φ(x2) □□□□=|φ(x1)−φ(x2)| |φ(x1)|·|φ(x2)|, it follows that π is such that the sum of the intervals on which the oscillation of 1 φ(x) is greater than σ m2 is less than λ, and 1 φ(x) is integrable according to Theorem 127. A second proof may be made by comparing the integral oscillations of f (x) and φ(x) with those of the functions (1) and (2) and applying Theorem 125.11 Theorem 137. If f (x) is an integrable function on an interval || a b , and if φ(y) is a continuous function on an interval | | Bf Bf , where Bf and Bf are the lower and upper bounds respectively of f (x) on || a b , then φ{f (x)}is an integrable function of x on the interval || a b .12 Proof. By Theorem 48 there exists for every σ > 0 a δσ such that for |y1−y2|< δσ, |φ(y1)−φ(y2)|< σ. (1) Since f (x) is integrable on || a b it follows by Theorem 127 that for every positive number λ there is a partition π such that the sum of the intervals on which the oscillation of f (x) is greater than δσ is less than λ. But by (1) this means that the sum of the intervals on which the oscillation of φ{f (x)}is greater than σ is less than λ. This, by Theorem 127, proves that φ{f (x)}is integrable. 11Cf. Pierpont, Vol. I, pp. 346, 347, 348. 12This theorem is due to Du Bois Reymond . It cannot be modified so as to read “an integrable function of an integrable function is integrable.” Cf. E. H. Moore , Annals of Mathematics , new series, Vol. 2, 1901, p. 153.

Chapter 9 IMPROPER DEFINITE INTEGRALS. § 1 The Improper Definite Integral on a Finite Inter- val. If f (x) is infinite at one or more points of the interval || a b , then, whatever may be the other properties of the function, the definite integral of f (x) defined in Chapter VIII cannot exist on the interval || a b . Definition.— If ∫b x f

(x)dx exists for every x, a < x < b , and if 1 L x .=a ∫b x f (x)dx exists and is finite, f (x) being unbounded on every neighborhood of x = a, then this limit is the improper definite integral on the interval || a b . If f (x) is unbounded in every neighborhood of x = a, and also in every neighborhood of x = b, but bounded on some neighborhood of every other point of the interval || a b , we consider two intervals || a c and || c b where c is any point a < c < b . If the improper definite integral exists on || a c and also on || c b , then the sum of these integrals is the improper definite integral on || a b . 1We will understand throughout this chapter that in the expression L x .=a ∫b x f (x)dx x approaches a on the interval | | a b . 153

154 INFINITESIMAL ANALYSIS. This definition can obviously be extended to the case where the function is unbounded in the neighborhood of a finite number of points. Such points are then considered as partition points, dividing the interval || a b into a set of subintervals. If the improper definite integral exists on each of these intervals, their sum is the improper definite integral on || a b . Theorem 138. If ∫b x f (x)dx exists for every x, a < x < b , then a necessary and sufficient condition that L x .=a ∫b x f (x)dx shall exist and be finite is that for every ε there exists a Vε ∗(a) such that for every two values of x, x1 and x2, on the interval || a b and on Vε ∗(a) □□□□ ∫x2 x1 f (x)dx □□□□< ε. Proof. This theorem is a special case of Theorem 27, since, by Theorem 110, ∫x2 x1 f (x)dx = ∫b x1 f (x)dx− ∫b x2 f (x)dx. Theorem 139. If ∫b x f (x)dx exists for every x, a < x < b , and if L x .=a ∫b x |f (x)|dx is finite, then L x .=a ∫b x f (x)dx exists and is finite. 2 Proof. By the necessary condition of Theorem 138 there is a Vε ∗(a) corresponding to any preassigned ε such that for any two values of x, x1 and x2, which lie on the segment || a b and on Vε ∗(a) □□□□ ∫x2 x1 |f (x)|dx □□□□< ε. 2The first part of the hypothesis in this theorem is not redundant, as is shown by the following example. Let f (x) = x− 1 2 for positive rational values of x and f (x) =−x− 1 2 for positive irrational values of x. In this case L x .=0 ∫b x |f (x)|dx exists and is finite, while ∫b x f (x)dx does not exist for any value of x on the interval a b , and consequently L x .=a ∫b x f (x)dx has no meaning since the limitand does not exist.

IMPROPER DEFINITE INTEGRALS. 155 But, by Theorem 107, □□□□ ∫x2 x1 |f (x)|dx □□□□□ □□□□ ∫x2 x1 f (x)dx □□□□, since, by the hypothesis and Theorem 105, ∫x2 x1 f (x)dx exists. Hence, by the sufficient condition of Theorem 138, L x .=a ∫b x f (x)dx exists and is finite. Theorem 140. If ∫b x f (x)dx exists for every x on the segment a b , and if (x−a)kf (x) is bounded on V∗(a) for some value of k, 0 < k < 1, then L x .=a ∫b x f (x)dx exists and is finite. Proof. By hypothesis (x−a)k|f (x)|□ M , i.e., |f (x)|□ M (x−a)k , where M may be taken greater than one. The proof of the theorem consists in showing that for every ε there exists a δε such that if 0 < x 1−a < δ ε, 0 < x 2−a < δ ε, x1 < x 2, then □□□□ ∫x2 x1 f (x)dx □□□□< ε. By Theorems 105 and 113, □□□□ ∫x2 x1 f (x)dx □□□□<= ∫x2 x1 |f (x)|dx <= ∫x2 x1 M (x−a)k dx = M 1−k { (x2−a)1−k−(x1−a)1−k} . That the last term of this series of inequalities is infinitesimal, the reader may verify by choosing δε = (ε(1−k) M) 1 1−k . This theorem may also be proved as a corollary of Theorem 143.

156 INFINITESIMAL ANALYSIS. Corollary.— If f (x) is integrable on || x b for every x of || a b , and is of the same or lower order than 1 (x−a)k for some value of k, 0 < k < 1, then L x .= a ∫b x f (x)dx exists and is finite. Theorem 141. If for any positive

number m and for any k □ 1 there exists a V∗(a) on which f (x) does not change sign, and on which (x−a)kf (x) > m for every x, then L x .=a ∫b x f (x)dx cannot exist and be finite. Proof. (1) In case ∫b x f (x)dx fails to exist for some value of x between a and b, L x .=a ∫b x f (x)dx fails to exist because the limitand function does not exist. (2) If ∫b x f (x)dx exists for every value of x between a and b, we proceed as follows: Let δ < 1 be the length of a V∗(a) on which f (x) does not change sign, and on which (x−a)kf (x) > m, and let x2 be the extremity of this neighborhood, which is greater than a. Then|f (x)|> m (x−a)k > m (x2−a)k for every x on this neighborhood. Take x1 so that (x2−a)k = 2(x2−x1). Then □□□□ ∫x2 x1 f (x)dx □□□□> m (x2−a)k (x2−x1) = 1 2 m. Hence, by the necessary condition of Theorem 138, L x .=a ∫b x f (x)dx cannot exist and be finite.

IMPROPER DEFINITE INTEGRALS. 157 Theorem 142. If L x .=a ∫b x f (x)dx exists and is finite and if f (x) approaches infinity monotonically as x .= a on some V∗(a), then L x .=a (x−a)·f (x) = 0 , or in other words f (x) has an infinity of order lower than 1 x−a .3 Proof. By means of Theorem 138 it follows from the hypothesis that for every ε there exists a Vε ∗(a) within V∗(a) such that for every x1 and x2 on || a b , and also on Vε ∗(a), □□□□ ∫x2 x1 f (x)dx □□□□< ε. Let x2 be any point of such a neighborhood and let x1 be so chosen that x1−a = x2−x1. Since x1 and x2 are on V∗(a), f (x1) > f (x2). It follows from Theorem 116 that □□□□ ∫x2 x1 f (x)dx □□□□>|f (x2)|·(x2−x1). But f (x2)·(x2−x1) = 1 2 f (x2)·(x2−a). Hence for x = x2, |f (x)|·(x−a) < 2ε. 3 L x .=a (x−a)·f (x) = 0 is not a sufficient condition for the existence of L x .=a ∫b x f (x)dx, as is shown by the following example. Consider a set of points x1, x2, x3, . . . , xn, . . . such that xn−a = 2(xn+1−a), x1−a being unity. Define f (x1) = 1, f (x2) = 4 3 , f (x3) = 2 , . . . , f (xn) = 2n n+1 , ... Let the function be linear from f (x1) to f (x2), from f (x2) to f (x3), etc. Then □□□□ ∫x2 x1 f (x)dx □□□□> 1 2 , □□□□ ∫x3 x2 f (x)dx □□□□> 1 3 , etc. Since these integrals are all of the same sign, their sum for any given number of terms is greater than the sum of the corresponding number of terms in the harmonic series. Also (xn−a)·f (xn) = 2 n+1 , whence L x .=a (x−a)·f (x) = 0.

158 INFINITESIMAL ANALYSIS. Since ε is arbitrary, and since x2 is any point in V∗(a), it follows that L x .=a f (x)·(x−a) = 0 . Corollary.— If ∫b x f (x)dx exists for every x between a and b, and L x .=a ∫b x f (x)dx exists and is finite, and if f (x) is entirely positive or entirely negative, then zero is a value approached by (x−a)·f (x) as x approaches a. Proof. Consider the case when the function is entirely positive. Suppose zero is not a value approached. Then there exists a pair of positive numbers ε and δ such that for every x, x−a < δ, (x−a)·f (x) > ε. On the interval, | | a a + δ, consider the function ε x−a. Since ∫b x ε x−a dx is a non-oscillating function of x, it follows from Theorem 25 that L x .=a ∫b x ε x−a dx exists, and by Theorem 142 this limit must be infinite. Since |f (x)|> ε x−a on the neighborhood under consideration, it follows from Theorem 107 and Corollary 2; Theorem 40, that L x .=a ∫b x f (x)dx exists and is infinite, which is contrary to the hypothesis.

IMPROPER DEFINITE INTEGRALS. 159 Theorem 143 4. If (1) f1(x) and f2(x) are of the same rank of infinity at x = a, or if f1(x) is of lower order than f2(x), (2) ∫b x f1(x)dx and ∫b x f2(x)dx both exist for every x on the segment a b , (3) There is a neighborhood of x = a on which f2(x) does not change sign, (4) L x .=a ∫b a f2(x)dx is

finite, 5 then it follows that L x .=a ∫b x f1(x)dx exists and is finite. Proof. Since from the hypothesis L x .=a ∫b x f2(x)dx exists and is finite, we have by Theorem 138 that for every ε there exists a V∗ ε (a) such that for every x1 and x2 on segment a b and on V∗ ε (a) □□□□ ∫x2 x1 f2(x)dx □□□□< ε. Consider x1 and x2 on a neighborhood of x = a for which □□□f1(x) f2(x) □□□< M and for which f2(x) does not change sign. Then, by Theorem 113, □□□□ ∫x2 x1 f1(x)dx □□□□< M· □□□□ ∫x2 x1 f2(x)dx □□□□< M·ε. Since M·ε can be made small at will by making ε small, it follows by Theorem 138 that L x .=a ∫b x f1(x)dx exists and is finite. 4This is what Professor Moore in his lectures calls the relative convergence theorem. Theorems 143, 144, 151, 152 in this form are due to him. 5We notice that since under the hypothesis f2(x) does not change sign, L ∫b x f2(x)dx cannot fail to exist either finite or infinite, for it follows from this hypothesis that ∫b x f2(x)dx is a non- oscillating function of x and therefore, by Theorem 25 that the limit exists.

160 INFINITESIMAL ANALYSIS. An important special case of this theorem is when f1(x) is of the same or lower order of infinity than f2(x), i.e., L x .=a f1(x) f2(x) = K, a constant not zero. The reader should verify for himself that Theorem 140 is a corollary of Theorem 143. The other previous tests for the existence of the improper definite integral can all be reduced to special cases of Theorem 143. Cf., for example, the logarithmic test on page 410 of Pierpont. Theorem 144. If (1) f1(x) and f2(x) are of the same rank of infinity at x = a, or if f1(x) is of higher order than f2(x), (2) ∫b x f1(x)dx and ∫b x f2(x)dx both exist for every x on the segment || a b , (3) There is a neighborhood of x = a on which f1(x) does not change sign, (4) L x .=a ∫b x f2(x)dx is infinite (see note under Theorem 143), then L x .=a ∫b x f1(x)dx exists and is infinite or fails to exist. 6 Proof. This is a direct consequence of Theorem 143, since if L x .=a ∫b x f1(x)dx, which exists by the foot-note of Theorem 143, were finite, then L x .=a ∫b x f2(x)dx would exist and be finite. Theorem 145. If for a function f1(x) which does not change sign in the neighborhood of x = a there exists a monotonic function f2(x) infinite of the same rank as f1(x) as x approaches a, ∫b x f1(x)dx and ∫b x f2(x)dx both existing for every x on the segment a b , then a necessary condition that L x .=a ∫b x f1(x)dx shall exist and be finite is that L x .=a (x−a)·f1(x) = 0 . 6This is what Professor Moore calls the relative divergence theorem.

IMPROPER DEFINITE INTEGRALS. 161 Proof. By hypothesis L x .=a ∫b x f1(x)dx exists and is finite. Hence, by Theorem 143, L x .=a ∫b x f2(x)dx exists and is finite. Therefore, by Theorem 142, L x .=a (x−a)·f2(x) = 0 . Since □□□f1(x) f2(x) □□□is bounded as x approaches a, i.e.,|f1(x)|< M·|f2(x)|, we have (x−a)·|f1(x)|< M·(x−a)·|f2(x)|. But L x .=a M·(x−a)·|f2(x)|= 0. Therefore, by Corollary 4, Theorem 40, L x .=a (x−a)·|f1(x)|= 0, or by Corollary 2, Theorem 27, L x .=a (x−a)·f1(x) = 0 . § 2 The Definite Integral on an Infinite Interval. The integral over an infinite interval, viz., L x .=∞ ∫x a f (x)dx, has properties analogous to those of the improper definite integral on a finite interval discussed in the preceding section, and is likewise called an improper definite integral. The following theorems correspond to Theorems 138 to 145. Theorem 146. If ∫x a f (x)dx

162 INFINITESIMAL ANALYSIS. exists for every x, a < x , then a necessary and sufficient condition that L x .=∞ ∫x a f (x)dx exists and is finite, is that for every ε

there exists a Dε, such that for every two values of x, x1 and x2, each greater than Dε, □□□□ ∫x2 x1 f (x)dx □□□□< ε. Proof. The theorem is a direct consequence of Theorems 105 and 27. Theorem 147. If ∫x a f (x)dx exists for every x greater than a, and if L x .=∞ ∫x a |f (x)|dx is finite, 7 then L x .=∞ ∫x a f (x)dx exists and is finite. Proof. The proof is like that of Theorem 139. Theorem 148. If ∫x a f (x)dx exists for every x greater than a, and if (x−a)k·f (x) is bounded as x approaches infinity for some k, k > 1, then L x .=∞ ∫x a f (x)dx exists and is finite. Proof. If in the proof of Theorem 140 we write D1−k ε = ε(1−k) M instead of δ1−k ε = ε(1−k) M , and use Theorem 146 instead of 138, the proof of Theorem 140 will apply to Theorem 148. Theorem 149. If f (x) does not change sign for x greater than some fixed number D, and if for some positive number m and some number k □ 1, □□(x−a)k·f (x) □□> m for every x greater than D, then L x .=∞ ∫x a f (x)dx cannot exist and be finite. 7Note on page 154 shows that this hypothesis is not redundant.

IMPROPER DEFINITE INTEGRALS. 163 Proof. By making suitable changes in the proof of Theorem 141 so as to make x1 and x2 approach infinity instead of a, that proof applies to this theorem. Theorem 150. If L x .=∞ ∫x a f (x)dx exists and is finite, and if f (x) is monotonic for all values of x greater than some fixed number, then L x .=∞ (x−a)·f (x) = 0 . Proof. By making slight modifications of the proof of Theorem 142, that proof applies to this theorem. Corollary.— If ∫x a f (x)dx exists for every x greater than a, and L x .=∞ ∫x a f (x)dx exists and is finite, and if f (x) does not change sign for x greater than some fixed number, then zero is a value approached by (x−a)f (x) as x approaches∞. The proof is similar to that of the corollary of Theorem 142. Theorem 151. If (1) f1(x) and f2(x) are infinitesimals of the same rank as x approaches∞, or if f1(x) is of higher order than f2(x), (2) ∫x a f1(x)dx and ∫x a f2(x)dx both exist for every x, a < x , (3) f2(x) does not change sign for x greater than some fixed number, (4) L x .=∞ ∫x a f2(x)dx is finite, then it follows that L x .=∞ ∫x a f1(x)dx exists and is finite. 8 Proof. The proof is analogous to that of Theorem 143. 8See note under Theorem 143.

164 INFINITESIMAL ANALYSIS. Theorem 152. If (1) f1(x) and f2(x) are infinitesimals of the same rank as x approaches infinity, or if f1(x) is of lower order than f2(x), (2) ∫x a f1(x)dx and ∫x a f2(x)dx both exist for every x, a < x , (3) f1(x) does not change sign for x greater than some fixed number, (4) L x .=∞ ∫x a f2(x)dx is infinite, then L x .=∞ ∫x a f1(x)dx exists and is infinite or fails to exist. Proof like that of Theorem 144. Theorem 153. If for a function f1(x) which does not change sign in the neighborhood of x =∞there exists a monotonic function f2(x) such that f1(x) and f2(x) are infinitesimals of the same rank as x approaches infinity, ∫x a f1(x)dx and ∫x a f2(x)dx both existing for every x > a , then a necessary condition that L x .=∞ ∫x a f1(x)dx shall exist and be finite is that L x .=∞ (x−a)·f1(x) = 0 . The proof is like that of Theorem 145. § 3 Properties of the Simple Improper Definite Inte- gral. The following definition of the simple improper definite integral is equivalent in substance to that given on page 153, and in form is partly the definition of the general improper definite integral given on page 168. The definite integral of a function is said to exist properly at a point x1 or in the neighborhood of this point, on the interval || a b if there exists an interval on | | a1 b1 containing x1 as an interior point (or as an end point in case x1 = a or x1 = b) such that the proper definite integral of f (x) exists on

this interval. The integral is said to exist improperly at

IMPROPER DEFINITE INTEGRALS. 165 a point x1 on the interval || a b if f (x) has an infinite singularity at x1 and there exists an interval | | a1 b1 on || a b containing x1 as an interior point (or end point in case x1 = a or x1 = b) such that the improper definite integral exists on each of the intervals | | a1 x1 and | | x1 b1 . If on an interval || a b the definite integral exists properly at every point except a finite number of points, and exists improperly at each of these points, then the improper definite integral is said to exist simply on the interval || a b , or the simple improper definite integral is said to exist on the interval || a b . Let x1, x2, . . . , xn be the points of || a b at which the integral exists improperly. The simple improper definite integral on || a b is the sum of the improper definite integrals on the intervals || a x 1 , | | x1 x2 , . . . , | | xn−1 xn , || xn b. We denote the simple improper definite integral of f (x) on the interval || a b by S ∫b a f (x)dx. This symbol is used generically to include the proper as well as the improper definite integral. Theorem 154. If a < b < c , and if two of the three simple improper definite integrals S ∫b a f (x)dx, S ∫c b f (x)dx, and S ∫c a f (x)dx exist, then the third exists and S ∫b a f (x)dx + S ∫c b f (x)dx = S ∫c a f (x)dx. Proof. If b is a point at which the integral exists improperly, and if S ∫b a f (x)dx and S ∫c b f (x)dx both exist, then by the definition of S ∫c a f (x)dx the latter exists and is equal to the sum of the two former. If one of the two integrals, say S ∫b a f (x)dx,

166 INFINITESIMAL ANALYSIS. exists, and if S ∫c a f (x)dx exists, then S ∫c b f (x)dx exists since only in that case does S ∫c a f (x)dx exist. The equation S ∫b a f (x)dx + S ∫c b f (x)dx = S ∫c a f (x)dx likewise holds. If b is a point at which the integral exists properly, then the theorem follows from the above argument and the definition on page 164. Theorem 155. If S ∫b a f (x)dx exists, then S ∫a b f (x)dx exists and S ∫b a f (x)dx =− S ∫a b f (x)dx. Proof. In case the integral exists improperly only at one point of the interval, then the theorem is an immediate consequence of Theorem 108 and Corollary 1, Theorem 27. (If L x .=a f (x) = K, then L x .=a {−f (x)}=−K.) The theorem in the general case follows directly from this case and the definition of the simple improper definite integral. Theorem 156. If c is a constant and if the simple improper definite integral of f (x) exists on || a b , then the simple improper definite integral of c·f (x) exists on || a b and c S ∫b a f (x)dx = S ∫b a cf (x)dx. Proof. The theorem is a direct consequence of Theorems 111 and 34.

IMPROPER DEFINITE INTEGRALS. 167 Theorem 157. If the simple improper definite integrals of f1(x) and f2(x) both exist on || a b , then the simple improper definite integral of f1(x) + f2(x) and of f1(x)−f2(x) both exist and S ∫b a {f1(x)±f2(x)}dx = S ∫b a f1(x)dx± S ∫b a f2(x)dx. Proof. The theorem is a direct consequence of Theorems 112 and 34. Theorem 158. If the simple improper definite integrals of f1(x) and f2(x) both exist, and if f1(x) □ f2(x), then S ∫b a f1(x)dx □ S ∫b a f2(x)dx. Proof. The theorem is a direct consequence of Theorem 113 and Corollary 2, Theorem 40. Theorem 159. If S ∫b a f (x)dx exists, then S ∫x a f (x)dx is a continuous function of the limit of integration on the interval || a b . Proof. If x is a point at which the integral exists properly, the theorem is the same as 118. If x is a point at which the integral exists improperly, then the theorem follows from Theorems 138 and 27. Theorem 160. If S ∫b a f (x)dx exists, it does not follow that S ∫b a |f (x)|dx exists.

Proof. Let x1, x 2, x 3, . . . , x n, . . . be an infinite sequence of points on || 0 1 in the order indicated from 1 towards 0 such that ∫xn−1 xn dx x = 1 n .

168 INFINITESIMAL ANALYSIS. Consider a function f (x) defined as follows: f (x) = 1 x on x1 1, x3 x2, etc. f (x) =−1 x on | | x2 x1 , | | x4 x3 , etc. Obviously L x .=0 ∫1 x f (x)dx9 exists and is finite since the series 1 2−1 3 + 1 4 . . . is convergent, while L x .=0 ∫1 x |f (x)|dx is divergent since the harmonic series is divergent. § 4 A More General Improper Integral. The problem of defining and studying the properties of the improper integral when the set of points of singularity is infinite has been treated by many writers. 10 In this section we give a few properties of improper integrals as defined by Harnack and Moore. Denote by P0 any set of points of content zero on || a b , and by P the set of all points of || a b not points of P0. P and P0 are complementary subsets of || a b . Denote by I any finite set of non-overlapping intervals of || a b which contain no point of the set P0. The symbol m(I) stands for the sum of the lengths of the intervals of I. For the sake of brevity D will be used for |a−b|. The following conditions are assumed to be satisfied: 9That 0 is a limit point of the sequence of points is obvious since in case this sequence has a limit point greater than zero the proper definite integral of the function 1 x would fail to exist on some interval | | a b where 0 < a < b , which is impossible. 10A. Cauchy and B. Riemann studied the case of a finite number of singularities in papers which are to be found in these writers' collected works. The infinite case has been treated by A. Harnack , Mathematische Annalen , Vols. 21 and 24 (1883–84). O. H ¨older, Mathematische Annalen , Vol. 24 (1884). C. Jordan , Cours d'Analyse, Vol. 2 (1894, 2d ed.). O. Stolz , Grundz¨ uge der Differential- und Integralrechnung, Vol. 3. A. Schoenflies , Jahresbericht der Deutschen Mathematiker-Vereinigung , Vol. 8 (1900). V all´ee-Poussin, Liouville's Journal, Ser. 4, Vol. 8 (1892). E. H. Moore , Transactions of the American Mathematical Society , Vol. 2 (1901). J. Pierpont , Theory of Functions of Real Variables (1906).

IMPROPER DEFINITE INTEGRALS. 169 (a) The definite integral of f (x) exists properly at every point of P . The sum of the integrals of f (x) on the intervals of I is denoted by ∫b a I f (x)dx. (b) For every positive ε there exists a positive δε such that for any two sets, I′and I′′, of intervals none of which contain any point of P0 and for which |D−m(I′)|< δε and |D−m(I′′)|< δε, □□□□ ∫b a I′ f (x)dx− ∫b a I′′ f (x)dx □□□□< ε. It follows by Theorem 27 that L m(I) .=D ∫b a I f (x)dx exists and is finite. This limit is denoted by b ∫b a P 0 f (x)dx and is called the broad improper definite integral with respect to P0 of the function f (x) on the interval || a b . It is to be noticed that all the points of P0 need not be on || a b ; those which are not on || a b do not affect the existence of b ∫b a P 0 f (x)dx. Therefore if f (x) is improperly integrable on some sub-interval || a′b′of || a b , its integral may be denoted by b ∫b′ a′P0 f (x)dx. Theorem 161. If a < b < c and if of the integrals b ∫b a P 0 f (x)dx, b ∫c b P 0 f (x)dx, b ∫c a P 0 f (x)dx, either (a) b ∫b a P 0 f (x)dx and b ∫c b P 0 f (x)dx exist, or

170 INFINITESIMAL ANALYSIS. (b) b ∫c a P 0 f (x)dx exists, then all three integrals exist and b ∫b a P 0 f (x)dx + b ∫c b P 0 f (x)dx = b ∫c a P 0 f (x)dx. (1) Proof. Every set I of intervals on || a c may be regarded as composed of a set I on || a b and a set I on || b c , while, conversely, every pair of sets I and I constitute a set I. Hence ∫c

a I f (x)dx = ∫b a I f (x)dx + ∫c b I f (x)dx. (Note that both members of this equation are multiple-valued functions of m(I) and of m(I) and m(I)). The conclusion of our theorem follows in case (a) from Theorem 34. It remains to show that if b ∫c a P0 f (x)dx exists, then b ∫b a P0 f (x)dx and b ∫c b P0 f (x)dx exist, and in that case also equation (1) holds. Suppose that on some sequence of sets [I] one of the two expressions ∫b a I f (x)dx and ∫c b I f (x)dx, say ∫b a I f (x)dx, approaches two distinct values as m(I) approaches D. Since there is some sequence of sets of intervals{ I′ } on which ∫c b I f (x) approaches only one value, it follows that on the sequence of sets of intervals obtained by associating with each I an I′and with each I′an I′, ∫c a I f (x)dx approaches two distinct values as m(I) .= D, which is contrary to hypothesis. If ∫b a I f (x)dx approaches infinity, then clearly ∫c b I f (x)dx must approach infinity of the opposite sign. Hence, by the corollary of Theorem 51a, ∫c a I f (x)dx will approach both +∞ and−∞as m(I) .= D, which again contradicts the hypothesis that b ∫c a P 0 f (x)dx exists. The equality b ∫c a P 0 f (x)dx = b ∫b a P 0 f (x)dx + b ∫c b P 0 f (x)dx now follows from the identity of the limitands ∫c a I f (x)dx and ∫b a I f (x)dx + ∫c b I f (x)dx.

IMPROPER DEFINITE INTEGRALS. 171 Theorem 162. If b ∫b a P 0 f (x)dx exists, then b ∫a b P 0 f (x)dx exists and b ∫b a P 0 f (x)dx =− b ∫a b P 0 f (x)dx. Proof. By Theorem 108, for every I ∫b a I f (x)dx =− ∫a b I f (x)dx, whence b ∫b a P 0 f (x)dx =− b ∫a b P 0 f (x)dx. Theorem 163. If b ∫b a P 0 f (x)dx exists, then b ∫b a P 0 c·f (x)dx exists and b ∫b a P 0 c·f (x)dx = c· b ∫b a P 0 f (x)dx. Proof. This is a direct consequence of Theorems 111 and 34. Theorem 164. If b ∫b a P 0 f1(x)dx and b ∫b a P 0 f2(x)dx both exist, then b ∫b a P 0 (f1(x)±f2(x))dx exists and b ∫b a P 0 f1(x)dx± b ∫b a P 0 f2(x)dx = b ∫b a P 0 (f1(x)dx±f2(x))dx. Proof. This is a direct consequence of Theorems 112 and 34. Theorem 165. If f1(x) □ f2(x), then b ∫b a P 0 f1(x)dx □ b ∫b a P 0 f2(x)dx, provided these integrals exist. Proof. By Theorems 113 and 40. Theorem 166. If b ∫b a P 0 f1(x)dx and b ∫b a P 0 f2(x)dx both exist, b ∫b a P 0 f1(x)·f2(x)dx does not in general exist.

172 INFINITESIMAL ANALYSIS. Proof. Let f1(x) = f2(x) = 1√x . In this case the hypothesis of the theorem is verified but the product, 1 x , fails to be integrable on the interval || 0 1. Theorem 167. b ∫x a P 0 f (x)dx is a continuous function of x. Proof. If x is a point at which the integral exists properly, the continuity follows by Theorem 118. If x is a point of the set P0, then, by Theorem 26, we need to show that for every ε there is a δε, such that for every interval || a′b′containing x1 and of length less than δε, □□□□□b ∫b′ a′P0 f (x)dx □□□□□< ε. By definition there exists a δε such that for every I′and I″ for which|m(I′)−D|< δε and|m(I″)−D|< δε, □□□□ ∫b a I′ f (x)dx− ∫b a I″ f (x)dx □□□□< ε. Let || a′b′be an interval containing x1 such that |a′−b′|< δε 2 . Let I′be any set of intervals not containing any point of P0 and containing no point of || a′b′, and such that |m(I′)−D|< δε. Denote by I(a′b′) any set of non-overlapping intervals on || a′b′containing no point of P0, and let I″be the set of all intervals in I′and I(a′b′). Then |m(I″)−D|< δε and ∫b aI″ f (x)dx = ∫b aI′ f (x)dx + ∫b′ a′I (a′b′) f (x)dx and □□□□□ ∫b′ a′I (a′b′) f (x)dx □□□□□= □□□□ ∫b aI″ f (x)dx □□□□− □□□□ ∫b aI′ f (x)dx □□□□. Hence □□□□□b ∫b′ a′P0 f (x)dx □□□□□□ ε. Corollary.— For x1 any point on || a b L x .=x1 b ∫x x1 f (x)dx = 0.

IMPROPER DEFINITE INTEGRALS. 173 Theorem 168. If f (x) is integrable with respect to P0, and if P1 is a set of points of content zero, then f (x) is integrable with respect to the set P2 consisting of all points in P0 and in P1 and b ∫b a P 0 f (x)dx = b ∫b a P 2 f (x)dx. Proof. Obviously the set P2 is of content zero. Any set of intervals I not containing a point of P2 is also a set I not containing a point of P0. Hence any value approached by∫b a I f (x)dx as m(I) approaches D is a value approached by ∫b a I f (x)dx as m(I) approaches D. Hence b ∫b a P 2 f (x)dx exists and b ∫b a P 0 f (x)dx = b ∫b a P 2 f (x)dx. Theorem 169. If f1(x) is integrable with respect to P1 and f2(x) is integrable with respect to P2, then f1(x)±f2(x) is integrable with respect to the set, P3, of all points in P1 and P2 and b ∫b a P 1 f 1 (x)dx± b ∫b a P 2 f 2 (x)dx = b ∫b a P 3 (f1(x)±f2(x))dx. Proof. By Theorem 168 each of the functions f1(x) and f2(x) is integrable with respect to P3, and b ∫b a P 1 f1(x)dx = b ∫b a P 3 f1(x)dx, and b ∫b a P 2 f2(x)dx = b ∫b a P 3 f2(x)dx, and hence, by Theorem 164, f1(x)±f2(x) is integrable with respect to P3 and b ∫b a P 1 f1(x)dx± b ∫b a P 2 f 2 (x)dx = b ∫b a P 3 (f 1 (x)±f 2 (x))dx. The broad improper definite integral as here defined contains as a special case the proper definite integral, the integral in that case existing properly at every point of the interval || a b . It does not, however, contain as a special case the simple improper definite integral considered in §3. This may readily be shown by means of the function used on page 167 to show that the simple improper definite integral is not absolutely convergent. In the case of this function a sequence of sets of intervals Ia may be so chosen that ∫b a I a f (x)dx shall approach any value whatever as m(Ia) approaches D.

174 INFINITESIMAL ANALYSIS. An improper integral which includes both the simple and the broad improper integrals is obtained as follows: Every set I is to be such that if I′is its complementary set of segments on a b , then every segment of I′contains at least one point of P0. The limit of ∫b a I f (x)dx as m(I) approaches D, if existent, is called the narrow improper definite integral and is denoted by n ∫b a P 0 f (x)dx. It is evident that if the broad integral exists, then the narrow integral also exists. The narrow integral includes the simple improper definite integral of the preceding chapter. Hence it follows that the broad and the narrow integrals are not equivalent. 11 Theorems 161 to 167 hold of the narrow integral as well as of the broad integral. The proofs are identical with the above except that the sets I are limited as in the definition of the narrow integral. It may be shown by examples that Theorems 168 and 169 do not hold in the case of the narrow integral. To show that 168 does not hold consider the function defined in the proof of Theorem 160, where P0 consists of the point 0. Let P1 be the [xi] of that example. Then obviously the narrow integral n ∫1 0 P2 f (x)dx, where P2 contains all the points of P1 and P2, fails to exist. The same example shows that Theorem 169 does not hold of the narrow integral. § 5 Special Theorems on the Criteria of Existence of the Improper Definite Integral on a Finite Inter- val. The examples of this section are intended to give an idea of the possible singularities of improperly integrable functions, and to indicate the difficulty of obtaining more general criteria of the divergence or convergence of the simple improper integral than those given in§§1 and 2 of this chapter. Lemma.— For every function f1(x) which is unbounded in every neighborhood of x = a there is a function f2(x) which is infinitesimal as x approaches a, such that f1(x)·f2(x) is

unbounded in every neighborhood of x = a, and such that f2(x) x−a is monotonic increasing as x approaches a. Proof. Since f1(x) is unbounded in every neighborhood of x = a, it follows that for every point x1 of the segment a b there is a point x2 on the segment a x 1 such that |f1(x2)|> 2|f1(x1)|> 2M, 11The narrow integral is so called because it has fewer properties than the broad integral. It exists for a wider class of functions.

IMPROPER DEFINITE INTEGRALS. 175 and such that (x2−a) □ 1 2 (x1−a). Let x1, x2, x3, . . . , xn, . . . be a sequence of points dense only at a such that |f1(xn)|> 2|f1(xn−1)|> 2n−1·M, and such that |xn−a|□ 1 2|xn−1−a|. We define f2(x) as follows: f2(x) = 1 n on the points x1, x2, . . . , xn, . . . and f2(x) is linear between the points of the sequence x1, x2, . . . , xn, ... Then there are values of x on | | xn xn−1 such that |f1(x)|·f2(x) > 2 n n−1 ·M, whence f1(x)·f2(x) is unbounded in the neighborhood of a.12 Obviously f2(x) x−a is monotonic increasing as x approaches a. Theorem 170. For every function f1(x) which is unbounded in every neighborhood of x = a there exists a non-oscillating function f2(x) such that L x .=a f1(x) ∫b x f2(x)dx exists and is finite, while (x−a)·f1(x)·f2(x) is unbounded in the neighborhood of x = a. Proof. According to the lemma there exists a function f3(x) such that L x .=a f3(x) = 0 , while f3(x)·f1(x) is unbounded and the function f4(x) = f3(x) x−a 12In case L x=0 f1(x) =∞, f2(x) = 1√ f1(x) or f2(x) = 1 log f1(x) would satisfy the requirements of the lemma except that they need not make f2(x) x− a monotonic.

176 INFINITESIMAL ANALYSIS. is monotonic increasing as x approaches a. Since (x−a)f4(x)·f1(x) = f3(x)·f1(x), (x−a)·f4(x)·f1(x) is unbounded in the neighborhood of x = a. Let x1, . . . , x n, . . . be a sequence of points on || a b whose only limit point is a, such that f3(x)·f1(x) is unbounded on this set. In the sequence (x1−a)f4(x1), (x2−a)f4(x2), . . . , (xn−a)f4 (xn), (1) L n .=∞ (xn−a)f4(xn) = 0, since L x .=a (x−a)f4(x) = 0. Hence there is a value of n, n1, such that |(x1−a)f4(x1)|□ 2|(xn1−a)f4(xn1)|, and another value of n, n2 such that |(xn1−a)f4(xn1)|□ 2|(xn2−a)f4(xn2)|, etc., nm+1 being so chosen that |(xnm−a)f4(xnm)|□ 2|(xnm+1−a)f4(xnm+1)|. In this manner we select from the sequence (1) a set of terms forming the convergent series (x1−a)f4(x1) + (xn1−a)f4(xn1) + . . . + (xnm−a)f4(xnm) + ... (2) We then obtain a function f2(x) as follows: For the set of values of x xnm+1 < x □ xnm, f 2(x) = f4(xnm). Then (1) f2(x) is non-oscillating since f4(xnm) < f 4(xnm+1). (2) (x−a)f2(x)·f1(x) is unbounded on the set x1, xn1, xn2, . . . , x nm, . . . , since on this set f2(x) = f4(x). (3) L x .=a ∫b x f2(x)dx = ∞∑ m=1 (xnm−xnm+1)f4(xnm). But the terms of this series are numerically smaller than the corresponding terms of the convergent series (2). Hence L x .=a ∫b x f2(x)dx exists and is finite.

IMPROPER DEFINITE INTEGRALS. 177 Theorem 170 may be regarded as showing that L x .=a (x−a)f2(x) = 0 is a strong necessary condition that, under the hypothesis of Theorem 142, L x .=a ∫b x f2(x)dx shall exist and be finite. For, according to Theorem 170, it is impossible to modify the function (x−a) by any factor f1(x) which shall approach infinity so slowly that for every function f2(x) where L x .=a ∫b x f2(x)dx exists and is finite L x .=a (x−a)f1(x)·f2(x) = 0 . 13 Theorem 171. For every function f1(x) defined on the interval || a b there exists a function f2(x) such

that (1) f2(x) is continuous and does not change sign on a certain neighborhood of x .= a. (2) L x .=a ∫b x f2(x)dx exists and is finite. (3) For x on a certain set [x′] L x .=a f1(x′) f2(x′) = 0. Proof. Let x′ 1, x′ 2, . . . , x ′ n, . . . be a set of points of the interval || a b dense only at a. Let B1, B2, B3, . . . , B n, . . . be a set of numbers such that Bn·n|f1(x′ n)|□ 2·Bn+1(n + 1)|f1(x′ n+1)|. (n = 1, 2, 3, . . .) On the x axis lay off a set of segments [σn] such that σn is of length Bn and xn is its middle point. On the segments σn as bases construct isosceles triangles on the positive side of the x axis whose altitudes are n·|f1(x)|. The measures of areas of these triangles form a convergent series. Let f3(x) be any continuous, monotonic, unbounded function such that L x .=a ∫b x f3(x)dx exists and is finite. We then define f2(x) as the function represented by the following curve: 13See Pringsheim, Mathematische Annalen, Vol. 37, pp. 591–604 (1890).

178 INFINITESIMAL ANALYSIS. (1) Those parts of the boundaries of the isosceles triangles just described which lie above the curve defined by f3(x). (2) Those parts of the curve defined by f3(x) which lie outside the triangles or on their boundary. Obviously the function so defined has the properties specified in the theorem, the points x′ 1, x′ 2, . . . , x ′ n, . . . being the set [x′] specified by (3) of the theorem. Theorem 171 means that from the hypothesis that the improper definite integral of f (x) exists on || a b it is impossible to obtain any conclusion whatever as to the order of infinity or the rank of infinity of f (x) at x = a. This is what one would expect a priori , since the definite integral is a function of two parameters, while the necessary condition in terms of boundedness would be in terms of only one of these. § 6 Special Theorems on the Criteria of the Existence of the Improper Definite Integral on the Infinite Interval. Theorem 172. For every function f1(x) which is unbounded as x approaches∞there exists a non-oscillating function f2(x) such that L x .=∞ ∫x a f2(x)dx exists and is finite, while (x−a)f1(x)·f2(x) is unbounded as x approaches∞. Proof. Obviously the lemma of Theorem 170 can be stated so as to apply to the case where x approaches∞instead of a. If then in the proof of Theorem 161 the set of points x1 . . . x n . . . is so taken that L n .=∞ xn =∞ instead of a, the proof of Theorem 161 applies with the exception that f2(x) is non- oscillating decreasing instead of non-oscillating increasing. Theorem 173. For every function f1(x) defined on the interval | | a∞there exists a function f2(x) such that (1) f2(x) is continuous and does not change sign for x greater than a certain fixed number. (2) L x .=∞ ∫a x f2(x)dx exists and is finite.

IMPROPER DEFINITE INTEGRALS. 179 (3) For x on a certain set [x′] L x .=∞ f1(x′) f2(x′) = 0. Proof. Such a function f2(x) may be defined in a manner analogous to that of the proof of Theorem 171. The remarks as to the meaning of Theorems 170 and 171 apply with obvious modifica- tions to Theorems 172 and 173.

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